

CLEAN ENERGY  
TECHNOLOGY  
OBSERVATORY



Nuclear Power in the European Union

*Status Report on Technology Development,  
Trends, Value Chains and Markets*

2025

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#### The Joint Research Centre: EU Science Hub

<https://joint-research-centre.ec.europa.eu>

JRC144653  
EUR 40605

PDF ISBN 978-92-68-35380-6 ISSN 1831-9424 doi:10.2760/5077071 KJ-01-26-023-EN-N

Luxembourg: Publications Office of the European Union, 2026

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How to cite this report: Kopecky, S., Albora, G., Georgakaki, A., Mountraki, A., Piela, K. et al., *Clean Energy Technology Observatory: Nuclear Power in the European Union - 2025 Status Report on Technology Development, Trends, Value Chains and Markets*, Publications Office of the European Union, Luxembourg, 2026, <https://data.europa.eu/doi/10.2760/5077071>, JRC144653.

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## Abstract

Nuclear power plays a crucial role in Europe's energy landscape, providing nearly a quarter of the electricity produced in the region. With approximately 100 active nuclear power reactors Europe operates almost a quarter of the global power reactors. Meanwhile, China is advancing rapidly with around 30 ongoing projects, positioning itself to soon overtake France and, eventually, the US to hold the largest nuclear reactor fleet worldwide.

Like other areas of innovation, the market for nuclear reactors in Europe faces challenges due to persisting specific national regulator prerogatives and constraints, preventing the development of a fully integrated single market. Stakeholders are addressing these challenges, aiming to unlock the full potential of the European nuclear industry.

Europe boasts competitive technology and industrial capacities, even on a global scale. European enterprises handle most components of the nuclear supply chain, although the global export market for large-scale nuclear power plants remains dominated by Russia.

Looking to the future, the development of small modular and advanced modular reactors presents a promising opportunity for the European industry. However, domestic start-ups face intense competition from predominantly US enterprises, which appear to have secured earlier and more support from venture capitalists. In Europe, on the other hand, venture capital is scarcer and nuclear has traditionally been more dependent on public as opposed to private funding.

The nuclear sector is a significant source of employment in Europe, supporting approximately 500 000 direct and indirect jobs, even in Member States without their own nuclear power plants. However, the age structure of the workforce presents a challenge. Over the next two decades, the sector will require up to 250 000 new recruits across a wide range of STEM<sup>1</sup> professions, necessitating proactive strategies to attract and train the next generation.

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<sup>1</sup> Science, Technology, Engineering, or Mathematics

## Foreword on the Clean Energy Technology Observatory

The European Commission set up the Clean Energy Technology Observatory (CETO) in 2022 to help address the complexity and multi-faceted character of the transition to a climate-neutral society in Europe. The EU's ambitious energy and climate policies create a necessity to tackle the related challenges in a comprehensive manner, recognizing the important role for advanced technologies and innovation in the process.

CETO is a joint initiative of the European Commission Joint Research Centre (JRC), who run the observatory, and Directorate-Generals Research and Innovation (R&I) and Energy (ENER) on the policy side. Its overall objectives are to:

- monitor the EU research and innovation activities on clean energy technologies needed for the delivery of the European climate change mitigation actions
- assess the competitiveness of the EU clean energy sector and its positioning in the global energy market
- build on existing Commission studies, relevant information & knowledge in Commission services and agencies, and the Low Carbon Energy Observatory (2015-2020)
- publish update reports on the Strategic Energy Technology Plan ([SET-Plan](#)) Information System (SETIS) online platform

CETO provides a repository of techno- and socio-economic data on the most relevant technologies and their integration in the energy system. It targets in particular the status and outlook for innovative solutions as well as the sustainable market uptake of both mature and innovative technologies. The project serves as primary source of data for the Commission's annual progress reports on [competitiveness of clean energy technologies](#). It also supports the implementation and development of EU research and innovation policies.

The observatory produces a series of annual reports addressing the following themes:

- Clean Energy Technology Status, Value Chains and Market: covering advanced biofuels, batteries, bioenergy, carbon capture utilisation and storage, concentrated solar power and heat, geothermal heat and power, heat pumps, hydropower & pumped hydropower storage, novel electricity and heat storage technologies, ocean energy, photovoltaics, renewable fuels of non-biological origin (other), renewable hydrogen, solar fuels (direct) and wind (offshore and onshore).
- Clean Energy Technology System Integration: building-related technologies, digital infrastructure for smart energy system, industrial and district heat & cold management, standalone systems, transmission and distribution technologies, smart cities and innovative energy carriers and supply for transport.
- Foresight Analysis for Future Clean Energy Technologies using Weak Signal Analysis
- Clean Energy Outlooks: Analysis and Critical Review
- System Modelling for Clean Energy Technology Scenarios
- Overall Strategic Analysis of Clean Energy Technology Sector

More details are available on the [CETO web pages](#)

## **Acknowledgements**

The author is very grateful for the comments and contributions received from the following colleagues:

U. Engelmann, W. Janssens, V. Rondinella

The JRC.C.6 energy system modelling team for providing the *Global CETO 2°C scenario 2025* generated by the POLES-JRC model: B. Schade, K. Keramidas, F. Fosse, P. Dowling, P. Russ., J. Aycart

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## Executive summary

### EU objectives and challenges / policy context

Aligned with the European Union's climate neutrality objective and the REPowerEU strategy, the European Commission has underscored the importance of low-carbon energy sources in its February 2024 Communication entitled '*Securing Our Future: Europe's 2040 Climate Target and Path to Climate Neutrality by 2050 – Building a Sustainable, Just and Prosperous Society*' [1].

In January 2025, the European Commission adopted the Communication entitled '*A Competitiveness Compass for the EU*' [2], presenting a strategic roadmap to revitalise Europe's economic dynamism and foster sustainable growth. Central to this initiative is the objective of enhancing the European Union's global competitiveness by closing the innovation gap, accelerating the decarbonisation of the economy, and reducing strategic dependencies. It outlines a strategic shift towards clean and affordable energy, notably through the introduction of the Clean Industrial Deal [3]. This initiative aims to support the reduction of carbon emissions—particularly within energy-intensive sectors—by facilitating the adoption of low-carbon technologies. Complementing this effort, an Affordable Energy Action Plan [4] is developed to help lower energy prices and reduce overall energy costs. Furthermore, the EU strengthens its innovation ecosystem through a dedicated EU Start-up and Scale-up Strategy [5]. This strategy will focus on promoting innovation-friendly regulation, enhancing access to finance, accelerating market deployment and expansion, attracting and retaining top talent, and improving access to essential infrastructure, networks, and services. In May 2025, the European Commission presented the REPowerEU Roadmap [6] to gradually phase out Russian oil and gas and restrict imports of uranium, enriched uranium, and other nuclear materials derived from Russia, while simultaneously accelerating the transition to clean energy.

In early July, the European Commission proposed an amendment to the EU Climate Law [7] establishing a 2040 target to reduce net greenhouse gas emissions by 90% relative to 1990 levels. This ambitious objective aims to provide long-term certainty for investors and innovators, reinforce the global industrial leadership of EU enterprises, and enhance Europe's energy security. The EU climate target for 2040 [8] emphasises the role of nuclear energy alongside other clean technologies.

The Net-Zero Industry Act (NZIA) [9], adopted last year, will be instrumental in facilitating and accelerating the manufacturing of nuclear components within the EU and could be a powerful legislative tool to strengthen and revitalise the nuclear supply chain development in the coming years. The Nuclear Illustrative Programme (PINIC) [10] adopted by the Commission in June 2025 provides an up-to-date, comprehensive, fact-based overview of nuclear energy investment needs across the EU, underlining the role of nuclear to '*supply clean power, suitable for low-carbon baseload electricity, also enhancing system integration and providing flexibility, facilitating further roll-out of other clean technologies*'.

As announced in the Affordable Energy Action Plan, a Communication on the SMR strategy is expected to be adopted in 2026.

More than 20% of the electricity production in Europe, amounting to over 600 TWh annually, is generated by 100 nuclear reactors. To achieve climate neutrality by 2050, the demand for electricity from low-carbon technologies is expected to rise, along with the need to supply process heat for industries generated by low-carbon energy sources. Current nuclear reactors primarily produce electricity, and their applications in providing process heat are limited. However, with the



development of Small Modular Reactors (SMRs) and Advanced Modular Reactors (AMRs), there is hope that the need for heat for industrial processes can also be addressed.

AMRs are still in the technical developmental phase, and clear financial and regulatory frameworks have yet to be established for SMRs. Additionally, the lack of new nuclear builds over the past decades has negatively impacted the nuclear supply chain across Europe.

### **Technology status**

The technology status in the nuclear domain varies significantly.

Large Generation III+ nuclear power plants are already in operation, but the situation for SMRs differs.

SMR designs using light water as a coolant share high readiness levels due to their similarity to large plants. The lower Technology Readiness Levels (TRL) of SMRs are a results of design concepts and choices that have not undergone the licensing process. For more advanced modular reactor concepts, TRLs vary widely, requiring extensive validation of novel designs, fuels, and safety systems before commercial readiness.

Numerous studies on nuclear energy production costs show a wide range of estimates for new plant builds, but generally, these costs are competitive with those of many renewable energy sources. There is much more consensus on the cost of the long-term operation of existing reactors, marking it as one of the most cost-effective sources of low-carbon electricity.

### **Investment and funding**

Research and innovation in nuclear safety are primarily funded by public sources, with funding exceeding EUR 1 billion annually and steadily increasing over the past five years. In terms of venture capital, Europe lags the US both in total funding amounts and timing. While early-stage funding is growing in Europe, the US is shifting its focus to later-stage financing.

### **Value chain**

The nuclear industry plays a significant role in the economy by providing approximately 500 000 jobs [11], both in energy production and throughout the nuclear supply chain. Its total economic impact is estimated at around EUR 250 billion annually. Of this amount, approximately EUR 190 billion is directly associated with energy production and the associated supply chain.

### **Sustainability**

Nuclear energy performs well compared to other low-carbon energy sources across several indicators, including greenhouse gas emissions, land use, water consumption, and pollution. However, public discussions often focus on the impacts of radiation impacts, potential consequences of accidents and waste management.

The average annual public radiation exposure from nuclear energy-based electricity is ten thousand times lower than the dose from natural background radiation. The *‘Technical assessment of nuclear energy with respect to the ‘do no significant harm’ criteria of Regulation (EU) 2020/852’* [12] finds fatality rates from severe accidents are notably lower than those from fossil-fuel plants. It finds the overall human health impact from radiological and non-radiological factors across the whole nuclear chain is of the same order as that of other low-carbon energy sources.

## EU positioning and global competitiveness

The nuclear market has been strongly nationally focused for a long time, and although supply chains are now global, some traces of the past still linger. The European nuclear industry appears competitive in the global marketplace. By omitting the import of raw materials, particularly natural uranium as the fuel base, the nuclear industry has created a trade surplus of approximately EUR 1 billion annually in recent years.

Europe's reliance on imported natural uranium is managed by the Euratom Supply Agency, which aims to diversify supply sources and establish secure supply chains to ensure consistent and reliable access while reducing dependence on a single supplier.

## SWOT analysis

**Table 1** CETO SWOT analysis for the competitiveness of nuclear fission technology

|                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Strengths</b> <ul style="list-style-type: none"><li>– Established technology</li><li>– Industrial capacities in Europe</li><li>– Dispatchable energy</li><li>– Contribution to security of supply</li></ul>                                                            | <b>Weaknesses</b> <ul style="list-style-type: none"><li>– Not fully integrated single market due to differences in national licensing requirements</li><li>– Workforce needs</li><li>– Almost all natural uranium is imported</li></ul> |
| <b>Opportunities</b> <ul style="list-style-type: none"><li>– Increased demand for low-carbon energy</li><li>– Development of SMR and AMR</li><li>– Increased electricity demand for datacentres and AI gigafactories</li><li>– Ensure technological sovereignty</li></ul> | <b>Threats</b> <ul style="list-style-type: none"><li>– Poor management of construction costs</li><li>– Social acceptance</li></ul>                                                                                                      |

Source: JRC 2025

# **1. Introduction**

## **1.1 Scope and context**

This report on nuclear power is part of the annual series of reports from the Clean Energy Technology Observatory (CETO). This report is the first CETO report on this topic. It provides an overview of the current state of nuclear power including large scale nuclear power plants, small modular and advanced modular reactors, and some of the aspects required for the deployment of these technologies. The back end of the fuel cycle will not be addressed in detail in this edition, neither will be Generation IV reactor designs.

Nuclear power can play a crucial role for achieving the climate neutrality objective and the REPowerEU strategy. The EU has set ambitious targets for carbon neutrality, and nuclear power is expected to contribute significantly to these efforts.

The report is organised into five main chapters. Chapter 2 examines the state of the art and future developments of nuclear fission technology, focusing on advancements in technology readiness, energy capacity, costs, and research funding. Chapter 3 focuses on the value chain analysis, covering economic contributions, sustainability, and the role of EU companies in the market. Chapter 4 provides an overview of the EU's global position and competitiveness in the nuclear industry, analysing market status, trade dynamics, and resource efficiency. Chapter 5 concludes the report by synthesizing key findings and highlighting strategic opportunities and challenges.

### **1.1. Methodology and data sources**

The present report follows the general structure of all CETO technology reports and is divided into four sections with several indicators aiming to present and evaluate the EU fission technology along its value chain:

- Technology State of the art and future developments and trends;
- Value chain analysis;
- EU position and global competitiveness.

The report uses the following information sources:

- Eurostat data;
- Existing studies and reviews published by the European Commission and international organisations;
- Information from EU-funded research projects;
- EU and international databases;
- EU trade data, trade reports, market research reports and others;
- JRC own review and data compilation;
- Stakeholders' input.

## 2. Technology status and development trends

Approximately one quarter of the final energy consumed in the European Union comes from electricity, and approximately two-thirds of the EU's electricity is generated from low-carbon sources [13]. In 2024 the EU and Brazil were the only major electricity markets generating more than 50% of their electricity from low-carbon sources, with the EU's contribution exceeding 70%. [14]

In terms of electricity production, nuclear power plants accounted for around 23.4% total electricity generated in 2024 [15]. France had the highest share of nuclear power in its electricity mix, at 64.1%, followed by Slovakia at 57.8% and Hungary at 43.3% [16]. In 12 of EU Member States are 101 reactors operating [17]. These reactors have a combined capacity of approximately 98 gigawatts electric (GWe). Notably, over half of the EU's nuclear electricity is produced in France. Most the EU's nuclear reactors are light water reactors, with 92 Pressurised Water Reactors (PWRs)<sup>2</sup> and 7 Boiling Water Reactors (BWRs) in operation. Additionally, two heavy water moderated reactors of the CANDU type, a design of a pressurised heavy water reactors (PWRH), are operating in Romania.

**Table 2:** Number of nuclear power plants and electricity capacity status 31.12.2024

|             | PWR    |        | BWR    |       | PWRH   |       | total  |        |
|-------------|--------|--------|--------|-------|--------|-------|--------|--------|
|             | number | MWe    | number | MWe   | number | MWe   | number | MWe    |
| Belgium     | 5      | 3 908  |        |       |        |       | 5      | 3 908  |
| Bulgaria    | 2      | 2 006  |        |       |        |       | 2      | 2 006  |
| Czech       | 6      | 3 934  |        |       |        |       | 6      | 3 934  |
| Finland     | 3      | 2 614  | 2      | 1 780 |        |       | 5      | 4 394  |
| France      | 57     | 63 000 |        |       |        |       | 57     | 63 000 |
| Hungary     | 4      | 1 916  |        |       |        |       | 4      | 1 916  |
| Netherlands | 1      | 482    |        |       |        |       | 1      | 482    |
| Romania     |        |        |        |       | 2      | 1 380 | 2      | 1 380  |
| Slovakia    | 5      | 2 308  |        |       |        |       | 5      | 2 308  |
| Slovenia    | 1      | 688    |        |       |        |       | 1      | 688    |
| Spain       | 6      | 6 059  | 1      | 1 064 |        |       | 7      | 7 123  |
| Sweden      | 2      | 2 211  | 4      | 4 733 |        |       | 6      | 6 944  |
| Total EU    | 92     | 89 126 | 7      | 15 77 | 2      | 1 380 | 101    | 98 083 |

Source: IAEA, 2025 [17]

The European Union's nuclear electricity production has experienced mixed trends over the past decade, with some key producers reducing their nuclear output. In April of 2023, Germany shut down its last operating nuclear power plants. In Belgium, Doel-3 and Tihange-2 were closed in 2022 and 2023, respectively, with a combined capacity of 2 000 MW. Between 2016 and 2020, Sweden closed the reactors Oskarshamn-1, Oskarshamn-2, Ringhals-1, and Ringhals-2, with a combined capacity of approximately 2 800 MW.

France has maintained its nuclear capacity despite closing Fessenheim 1 and 2 in 2019 and 2020, which had a combined capacity of approximately 1 700 MW. This loss was offset by the equivalent

<sup>2</sup> The 91 PWRs includes Western-type PWRs and Russian-type VVERs.

capacity of the new reactor Flamanville-3, which was connected to the grid in 2024 [18]. Finland increased its production capacity when Olkiluoto-3, with a capacity of 1 600 MW, was connected to the grid in 2023 [19]. In Slovakia Mochovce-3, a VVER-440 reactor, connected the grid in 2023 [20], and at Mochovce-4, a reactor of the same type, began hot testing in early 2025 [21], with grid connection is expected late 2025 or early 2026.

These developments align with the forecasts of the Euratom Treaty's nuclear illustrative programme (PINC) of 2017 [22]. If nuclear reactors are not granted lifespan extensions, nearly all existing units are expected to cease operations before 2050. According to the latest PINC report of 2025 [23], if lifespan extensions are assumed, the capacity of current reactors is projected to decrease from approximately 98 GWe today to around 50 GWe by 2050. Adding new reactors, which could contribute roughly 60 GWe, would potentially bring the total installed capacity to approximately 109 GWe by 2050—an 11% increase compared to current levels. This capacity may temporarily exceed 109 GWe if new reactors come online earlier than planned. By 2050, these reactors are anticipated to supply about 12% of the European Union's electricity generation. In addition to traditional large-scale reactors, the scenario may be supplemented with SMRs. The European SMR Industrial Alliance is working on setting out a strategic plan to achieve the first SMRs in commercial operation in the early years of the next decade. In 2023, in the preparatory phase of the European SMRs Industrial Alliance, a preliminary evaluation by the sector organisations resulted in projections of SMR capacity ranging from 17 GWe to 53 GWe by 2050 [24]. Such projections are consistent with other more recent reports [25] [26].

Nuclear capacities anticipated in this 'base case' scenario will require investments of around EUR 241 billion in present value terms. New-build large-scale reactors account for EUR 205 billion of these investments, while lifetime extensions account for EUR 36 billion [23]. Thus, while the actual achievement of lifespan extensions can significantly impact the installed capacity by 2050, they represent only a minor share of the total investment needed. Additional investments are needed in Small Modular Reactors (SMRs), Advanced Modular Reactors (AMRs), and microreactors for the longer term.

Currently, six European Union member states - Bulgaria, the Czech Republic, Hungary, Poland, Romania, and Sweden - are actively planning the construction or finalisation of new nuclear power plants. Additionally, the Czech Republic, Poland, Slovakia, Slovenia, Sweden, Netherlands, Romania and France have proposed new large nuclear power plant projects, the maturity of the plans vary.

Furthermore, several member states are exploring the deployment of SMRs, not only to generate electricity but also to provide energy for non-electric applications such as process heat or district heating. (Presently, several "conventional" nuclear power plants serve both as a source of electricity and as a provider of direct heat). Some of the observed changes in the stance of governments vis-à-vis nuclear energy have been triggered by energy security considerations after the start of the Russian invasion in Ukraine.

Some of these geopolitical considerations were also considered in the report of the competitiveness of Europe, i.e. the Draghi report [27]. Following the Draghi report to maximise the potential of nuclear energy, the EU should focus on three key areas. Firstly, extending the lifespan of existing reactors, provided that a robust safety case can be demonstrated, can help to maintain a low-carbon energy supply and ensure that the EU's climate goals are met. Secondly, building new reactors using established technologies can help to increase the EU's nuclear capacity, but it is essential that costs and schedules are kept under control to make nuclear power a competitive option.

Thirdly, the EU should prioritise the development and deployment of next-generation nuclear reactors, such as SMRs. These innovative technologies have the potential to play a significant role in integrated energy systems with high renewable penetration, providing flexible generation and enhancing overall energy system efficiency. While the deployment of these new reactors is expected to have a medium-term impact, with most plans in Europe scheduled to begin in the next decade, they can help build a competitive technological supply chain in the EU and ensure the EU remains a leader in the nuclear industry. By pursuing these three areas of action, the EU can unlock the full potential of nuclear energy and ensure a secure, reliable, and low-carbon energy supply.

## **2.1. Long term operation**

The long-term operation (LTO) of nuclear reactors is crucial for decarbonisation strategies, as 444 reactors worldwide providing a significant source of low-carbon electricity. As the nuclear industry evolves, addressing technological challenges is essential to ensure the safe and efficient operation of reactors beyond their initial design life.

Recent analysis has revealed that refurbishing existing nuclear facilities to extend their operational lifespan can significantly lower costs compared to constructing new ones [28]. This approach, which leverages existing infrastructure, yields cost savings even when the facilities are not operating at full capacity, a scenario that may become more common as renewable energy sources become increasingly prevalent. In fact, the costs associated with these refurbishments are often lower than those of investing in other low-carbon energy technologies, even when those technologies have long lifespans, such as hydroelectric power plants.

LTO is supported by a strong scientific foundation and extensive operating experience, enabling ageing management, maintenance, refurbishment, and replacement of structures, systems, and components (SSCs) where needed. Most components of a nuclear power plant can be replaced during regular maintenance outages or LTO refurbishments, which also present opportunities for safety upgrades and modernisations, such as replacing analogue equipment with digital systems to enhance monitoring and control, or implementing power uprates to boost electricity generation capacity.

However, challenges related to the ageing and obsolescence of SSCs need attention. As reactors age, the risk of component failure increases, and the availability of spare parts and qualified suppliers can become a concern. Integrating new technologies, such as digital systems and advanced materials, also presents challenges in terms of compatibility and validation.

To overcome these challenges, the nuclear industry is developing advanced ageing management programmes that involve continuous monitoring and assessment of reactor components. These programmes also use advanced materials and technologies to mitigate the effects of ageing. For instance, advanced non-destructive testing techniques help detect defects in components, while coatings and surface treatments enhance corrosion resistance and reduce maintenance needs.

Already in 2008, a study published by the electric power research institute (EPRI) found that about 20% of identifiable plant equipment was obsolete [29], meaning it was no longer commercially available for purchase. A report published by FORATOM in 2020 [30] suggests this figure is now considerably higher. European nuclear fleet operators are particularly concerned about the obsolescence of SSCs and their ability to replace them. Licensees operating large fleets of standardised reactors and participating in international cooperative efforts face fewer concerns. The JRC supply chain initiative [31] has highlighted the obsolescence and supply chain issues faced by several licensees from various countries.

To support technological development, ongoing research in advanced materials, digitalisation, and artificial intelligence is essential. International cooperation and knowledge sharing among nuclear operators, regulators, and research institutions help develop best practices and standards for LTO. Efforts are also being made to foster international collaboration for qualifying industrial equipment from sectors with similar safety needs, such as oil, chemical, and aeronautics, for use in the nuclear industry.

## **2.2. Power Plants – Generation III+**

Most of European Union Member States planning to build large power plants are expected to deploy Light Water Reactors (LWRs). The European Utility Requirements (EUR) organisation, composed of European utility companies, aims to create common European positions on design requirements for future Nuclear Power Plants (NPPs). This organisation was founded to promote harmonisation of requirements toward standard bodies across Europe. Its main goal is to ensure NPP designs comply with safety and technical requirements set by national nuclear authorities and to harmonise and make design requirements more transparent, thus facilitating the entry of NPP vendors and designs into the European market. A report submitted to the Directorate-General for Energy states that the EUR organisation's criteria align with the latest recommendations from the Western European Nuclear Regulators' Association (WENRA) and the International Atomic Energy Agency (IAEA) [32]. Currently focused on LWRs, the organisation plans to extend its reach to SMRs in the future.

For a reactor design to be considered for assessment, it must have the backing of both utility companies and vendor organisations. Consequently, most reactor types involved in European projects have engaged with the EUR organisation to some extent. According to the World Nuclear Association, there are currently 12 vendors of large-scale reactors, one of which is from the EU (Framatome). The following reactor designs have been accepted as meeting the EUR standards [33]

- AP1000 developed by Toshiba-Westinghouse (1100 MWe, the version certified in USA)
- AES-92 (1 000 MWe PWR), the VVER design developed by AtomEnergoprojekt
- VVER-TOI designed by AtomEnergoprojekt
- “Standard EPR” (1 600 MWe) developed by Areva
- EU-APWR by Mitsubishi. (1 700 MWe PWR) This design had been in a tendering process for a Czech power plant, but has subsequently withdrawn from the process. Presently, no project in Europe is considering this design.
- EU-APR (1 400 MWe) by Korean Hydro & Nuclear Power (certified in 2017). However, the Korean company KHNP has withdrawn from all but two projects in Europe, citing strategic reasons.
- EU HPR1000 (Hualong One 1 200 MWe) by China General Nuclear Power Corporation, has been tendering for projects in Europe, and has been excluded in some countries citing geopolitical considerations.

## **2.2.1. Status of ongoing new-build projects in Europe:**

### **2.2.1.1. Bulgaria:**

In Bulgaria, a project to develop two reactors, each with a capacity of 1200 MW, at Kozloduy is ongoing. The Westinghouse AP1000 design has been selected for this project. The call for tenders for the construction began in 2024. The goal is to have both units operational by 2035 and 2037, respectively.

### **2.2.1.2. Czech Republic**

The Czech Republic has been planning new nuclear power plants for a significant period. The current plans involve building at least one, with binding tenders for up to four new units. In April 2021, the Czech government announced the exclusion of Russian and Chinese companies from the tendering process.

The four units being considered include two in Temelín and two in Dukovany, with a binding offer for Dukovany 5. In June 2024, the Czech government selected the South Korean APR1000 design from KHNP, at a cost of approximately USD 9 billion per unit, and began contract negotiations. However, in May 2025, a Czech court blocked the electricity firm CEZ from signing a contract worth at least USD 18 billion with South Korea's KHNP for a new nuclear power plant, pending the resolution of a complaint from a losing bidder, France's EDF. In June 2025 hours after the Supreme Administrative Court annulled a regional court's injunction, the contract was signed.

### **2.2.1.3. Hungary**

The Paks II nuclear power plant project in Hungary began to take shape by the intergovernmental agreement of January 2014 between Hungary and Russia on cooperation in peaceful uses of nuclear energy. In accordance with that agreement, Russia would grant a state loan to Hungary, enabling it to finance the development of two new power reactors, while Hungary would entrust, by means of a direct award, the construction of the reactors to the Russian company Nizhny Novgorod Engineering.

The planned units, each with a capacity of up to 1 200 MWe, were expected to cost around EUR 12 billion and start operating in 2025/2026. In December 2014, MVM Paks II, Hungarian State-owned company intended to become, free of charge, the owner and operator of the two new reactors signed three implementation agreements with NIAEP-ASE of Nizhny Novgorod, which outlined the design, procurement, and construction parameters for the new units.

Hungarian-Russian cooperation regarding the Paks-II project was intended to also cover operation and maintenance support, fuel supply, and the handling and storage of used nuclear fuel.

The fuel supply contract between MVM Paks II and a supplier from the Russian, Rosatom Group was initially opposed by the Euratom Supply Agency, backed by the European Commission, as its clauses tended to establish excessive dependence on a single external supplier, and had to be amended to cover supply from the same non-EU supplier only for the first 10 years of operation.



Thereafter, the project progressed, and in January 2022, MVM submitted an application to build the containment building for the first unit at Paks II. The Hungarian Atomic Energy Authority (HAEA) issued a construction license for the two VVER-1200 PWRs at Paks II in August 2022, with Rosatom as the contractor. However, in January 2023, Energy Minister Csaba Lantos announced a two-year delay in the construction of Paks II, pushing the completion date to 2032. Despite this setback, the project continued to advance, and in November 2024, the HAEA approved the preliminary safety report required for the first concrete pouring. Although technical problems in the construction pit have caused a postponement, a permit to commence the pouring of the first concrete early 2026 has been received in autumn of 2025.

In September 2025 the European Court of Justice has annulled a European Commission decision approving state aid from Hungary for the construction of two new nuclear reactors at the Paks II power station [34]. The impact of this court decision on the project is unclear at the time of writing.

#### **2.2.1.4. Poland**

In 2011, the country established a regulatory framework with the passage of the Nuclear Energy Law. This was followed by a bilateral agreement with the USA in 2019, paving the way for further cooperation. In 2021, the U.S. Trade & Development Agency provided a grant to support front-end engineering and design studies for a potential nuclear power plant. Several companies, including EDF, Korea Hydro & Nuclear Power, and Westinghouse, submitted offers to build nuclear power plants in Poland.

In October 2022, Westinghouse was chosen to build the country's first nuclear plant in the Pomerania province. An agreement was signed between Westinghouse and Polskie Elektrownie Jądrowe (PEJ) in December 2022, outlining the next steps for the project. The project has continued to progress, with the signing of an Engineering Development Agreement in April 2025. This agreement sets the framework for crucial preparatory efforts, including site development and regulatory documentation, needed to secure permits and begin construction. The partners expect the EPC contract to be signed late 2025.

Additionally, a separate project is planned for the Pątnów site in central Poland, where energy company ZE PAK has signed a letter of intent with PGE and Korea Hydro & Nuclear Power to build a nuclear plant based on APR1400 technology. However, the status of this project might be uncertain due to both political support issues and technical concerns, especially given Korea Hydro & Nuclear Power's recent withdrawal from most European projects.

Poland also plans to construct an HTGR demonstrator (HTGR-POLA) on the NCBJ site in Świerk. This project aims to facilitate later deployment, particularly for providing process heat in energy-intensive industries.

ORLEN Synthos Green Energy (OSGE), Synthos Green Energy (SGE), and GE Vernova Hitachi Nuclear Energy (GVH), led the BWRX-300 SMR initiative in Europe. They jointly drive the development, licensing, and deployment of the technology. OSGE is actively advancing SMR projects within Poland, while SGE and GVH are leading efforts across Europe. The project intends to contribute to the decarbonisation of the EU energy system by providing energy for district and industrial heat, H<sub>2</sub> production, and data centres.

### **2.2.1.5. Romania**

As early as 2002, the Romanian government expressed its interest in completing the two additional CANDU reactors at the Cernavoda site, a project that began in the 1980s but was halted after the fall of the Ceaușescu regime. Following several restructurings and changes in financial partners, an EPC contract was signed in November 2024 for the completion of Units 3 and 4, with a delivery date set for the early 2030s.

RoPower Nuclear S.A., established on the 27th of September 2022, has two principal shareholders holding equal shares: SN Nuclearelectrica SA (the national nuclear power operator, majority-owned by the State) and Nova Power&Gas SRL, which owns the site considered for the development of a Small Modular Reactor (SMR) Nuclear Power Plant (NPP) using US NuScale VOYGR™ Power Modules.

### **2.2.1.6. Sweden**

In November 2023, the government announced a roadmap aimed at enhancing nuclear energy production. This plan aims for the construction of two large-scale nuclear reactors by 2035 and developing the equivalent of 10 new reactors, including small modular reactors, by 2045. This initiative is supported by newly approved legislation that removes previous restrictions, allowing for more than ten reactors and enabling construction beyond currently existing nuclear sites. Since then, the government has introduced several bills to create the financial framework to support the development of new nuclear capacity.

Vattenfall has been progressing its plans for expanding nuclear capacity near the Ringhals plant. Already in 2020, the company began a pilot study to evaluate the construction of at least two Small Modular Reactors (SMRs) adjacent to the plant. By November 2023, Vattenfall applied for planning permission to develop new nuclear generating capacity on the nearby Värö Peninsula, aiming for deployment in the early 2030s. In August 2025, the company confirmed its decision to utilise SMRs for 1 500 MW of new capacity at this site, shortlisting GE Vernova Hitachi's BWRX-300 and the Rolls-Royce SMR as potential technologies.

### **2.2.1.7. The Netherlands**

In 2024, the Dutch government announced its intention [35] to build two new large reactors in addition to the existing Siemens-built 485 MWe Borssele PWR, which will remain operational. Preparation work for the first two reactors and site selection is ongoing. Initially, EdF, Westinghouse, and KHNP were considered as tenderers, but KHNP has officially withdrawn. At a later stage, more reactors, possibly including Small Modular Reactors (SMRs), will be considered. The Dutch government resigned in June 2025, and elections have taken place on 29 October 2025. At the time of writing the impact of elections results on the plans for deployment of nuclear power is not known.

The Netherlands has an active nuclear supply chain, particularly with fuel supplier Urenco, and a developing SMR landscape. ULC has proposed constructing a Rolls-Royce SMR, Thorizon is developing a molten salt reactor, and Allseas plans to construct a pebble bed HTGR for maritime applications.

### 2.2.2. Construction costs

The construction of new nuclear power plants in Western Europe and the US has been plagued by significant cost overruns and delays, which must be addressed to ensure the viability of future projects. A review of recent builds from around the world reveals a striking disparity in costs between Western countries and their Asian counterparts [36]. While labour costs undoubtedly contribute to this disparity, they do not provide a complete explanation for the differences observed.

**Table 3:** Comparison of building cost of nuclear power plants

| Type    | Country | Unit        | Start of construction | Planned duration | Real duration | Estimated cost | Realised costs |
|---------|---------|-------------|-----------------------|------------------|---------------|----------------|----------------|
|         |         |             |                       | years            | years         | (USD/kWe)      | (USD/kWe)      |
| AP1000  | China   | Sanmen      | 2009                  | 5                | 9             | 2 044          | 3 145          |
|         | US      | Vogtle      | 2013                  | 4                | 10            | 4 300          | 8 600          |
| APR1400 | Korea   | Shin Kori   | 2012                  | 5                | 8/10          | 1 828          | 2 410          |
| EPR     | Finland | Olkiluoto   | 2005                  | 5                | 17            | 2 020          | 5 723          |
|         | France  | Flamanville | 2007                  | 5                | 17            | 1 886          | 8 620          |
|         | China   | Taishan     | 2009                  | 4.5              | 9             | 1 960          | 3 222          |

Source: NEA, 2020 [36]

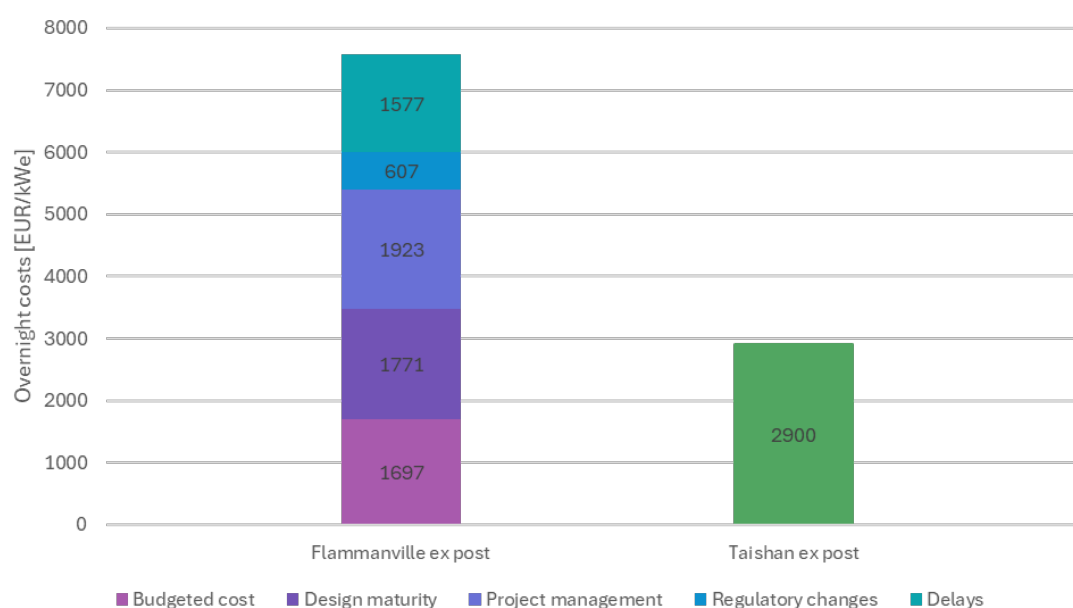
A more detailed analysis of the cost structures of Asian and Western builds reveals that project overruns are a significant factor in the increased financial cost of nuclear power plant construction in the West. The escalation of indirect costs can be attributed to various factors, including changes to designs during the construction phase, whether driven by evolving regulatory requirements or the challenges inherent in first-of-a-kind designs [37]. Furthermore, weaknesses in the supply chain and project management have also been identified as significant contributors to project overruns and increased building costs [38].

The experience of recent European and US nuclear power projects offers valuable lessons, and a closer examination of the European Pressurised Reactor (EPR) projects in Europe can provide some insights. The EPR was conceived in the 1990s by a French-German partnership (Framatome/Areva and Siemens) to combine the safety and cost-efficiency features of France's N4 and Germany's Konvoi reactors. Although a design report was approved in 1997, progress stalled due to declining public support for nuclear energy in both countries. In 2003, Areva and Siemens secured a EUR 3 billion contract to build the first EPR at Finland's Olkiluoto, with construction commencing in 2005 despite incomplete design work and an unestablished supply chain. Similarly, EDF began constructing a second EPR at Flamanville, France, in 2007, with less than 40% of the detailed design finalised. These projects subsequently faced severe delays and cost overruns due to immature designs, regulatory differences between Finland and France, and over 4,500 design modifications.

Another aspect, highlighted in an analysis of the Flamanville [39] construction, is the long absence of new builds in France, and in the rest of Europe. In fact, no new reactor had been built for approximately 2 decades, resulting in a loss of people, knowledge and capabilities across the supply chain and even with regulators. This has affected not only project management and technical expertise but also the skills of industrial component manufacturers, welders, and other players, leading to a shortage of qualified personnel and a lack of technical monitoring. The supply chain has been particularly impacted, with many companies experiencing a significant reduction in their nuclear-related workload, leading to a decline in their expertise and capabilities.

For the US projects, the conclusions drawn are very similar to the European cases. The recent US projects started [40] with incomplete designs, particularly regarding key component specifications. In one case, the inexperience of the project management team further complicated matters. These teams had to establish new supply chains and work with companies that had limited experience in the nuclear field. Additionally, the reliance on new construction techniques made these issues even more problematic.

**Figure 1:** Key drivers of Flammanville cost overruns compared to Taishan realised costs



Source: NEA, 2020 [36], derived on data from [39]

In contrast, the Taishan EPRs in China leveraged EDF's experience [39] and the simultaneous construction of two units on the same site allowed for shared resources, continuous feedback, and a "spare parts warehouse" to address unforeseen problems. The configuration of the construction site, with ample space for companies' premises, workshops, and parts storage, also contributed to the project's efficiency. The on-site presence of the project team and significant design and engineering resources from the outset enabled swift decision-making. Additionally, China's two-decade experience in constructing nuclear power plants, having created a stable regulatory framework, having invested in its domestic supply chain and its ability to mobilise large numbers of skilled workers helped to drive the project's success [41].

Allocating the risk of cost overruns and delays appropriately between the project developer and the lead EPC contractor may provide an important lever to incentivise efficient construction. Key questions are how to allocate cost overruns and potential cost savings [42] between the two parties. The deal between the US government and Westinghouse announced in late 2025 for the construction of AP1000 reactors in the US includes contingent claims by the US government [43]. The optimal structuring of such contingent claims to incentivise efficient construction, considering contract incompleteness and default risk of both parties, seems under-researched by the academic community.

## 2.3. Small modular reactors

### 2.3.1. History and Context

Small Modular Reactors are compact nuclear reactors with a capacity of less than 300 Megawatts electric (MWe) and are designed in modular units. SMRs have a long history, especially in naval applications. Approximately 100 nuclear reactors power 86 submarines and aircraft carriers today, providing electricity, heat, and propulsion [44]. The U.S. Navy, for instance, has been refining SMR technology for over 75 years, beginning with the USS Nautilus, the first nuclear-powered submarine, launched in 1955. Many other countries, including the UK, France, Russia, and China, have developed their nuclear fleets, showcasing extensive experience with modular reactors.

The main reasons they gained a lot of attention in recent years are their key advantages over conventional large-scale nuclear power plants.

- Lower upfront investment costs:  
SMRs require a lower upfront capital investment due to their smaller size. It is expected that they will achieve cost efficiency through design simplification, factory fabrication, and potential regulatory harmonisation, making their electricity costs competitive with traditional reactors.
- Quicker deployment:  
Traditional nuclear plants can take many years to become operational due to site-specific designs and significant on-site labour. The aim is for SMRs to be primarily manufactured in factories, minimising on-site labour, which could reduce deployment timelines.
- Siting flexibility and land efficiency  
Their compact size and modular design provide greater siting flexibility compared to traditional reactors. SMRs might be integrated into conventional industrial sites to supply process heat or produce industrial gases, such as hydrogen, effectively utilising land.

Several different SMR technologies are envisaged, ranging from technologies already employed in large nuclear reactors, to more innovative designs, e.g. Generation IV nuclear reactors. Therefore, there is a significant variation in technology readiness levels, with more mature light water concepts, gas-cooled, and sodium-cooled concepts which have been built before, and thus at the higher end of the range, while some of the more advanced concepts, such as liquid lead or molten salt reactors, are at a lower level.

- Boiling water reactor (BWR)  
Smaller, modular versions of traditional reactors using water as a coolant and neutron moderator. The technology is well established, with licensing, fabrication, and supply chain close to demand of existing power plants.

- Pressurised water reactor (PWR)

Smaller, modular versions of traditional reactors using water as a coolant and neutron moderator. The technology is well established, with licensing, fabrication, and supply chain close to the demand of existing power plants.

- Integral pressurised water reactor (iPWR)

This is a self-contained design integrating all necessary components into a single vessel, offering enhanced safety and simplified operation. The technology is similar to classical PWRs, but due to integrated designs, some developments of modelling tools for the licensing process, as well as new fabrication, and supply chains will have to be established.

- High-temperature gas reactor (HTGR)

These use helium gas as a coolant and graphite as a moderator, suitable for industrial high-temperature applications. Demonstrators for this technology are being operated in China (HTR-10 and HTR-PM, 2 HTR-PM600 under construction) and in Japan (HTTR); other designs in the west are still in the design phase (e.g. Xe-100 or KRONOS in the US). In Europe, although the technology's original home, the supply chain for fuel, materials and components needs to be revived.

- Liquid metal-cooled reactor (LMR)

Liquid metals such as sodium and lead are utilised as coolants, offering potential for both electricity generation and nuclear propulsion. While these are often grouped together, they represent two distinct concepts currently under consideration.

For sodium-cooled fast reactors, there is significant experience in Europe, particularly with the operation of reactors like Phénix and Superphénix. Although the technology is well-established, further research and advancements in materials may be necessary.

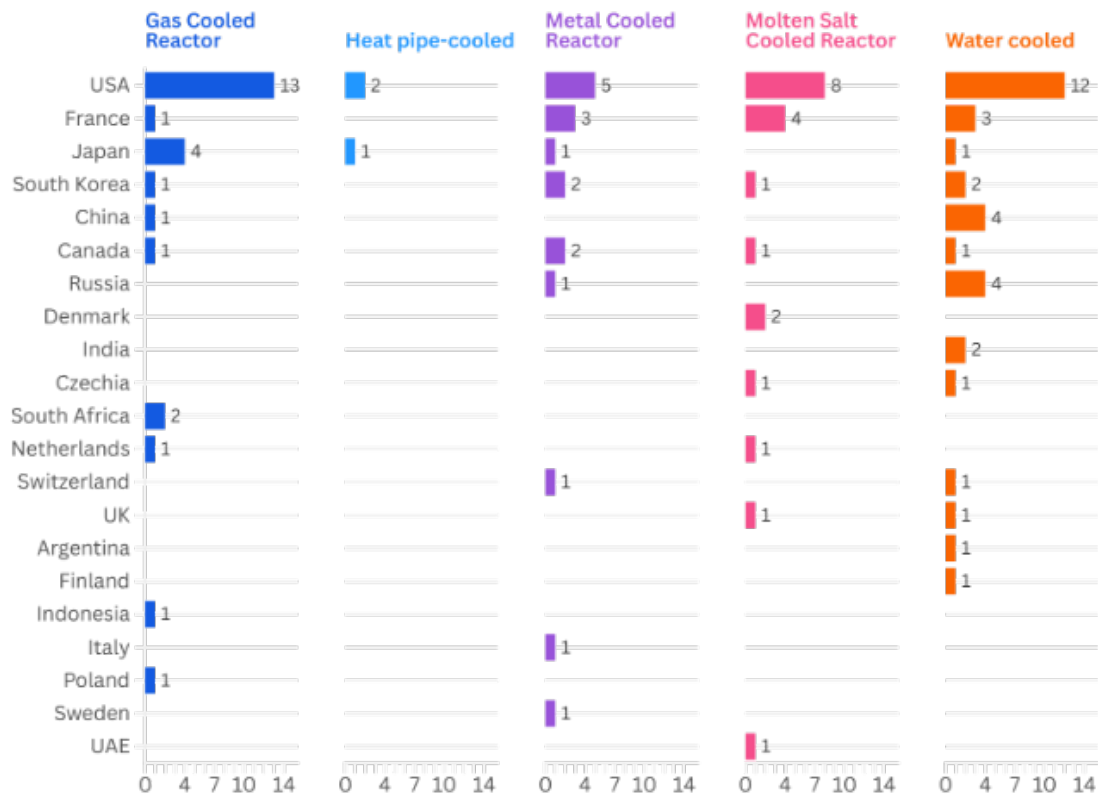
In the case of liquid lead-cooled reactors, the technological situation in Western Europe is similar, with ongoing material research required. However, unlike sodium-cooled reactors, there is little operational experience outside of Russia.

For both concepts, the nuclear fuel typically used is Mixed Oxide (MOX) fuel, which contains a higher concentration of plutonium than in Light Water Reactors (LWRs). Currently, production capacities are not sufficient for large-scale deployment, and the available plutonium stock is very limited.

- Molten salt reactor (MSR)

With the use of liquid fuel, generally a mixture of fluoride or chloride salts serving as both fuel and coolant, these reactors are ideally suited for operations at relatively high temperatures. Yet, several crucial challenges must be addressed before MSRs can be deployed on a wide scale. These challenges mainly concern safety, fuel management (including online processing if necessary), and material compatibility, which involve the establishment of standards for design safety, managing the transportation of fuel salt, building dependable supply chains for specialised components, and conducting in-depth analyses of unique accident scenarios. Additionally, important research areas include understanding and managing the long-term behaviour of fuel salt under irradiation and ensuring effective radionuclide retention.

**Figure 2:** Number of SMR designs according to technology and geographical location of company headquarters



Source: JRC 2025, using data from [45] [46] [47] [48] [49]

## 2.3.2. Status of commercial SMR's worldwide

As can be seen in Figure 2, a large number of SMR projects are ongoing worldwide. They engage a variety of technologies and designs. Since addressing all of them would exceed the scope of this report, we will only briefly examine projects that have already reached the construction phase.

### 2.3.2.1. SMR in operation

#### 2.3.2.1.1. Russian Federation:

The Akademik Lomonosov floating nuclear power plant is equipped with two KLT-40S reactors, each providing 35 MW(e) of power. This facility started commercial operations in May 2020 and underwent its first refuelling in 2023. It supplies heat and electricity to the town of Pevek in the Chukotka region.

#### 2.3.2.1.2. China:

The HTR-PM, a high temperature gas-cooled reactor demonstration plant at the Shidaowan site commenced commercial operations in December 2023. It operates two reactors connected to a single power turbine, collectively generating 200 MWe at full capacity.

### **2.3.2.2. SMRs in the Construction Phase**

#### **2.3.2.2.1. Argentina**

The CAREM-25, a pressurised water reactor, a prototype is currently under construction. In 2024, construction was stopped, and its status is not clear. However, the previously announced grid connection for 2028 seems rather unlikely [50].

#### **2.3.2.2.2. Canada**

Ontario will be the site of the world's first GE Hitachi BWRX-300 small modular reactor at the Darlington nuclear plant, aiming to address a projected 75% increase in electricity demand by 2050 [51]. Construction has started in 2025 and is set to finish by 2029, operations are estimated to begin in 2030 [52].

#### **2.3.2.2.3. China:**

The ACP100 demonstration plant, known as Linglong One, has been under construction in Changjiang, Hainan province since 2021. In August 2023, the reactor core was installed. [53] This multi-purpose pressurised water reactor is planned to produce 125 MW(e) by 2026. It is the world's first commercial small modular reactor that has passed the International Atomic Energy Agency general safety review. Cold functional testing was completed in October 2025, and it is expected to be connected to the grid by the end of 2026 [54].

There is also the construction of 2 HTR-PM600 plants in the Shandong Province. One HTR-PM600 plant consists of 6 HTR-PM modules producing in total 600 MWe.

#### **2.3.2.2.4. Russian Federation:**

In June 2021, construction began on the BREST-OD-300, a lead-cooled fast reactor, in Seversk, Siberia [55]. The installation of the 165-tonne steel reactor base plate in 2024 marked the beginning of significant installation work. This pilot demonstration reactor aims to be operational by 2026, with a planned output of 300 MWe.

#### **2.3.2.2.5. United States of America:**

In July 2024, ground work began on Kairos Power's Hermes Low-Power Demonstration Reactor in Oak Ridge, Tennessee [56] and in May 2025 first nuclear safety-related concrete was poured. This is the first Generation IV reactor to obtain a construction license from the U.S. Nuclear Regulatory Commission and the first non-light-water reactor approved in the USA in over 50 years. The Hermes reactor is designed to demonstrate affordable heat production using a fluoride salt-cooled, high-temperature design, but it will not generate electricity.

In July 2025, BWX Technologies, Inc. announced [57] that it would begin fabricating the reactor core for the Pele microreactor, a transportable nuclear reactor that utilises HALEU TRISO fuel. The first module is set to be tested at the Idaho National Laboratory (INL) site, with funding provided by the U.S. Department of Defence.

In summer of 2025 the U.S. Department of Energy's launched the 'Reactor Pilot Program' (RPP) [58] and the 'Fuel Line Pilot Program' (FLPP) [59], aiming to accelerate the development and deployment of advanced nuclear reactors by building a robust domestic supply chain. These initiatives prioritise enabling at least three new reactor designs to achieve criticality by mid-2026 outside federal research labs. The RPP identifies and partners with private companies to demonstrate cutting-edge

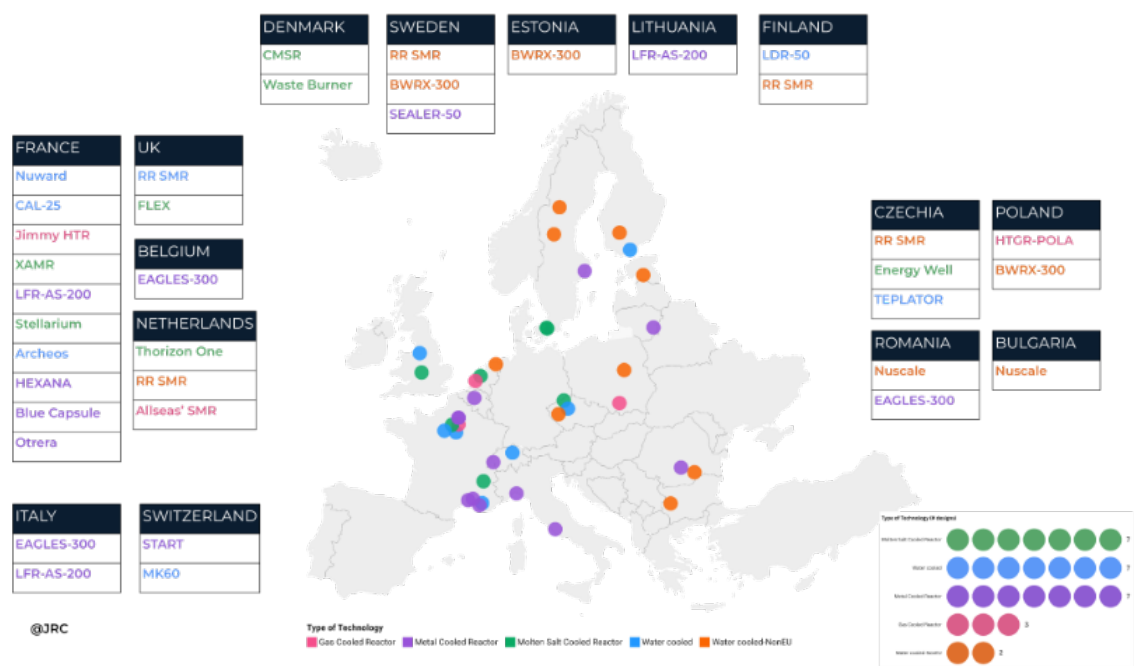


reactor technologies, while the FLPP focuses on scaling production of advanced nuclear fuels—such as TRISO fuel utilizing high-assay low-enriched uranium—to support these reactors. Together, the programmes streamline licensing processes, foster commercialisation, reduce dependence on foreign supply chains, and expand US. nuclear energy capacity.

### 2.3.3. Projects in the European Industrial Alliance on SMR

The European Union is advancing its agenda to decarbonise energy systems through the development and deployment of Small Modular Reactors (SMRs). This endeavour is being led by the European Industrial Alliance on SMRs [60], launched in February 2024, which aims to support deployment of SMR projects across Europe by the early 2030s. The Alliance's objectives include identifying promising SMR technologies, strengthening the nuclear supply chain, addressing investment barriers, engaging potential industrial users, and pinpointing research and skills needs. The initiative seeks to create a comprehensive framework supporting the widespread adoption of SMR technologies. The European Industrial Alliance on SMRs adopted its first Strategic Action Plan at their second General Assembly on 1st September 2025 [61]. The plan provides a comprehensive and detailed overview of its planned activities over the next five years to facilitate the development, demonstration, and deployment of SMRs in Europe by the early 2030s. The timely deployment of SMRs is crucial for maintaining the competitiveness of European industry, driving the energy transition towards a carbon-neutral future by 2050, and enhancing the EU's strategic autonomy in the energy sector.

Figure 3: SMR designs in Europe



Source: JRC, 2025, using data from [45] [46] [47] [48] [49]

Globally and within Europe, the SMR landscape is very lively, with new approaches regularly emerging and some startups advancing or disappearing. Given this dynamic environment, providing a comprehensive overview of all SMR projects and actors is beyond the scope of this report. Furthermore, a number of documents [45] [49] [62] [63] provided by various actors offer

information on many of these concepts. Here, we focus on describing nine projects [64] selected in the first phase of proposals from the Industrial Alliance according to their Terms of Reference [60]. These projects represent Europe's multiple strategies for SMR development, each emphasizing distinct reactor technologies for specific roles and applications. In 2025, the Alliance decided that all projects would undergo regular checks to ensure they still comply with the criteria set out in the Terms of Reference. The inclusion of these projects for more detailed discussion in this report should not be interpreted as a judgment on their relative importance or significance. Rather, it reflects an opportunity to showcase specific aspects of the diverse and dynamic landscape of SMR development in Europe.

The technologies used are:

- PWR: CityHeat (Calogena, Steady Energy), Nuward (EDF), NuScale VOYGR™ SMR (RoPower Nuclear S.A), Project Quantum (Last Energy), Rolls-Royce SMR (Rolls-Royce SMR Ltd),
- BWR: European BWRX-300 SMR (OSGE)
- LFR: EAGLES (Ansaldo, ENEA, RATEN, SCK-CEN), European LFR AS Project (newcleo)
- MSR: Thorizon One (Thorizon)

It should be noted that currently, neither SFRs nor HTRs have been selected yet, although development and demonstration for both technologies continue to be performed in the EU. A second call for PWGs is planned for early 2026.

### **2.3.3.1. CityHeat Project**

Led by Calogena<sup>3</sup> and Steady Energy<sup>4</sup>, this project focuses on using small modular reactors (SMRs) for urban heating solutions, aiming to deliver sustainable heating options and reduce carbon footprints in city environments [65]. Both companies are European startups. Calogena is a French company founded by former nuclear industry executives, while Steady Energy is a spin-out of VTT, Technical Research Centre of Finland.

Both companies are developing reactor designs primarily for district heating applications, targeting water temperatures around 120°C. Steady Energy aims for a thermal power output of 50 MW, while Calogena targets 30 MW. The design philosophies for both reactors are quite similar: they utilise a PWR core, enclosed in a double shell with natural convection for core cooling, located inside a pool. The primary circuit is fully enclosed within the pressure vessel and uses 'clean' water, which means boron is excluded for reactivity control. Due to lower operating temperatures than those of conventional PWRs used for electricity production, the operating pressures are well below 1 MPa, resulting in reduced pressure vessel wall thickness and less stringent engineering constraints.

The reactors are intended to use conventional low enriched uranium fuels, similar to those used in traditional PWRs, with a refuelling cycle of approximately 2 years. Due to the low energy density and robust design, significant fission gas release from the nuclear fuels is not expected even under accidental conditions. Additionally, the decay heat after shutdown is significantly below 1 MWth, allowing for passive cooling through natural convection in the pool during this phase. The inclusion

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<sup>3</sup> <https://www.calogena.com/en/>

<sup>4</sup> <https://www.steadyenergy.com/>

of a pool around the PWR shell enhances passive cooling in various accident scenarios, emphasising the passive safety features crucial for deployment in or near residential areas.

Steady Energy also plans to start building a full-scale non-nuclear model in 2025 to test all safety, engineering, and design choices. This model will use an electric heater as a heat source at the Salmisaari B coal power plant in Finland. Meanwhile, Calogena reached an agreement with CEA Cadarache in 2025 to build a full-scale nuclear demonstration plant at their site.

In June 2025, the Finnish Radiation and Nuclear Safety Authority (STUK) [66] reviewed Smart Energy's plant description and assessed its safety using the model proposed in the new nuclear energy law. They concluded that the plant can be designed to meet safety requirements.

A model, LDR-Lite, developed by VTT, is closely linked to the LDR-50 concept. However, it excludes certain 'business-relevant' design choices and is used in various European projects (e.g., EASI-SMR) to study accident scenarios [67], develop a better understanding of design choices, and establish licensing criteria.

Relevant for this project is the announcement in 2024 of the Helsinki's city energy company, HELEN OY [68] to initiate a nuclear energy programme. This programme aims to build a business case and select a site for the possible deployment of SMRs for district heating applications. The goal is to have the first SMR for district heating operational by the early 2030s.

A policy change has facilitated the deployment in an urban environment. Starting in 2024, the Finnish Radiation and Nuclear Safety Authority (STUK) will no longer mandate a fixed 5-kilometer Precautionary Action Zone (PAZ) or a 20-kilometer Emergency Planning Zone (EPZ) around all nuclear power plants. Instead, the definitions of both PAZ and EPZ will be determined on a case-by-case basis, considering specific plant characteristics and risk assessments [69].

#### **2.3.3.2. Nuward<sup>5</sup>**

A collaborative project led by EDF is focused on designing a compact and modular reactor called Nuward. Originally, NUWARD was an integrated Pressurised Water Reactor designed to generate 340 MWe from two independent units. It was aimed at replacing fossil-fired plants and supporting various applications, including electricity and heat cogeneration, hydrogen production, district heating, and seawater desalination. The design featured a robust, simple architecture with innovative components and safety systems, emphasising passive safety to allow operation for over three days without external power. Reactivity is controlled using control rods and burnable poisons, adopting a boron-free approach to minimise effluents. Safety features include passive and active systems to manage both normal and severe conditions, utilising natural circulation heat transfer to a reactor pool, and measures for hydrogen management.

This original design underwent a joint early review by six European regulatory authorities in 2022-2023. However, in 2024, EDF announced a new strategy for the design to better align with utility and industry expectations. The updated NUWARD SMR will deliver 400 MWe of power and offer cogeneration capabilities of up to about 100 MWth. It will utilise established technological components from the nuclear sector to ensure safety and meet market needs. The company emphasises simplicity, modularity, and prefabrication as key values, aiming to enhance

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<sup>5</sup> <https://www.nuward.com/en>

competitiveness and reduce construction times. The goal is to finalise the reactor's conceptual design by mid-2026 and release a market-ready product by the 2030s, with the first-of-a-kind reactor to be built in France. NUWARD has developed five potential business cases for low-carbon applications, including utilities, data centres, industries and hydrogen production.

#### **2.3.3.3. NuScale<sup>6</sup> VOYGR™ SMR**

RoPower Nuclear S.A. leads this initiative, featuring the VOYGR™ SMR, a flexible power plant concept consisting of multiple small reactors that can be adapted to meet varying energy demands.

NuScale Power is a US-based company established in 2007, initially rooted in nuclear research at Oregon State University. The company focuses on advancing nuclear technology through the development of SMR designs. In 2020, NuScale announced a collaboration with the Romanian government to pursue an SMR project in Romania, marking a key step in its development. One of its notable projects is the NuScale Voyager reactor, an integral pressurised water reactor designed to produce 77 MWe (gross) per module. The design integrates the reactor vessel, steam generator, and pressuriser into a single unit, enabling factory fabrication and on-site assembly.

The NuScale Voyager reactor uses standard LWR fuel, consisting of uranium dioxide pellets enriched to less than 5% uranium-235, encapsulated in zircaloy cladding. The fuel assembly is arranged in a 17x17 lattice, approximately 2 meters in length, with operational cycles planned to be 18-21 months.

#### **2.3.3.4. Project Quantum**

Spearheaded by Last Energy<sup>7</sup>, the Quantum project seeks to commercialise SMR units by focusing on modular construction techniques and providing cost-effective nuclear energy solutions.

Last Energy, a US-based nuclear startup, is focused on developing Pressurised Water Reactor technology to provide energy for data centres. The company's main goal is to deliver prefabricated units of 20 MWe. To achieve this, Last Energy relies on established PWR technology, minimising the introduction of new concepts in fuel design or reactor physics. In August 2025, it was selected by DOE in the Nuclear 'Reactor Pilot Program', aiming to create *"a pathway toward fast-tracking commercial licensing activities"* [70].

In the second half of 2024, the company received pre-authorisation from the Romanian Commission for the Control of Nuclear Activities, allowing it to enter the licensing phase. Similarly, in early 2025, Last Energy started a nuclear site licensing process in the UK and initiated pre-application procedures with the U.S. Nuclear Regulatory Commission.

The PWR-20 design emphasises modularisation, standardisation, and ease of construction, utilising off-the-shelf components and proven technology to reduce risks and costs associated with new reactor technology. The core uses 12 low-enriched uranium 17x17 fuel assemblies. The reactivity control system integrates both control rods and burnable poisons.

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<sup>6</sup> <https://www.nuscalepower.com/>

<sup>7</sup> <https://www.lastenergy.com/>

The reactor pressure vessel and its internals are designed as a compact, integrated unit that can be prefabricated in a factory setting. The Nuclear Steam Supply System features a single-loop configuration. The reactor coolant system employs forced convection. This is complemented by an air-cooled tertiary loop. The operating cycle is 72 months.

#### **2.3.3.5. Rolls-Royce SMR<sup>8</sup>**

The Rolls-Royce SMR has a 470 MWe output, making it one of the larger SMR designs, and it utilises a standard three-loop PWR configuration. The design entered formal regulatory assessment by UK regulators in 2022 and has successfully completed two steps of the three-step regulatory process. Completion is targeted in time for the construction of the first-of-a-kind power station to commence in 2027. The design aims to take advantage of proven technologies used in current PWRs, thereby minimizing regulatory risk and maximising modular build, standardisation, and commoditisation.

The SMR employs 121 standard 17x17 low-enriched PWR fuel assemblies. To compensate for reactivity variations throughout the fuel cycle, gadolinium is used as a neutron poison, with the fuel assemblies designed to have varying poison loadings. As the reactor approaches equilibrium, it operates on an 18-month fuel cycle, featuring a three-batch equilibrium core. Duty reactivity control is achieved through the movement of control rods, eliminating the need for soluble boron concentration in the primary coolant

#### **2.3.3.6. European BWRX-300 SMR<sup>9</sup>**

Managed by Orlen Synthos Green Energy, a company aiming to bring SMR technology to Poland, this project focuses on the BWRX-300, a small boiling water reactor designed by GE Vernova Hitachi to simplify nuclear power generation while maintaining cost efficiency. Construction is planned to start in the late 2020s. The BWRX-300 is an SMR with a capacity of 300 MWe.

GE Vernova Hitachi is an alliance created between the US company General Electric and Japan's Hitachi. The BWRX-300 reactor core is a vertical cylinder designed for efficient thermal-hydraulic stability. It utilises low-enriched uranium (< 4.95%) GNF2 fuel assemblies, which are commercially available. Fuel cycle lengths can range from 12 to 24 months. Reactivity is controlled using control rods that also contain burnable absorbers.

The core is encased in a Reactor Pressure Vessel (RPV) with a tall chimney, which facilitates efficient natural circulation for core cooling, eliminating the need for recirculation pumps. Within the RPV, water entrained in the steam generated in the core is removed by steam separators and the steam dryer before exiting the RPV and flowing to the steam turbine.

The same small reactor design is also considered the reference design for deployment by Fermi Energia in Estonia.

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<sup>8</sup> <https://www.rolls-royce-smr.com/>

<sup>9</sup> <https://www.gevernova.com/nuclear/carbon-free-power/bwr-x-300-small-modular-reactor>

### **2.3.3.7. EAGLES**

EAGLES is a 300 MWe lead-cooled fast reactor. The consortium, including Ansaldo Nucleare, SCK-CEN, ENEA, and RATEN, is working to develop lead-cooled fast reactors, valued for their enhanced safety and efficiency. This project aims to build a 300 MWth demonstrator named ALFRED in Romania. ALFRED is designed not only to serve as a lead-cooled SMR but also as a foundation for the development of larger-scale LFR nuclear reactors.

ALFRED is a pool-type reactor intended to operate at 300 MWth, with a temperature range from 400°C (inlet) to 520°C (outlet). Its role as a demonstrator reactor includes studying lead chemistry control and investigating the compatibility of lead with structural materials to prevent corrosion. The strategy for qualifying ALFRED and its materials involves a gradual increase in temperature gradients over several operational stages, ultimately achieving 125 Mwe.

The coolant in ALFRED [71] is pure lead, circulated within the primary system. Inside the reactor vessel, three Steam Generators (SGs) based on bayonet tubes are inserted.

The reactor core design features 56 inner and 78 outer mixed oxide (MOX) hexagonal fuel assemblies, each containing 126 fuel pins [72]. The fuel pins have a slightly larger diameter compared to 'standard' PWR pins and include a 2 mm open channel filled with helium in the middle. The plutonium enrichment in these assemblies varies between approximately 20% and 26% for the inner and outer assemblies, respectively, to maintain a uniform radial power distribution. Currently, the only large-scale commercial production of such fuel is in Russia.

Before building ALFRED, a smaller-scale device (LEANDREA) will be constructed by 2035 in Mol, Belgium. This smaller reactor is intended to demonstrate the technological and construction aspects of the LFR design. In September 2025, Belgian, Italian, and Romanian nuclear regulators have announced the launch of a pre-licensing assessment for the EAGLES-300 lead-cooled small modular reactor, supported by the IAEA as a pilot under its Nuclear Harmonisation and Standardisation Initiative.

### **2.3.3.8. European LFR AS Project**

Developed by Newcleo<sup>10</sup>, a French-Italian-UK startup, this project aims to promote Lead-Cooled Fast Reactor (LFR) technology further in Europe, emphasising innovative reactor designs to enhance safety and fuel efficiency.

The reactor design is based on a design developed by Hydromine Nuclear Energy [73], a company that Newcleo acquired in August 2021 [74]. The design features an open-pool lead-cooled reactor with a thermal power of approximately 480 MWth and an electrical output of 200 MWe. The overall design concepts of these two reactors are similar; however, significant differences in technical implementation exist, such as the size and shape of the reactor vessel and the use of helical coil steam generators.

According to publicly available design information, the reactor will use hexagonal MOX fuel assemblies, with 61 fuel assemblies in the core and an average plutonium enrichment of 19-23.2%. The fuel pin design and pitch are similar to other LFRs, but the grouping of fuel elements differs,

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<sup>10</sup> <https://www.newcleo.com/>

with Newcleo's fuel elements containing approximately 390 fuel pins [75]. Newcleo intends not to rely on third-party fuel supply but to develop and build the necessary MOX production plants, ensuring a self-sufficient fuel cycle.

Newcleo, in collaboration with ENEA, plans to operate the first non-nuclear simulator in 2026 to facilitate corrosion studies in the liquid lead environment. The first nuclear demonstrators are expected to be built by 2031, in collaboration with the French Alternative Energies and Atomic Energy Commission.

In 2025, Newcleo and Blykalla<sup>11</sup>, a Swedish startup and developer of the LFR project SEALER, signed an agreement for the joint research and development of materials for lead-cooled fast reactors (LFRs). The partnership involves the exchange of materials, results, and associated data to support each party's respective R&D programmes. By combining findings from their individual R&D programmes and industrialisation processes, the parties aim to expedite the commercialisation of LFR technology.

While the UK government accepted Newcleo's nuclear design for the voluntary Generic Design Assessment (GDA) in early 2025, the company halted its UK activities in June 2025 over insufficient funding and support in the country [76].

#### **2.3.3.9. Thorizon One Project**

Led by Thorizon, a Dutch-French nuclear startup, this project explores the use of thorium-based fuel cycles in SMRs, which could offer a more abundant and less waste-intensive nuclear energy option. The reactor's first stage is designed to burn plutonium and minor actinides, with further fuel developments including the potential to breed uranium from thorium.

This project is built around the patented concept of nuclear fuel cartridges [77]. Unlike traditional molten salt reactors, where the fuel/coolant mixture cycles in and out of the reactor vessel, this design uses individually sealed cartridges. The core salt volume is divided into subcritical assemblies, specifically using closed and contained cartridges, which prevent salt from entering or leaving and maintain low primary pressures under all circumstances. The reactor relies on pump action to remain critical, and if the pump fails, the core will automatically drain into a subcritical configuration. The design features double cartridge containment and primary containment integrity monitoring, along with passive decay heat removal from the cartridges' outer surfaces during station blackout scenarios.

Fuel development is conducted within a consortium that includes ORANO and Stellaria, a French MSR concept. Presently, the company is working with partners and regulators to complete the design phase and evaluate possible sites, aiming to start construction by 2030.

## **2.4. Technology readiness level**

The primary aspects of various nuclear technologies have been discussed previously. In Europe, technology for most large nuclear power plants is well-established, with facilities based on designs

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<sup>11</sup> <https://www.blykalla.com/>

already operating worldwide. However, the situation is different for SMRs, resulting in a lower Technology Readiness Level (TRL), as the TRL score is based on the design concepts.

For many technologies, the core technology is not the main barrier to deployment nor the sole factor in assigning a TRL, in fact for many considered (sub-)technologies test reactors are or were operating. While for all these technologies development and testing is ongoing in facilities worldwide, oftentimes the challenge remains to transform this experience into designs that are economically viable and comply with all regulatory requirements. Additionally, significant advancements in materials and processes might be necessary, alongside the development of a supply chain.

Within each technology category, TRLs for specific SMRs vary due to differences in design specifics and stages of regulatory design approval. By considering all these factors, we can derive an estimate of the overall TRL score for SMR technology, indicating its progress. Given the numerous projects, a range of TRLs emerges for each technology, reflecting our current assessment of the technologies. Industrial details and design choices that are not publicly available, along with evolving regulatory processes, can give rise to some difficulty in assigning TRL levels.

**Table 4:** Current TRL of nuclear fission technologies used in SMRs

| Sub-Technology             | TRL (Technology Readiness Level) |   |   |   |   |   |   |   |   |
|----------------------------|----------------------------------|---|---|---|---|---|---|---|---|
|                            | 1                                | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| NPP<br>(operating designs) |                                  |   |   |   |   |   |   |   |   |
| SMR                        |                                  |   |   |   |   |   |   |   |   |
| SMR - PWR                  |                                  |   |   |   |   |   |   |   |   |
| SMR - BWR                  |                                  |   |   |   |   |   |   |   |   |
| SMR - SFR                  |                                  |   |   |   |   |   |   |   |   |
| SMR - HTGR                 |                                  |   |   |   |   |   |   |   |   |
| SMR - LFR                  |                                  |   |   |   |   |   |   |   |   |
| SMR - MSR                  |                                  |   |   |   |   |   |   |   |   |

Source: JRC analysis based on data [45] [46] [47] [48] [49] [62] [63], 2025

## 2.5. Installed energy capacity and production/generation

In a previous section, we discussed some of the installed nuclear capacity in Europe. Overall, Europe's nuclear capacity has remained relatively stable over the past decade, with notable exceptions. In Germany and Belgium, several reactors were shutdown in recent years, leading to a decline in their nuclear output.

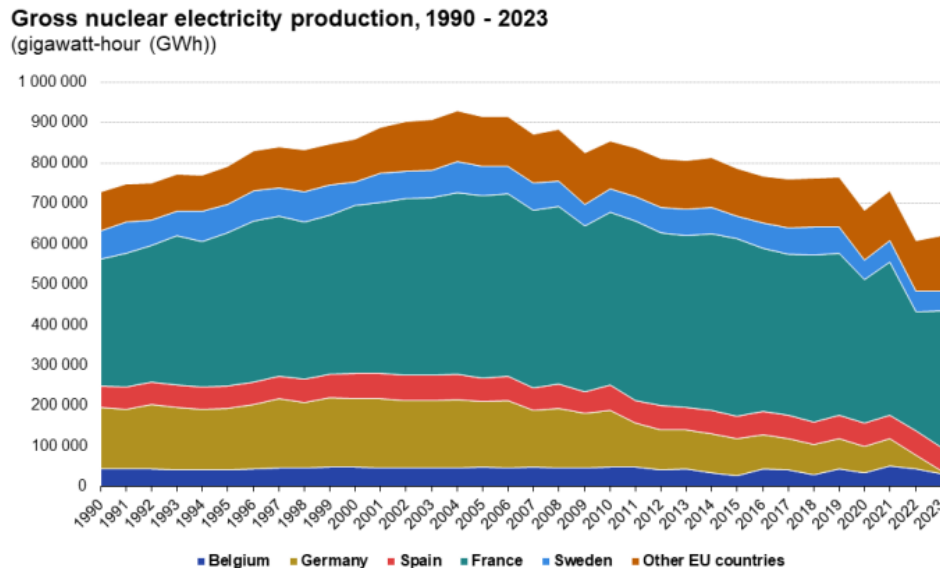
Over the last decade there has been decline in energy production, as can be seen in Figure 4, [15]. The decline is mostly caused by reduction of available installed capacity.

France has maintained its leading position in nuclear capacity. In 2022, France's nuclear fleet produced only 282 TWh, significantly below its ten-year average of 395 TWh [78]. This drop was primarily due to corrosion issues detected at Civaux 1 in December 2021, which required extensive repairs and inspections. Additionally, planned maintenance outages from 2021 were delayed or reduced due to the COVID-19 pandemic, and industrial strikes further disrupted operations.



In 2023, output partially recovered to 320 TWh as reactors returned to service after repairs. However, some units remained offline due to ongoing corrosion concerns. Full normalisation of output was reached in 2024.

**Figure 4:** Gross nuclear electricity production in the EU,



Source: Eurostat, 2025

## 2.6. Technology costs

The cost of electricity from different power sources can be divided into three main categories for a fair comparison: (1) the upfront cost of building the plant (capital costs), (2) the ongoing fixed costs of running it, such as maintenance and salaries (fixed O&M), and (3) the variable costs that change with the amount of electricity produced such as fuel costs, which for nuclear energy typically account for all costs in the front-end and the back-end of the fuel cycle.

Nuclear energy is characterised by high initial construction expenses. For a new nuclear plant, the upfront construction cost can even exceed 80% of the total energy cost [79]. The remaining expenses are allocated between operations and maintenance and fuel costs, with fuel cost typically accounting for 5-10% of the total costs [79]. Typically, more than 2/3 of the fuel costs cover the front-end, i.e. mining, milling, conversion, enrichment and fuel production, and less than 1/3 the back end, i.e. from the moment the fuel is unloaded from the core until disposal, including construction of the final repository. The given percentages vary depending on factors such as interest rates, construction delays, and the specifics of engineering contracts. It has been reported [80], that the choice of back-end strategy, direct disposal, partial recycling in a LWR, or multiple plutonium recycle in LWR and fast reactors, has only a minor effect on the magnitude of the fuel costs. Finally decommissioning costs, while depending on assumed interest rates, are in general well below 1% [81].

In comparison to gas turbines, nuclear power has significantly lower fuel costs, but its capital costs are up to six times higher, strongly depending on interest rate and capacity factor assumptions.

This, in addition to their flexibility to accommodate quickly large load changes, gas fired power plants are broadly used to adapt to fluctuating electricity demand, while nuclear power plants, which can accommodate load changes in a more limited way, are generally preferred for maintaining a consistent output.

Typically, the Levelised Cost of Electricity (LCOE) is used to compare different electricity production methods. However, as various technologies offer different services and roles within the energy system, their outputs can impose varying overall costs beyond the plant-level LCOE. This means that while LCOE provides a useful baseline for comparing costs, it does not fully capture the broader economic impacts and system-level costs associated with integrating different types of energy technologies (see e.g. [81]). Additional components that should be considered include:

- The extra costs of managing a grid with variable power sources, such as wind and solar.
- The timing of when electricity is produced, which affects its value.
- Harmful effects like carbon emissions and pollution (externalities).
- Strategic considerations, such as minimizing reliance on a single player for imports

Consideration of these factors provides a more comprehensive understanding of the true costs and impacts of different energy technologies.

System integration challenges have been addressed in the nuclear illustrative programme [23]. Evidence suggests that, although nuclear energy involves significant initial capital costs, its deployment can reduce overall system expenses by minimising the need for investments in transmission, distribution, and storage infrastructure. This is supported by studies across the EU [26] and France [82]. A Polish case study shows [83] that scenarios incorporating nuclear energy generally result in lower total system costs. However, it is important to note that the effectiveness of any solution is highly context-specific, requiring careful consideration of regional energy markets, infrastructure needs, and policy objectives [84].

### **2.6.1. Levelised cost of electricity**

The Levelised cost of electricity (LCOE) for nuclear power is primarily driven by construction-related capital costs. These costs are highly sensitive to factors such as assumed discount rates and specific building expenses for nuclear facilities. The limited number of new construction projects, coupled with cost overruns due to various factors previously discussed, makes it challenging to reliably estimate future costs. As a result, the LCOE for nuclear power is highly influenced by the assumptions and parameters used, leading to significant variation in reported LCOE figures across studies.

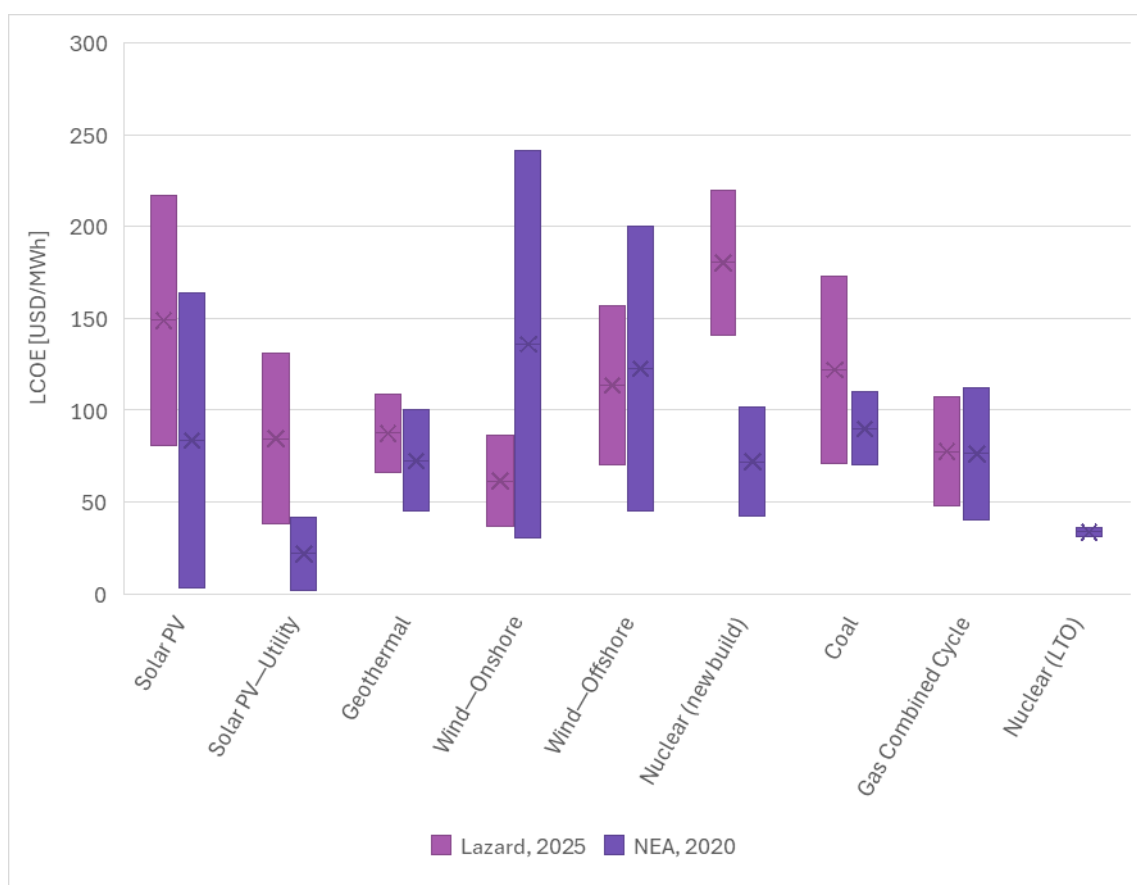
For new nuclear constructions, estimates of the LCOE vary widely. One source indicates a range of 40-100 USD/MWh [81], using data from eight builds worldwide. Another report [85] suggests an LCOE ranging from 80-100 £/MWh, based on data from [86] and projected construction costs. Meanwhile, another estimate suggests a range of 140-220 USD/MWh [79], using data from Vogtle 3 and assuming a 30% cost reduction for Vogtle 4 due to anticipated learning and efficiencies. The main factors for the LCOE of nuclear new builds are capacity factor and capital costs, i.e. the combination of overnight costs, capacity factor and discount rate. While for an individual study applied discount rates and capacity factors determine the ranges given, differences between studies often are caused by assumptions on overnight costs. Typically, variation in fuels

costs – including costs for waste disposal - induce changes hardly exceeding a few percent, i.e. far within the given ranges.

SMRs are expected to have LCOE figures within a similar range [87]. However, the economic parameters for SMRs could be positively affected by their capacity for cogenerating heat. Whether these benefits can be realised may depend on several factors including the regulatory framework, factory manufacturing processes, and modularity, among others. Furthermore, reliable estimates for building costs and the extent of cost reductions achievable between demonstrator models and full-scale industrial production remain areas of significant uncertainty and debate.

In contrast, the long-term operation (LTO) of existing nuclear facilities incurs much lower costs. This is because the investment required for upgrades is significantly lower, and the original construction costs have already been amortised. Typically, the LCOE for LTO is estimated to be around 30 USD/MWh, with much greater consistency across different studies.

**Figure 5:** Comparison of LCOE for different sources of electricity



Source: NEA, 2020 [81] and Lazard, 2025 [79]

However, comparing different studies—not only for one electricity production method but also across various sources—can be challenging. This difficulty arises because different definitions and assumptions are often mixed, a point also noted in a study [88]. According to [81], nuclear energy compares quite favourably to other sources, according to [79] the situation is less favourable for

nuclear energy. However, it's important to interpret all the numbers presented with the aforementioned caveats in mind.

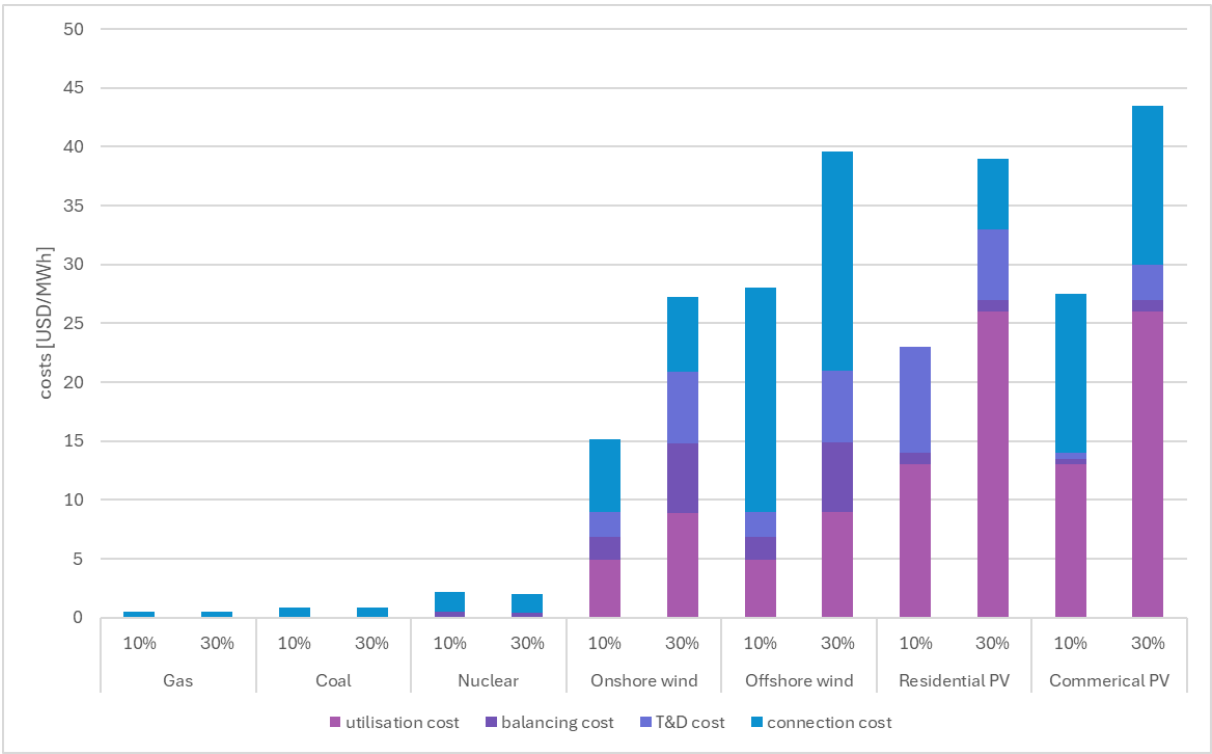
2.6.2. System costs

System costs can be divided into three main categories: utilisation costs, balancing costs, and grid costs. Grid costs reflect the impact on transmission and distribution networks due to location constraints. Connecting geographically constrained production sites to areas where electricity is needed may require additional infrastructure, such as building new lines or reinforcing existing ones. This is especially necessary when transporting electricity over long distances, which increases losses.

Utilisation costs are primarily associated with variable energy sources like solar and wind, whose output depends on weather conditions and time of day. This variability leads to a concentration of output that can reduce the market value of each megawatt-hour (MWh) produced. At high levels of variable renewable energy (VRE) penetration, multiple plants may generate power simultaneously, further decreasing the electricity's market value

Balancing costs arise from the need to adjust the system more frequently to balance supply and demand due to the uncertainty and variability in power production from VRE sources. The system requires greater reliance on spinning reserves, and conventional power plants must cycle and ramp their operations more frequently, leading to inefficiencies and higher overall costs.

Figure 6: Comparison of total system costs with two different amounts of variable renewable energy



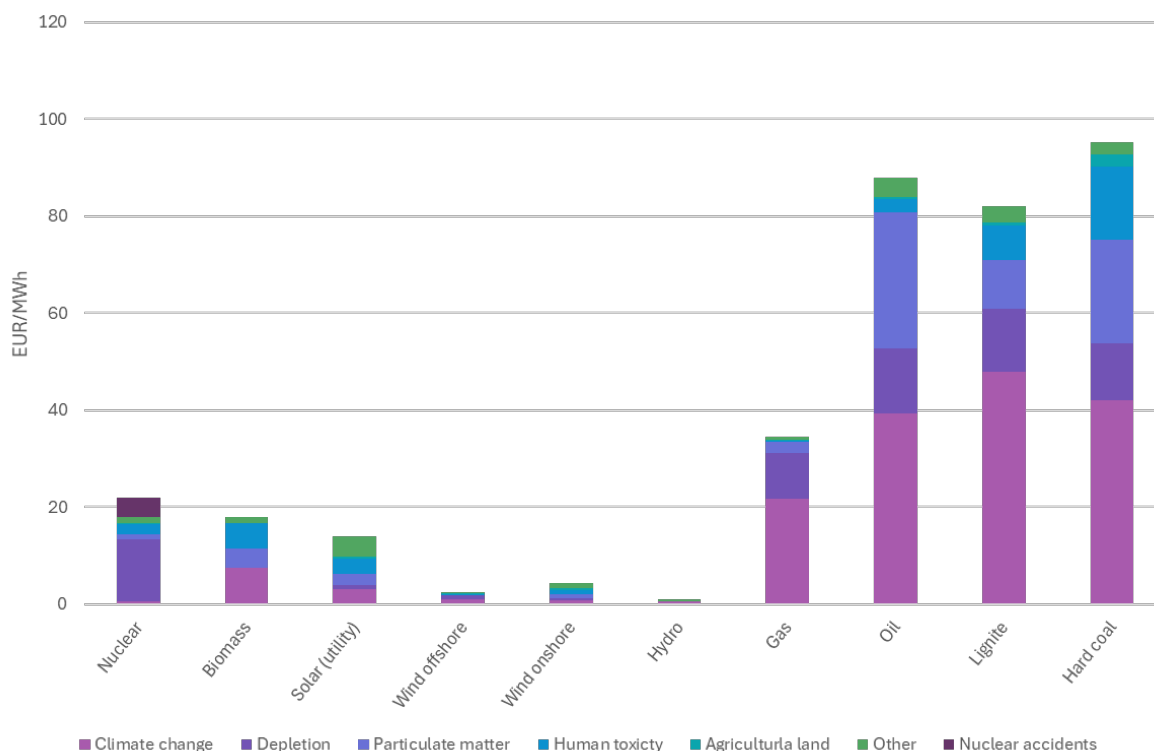
Source: NEA, 2018 [28]

The provided data illustrates an example of the overall costs [28] for various traditional and renewable energy technologies, considering different levels of usage. These estimates are not specific to any country but are rather general averages derived from multiple studies. It's important to note that the impact of energy systems can vary significantly by country, and the different components of the system are closely interconnected. This complexity means that simply aggregating parts from different studies might not yield an accurate total cost for a specific country. The figure, depicting cost for 10 and 30 percent variable renewable energy in the system, is intended to provide a rough idea of how the trends for different technologies compare in terms of costs, rather than offering an exact cost for any situation.

### 2.6.3. External costs

Following the methodology described in report [89], there are five major components contributing to the external costs of electricity production. For nuclear energy, the overall external costs are relatively low due to minimal greenhouse gas emissions, limited land use, and low chemical involvement throughout the process. The highest external costs for nuclear energy stem from uranium use and related mining activities. These costs could be reduced by utilising fast reactors and transitioning from a once-through to a recycling scheme. However, in the latter case, it would be essential to rigorously estimate the external costs associated with chemical usage and potential radioactive risks in reprocessing plants.

**Figure 7:** External costs for electricity production



Source: ECOFYS, [89]

Several studies, employing slightly different methodologies, estimate the external costs of various energy systems. While the exact costs differ depending on the methodology used (e.g. [90] [91]), the overall trends and figures are generally consistent. This consistency helps in understanding the broader impact of different energy sources, particularly nuclear energy, in terms of their external costs.

Specifically, for nuclear energy, differences in the external costs can also arise from how the costs of potential nuclear accidents are accounted for and whether the costs for final repositories and decommissioning are included in the Levelised Cost of Electricity or considered as external costs. In this report we prefer to consider decommissioning and disposal costs in the LCOE [81]. Another aspect is that carbon pricing is mostly included in the LCOE of fossil fuels, thereby internalising partially or fully the external costs related to climate change which are depicted in Figure 7. The figure, based on data from [89], is not intended to provide a comprehensive overview of all studies or an exhaustive review of all available discussions. Its sole purpose is to illustrate the order of magnitude and trends regarding the externalities of different technologies, the trends of costs are similar in different studies (e.g. [92] [93]).

## **2.7. Public RD&I funding**

The data covers investments by EU Member States and global investments and distinguishes between national investments and those made through EU framework programmes like Horizon 2020, which have been tracked since 2014. Data from 2023 are provisional and should be regarded as incomplete.

Several countries have revised their methods for collecting R&D data, leading to changes in reporting. Starting in 2021, Spain expanded its data coverage to include contributions from state and regional government, which resulted in a break in the time series between 2020 and 2021. France revised data from the National Centre for Scientific Research for the period from 2002 onwards, resulting in a reported increase of R&D expenditures by an average of 21% between 2002 and 2019. Meanwhile, Finland updated its data classification system in 2021, although not all institutions have yet submitted detailed data.

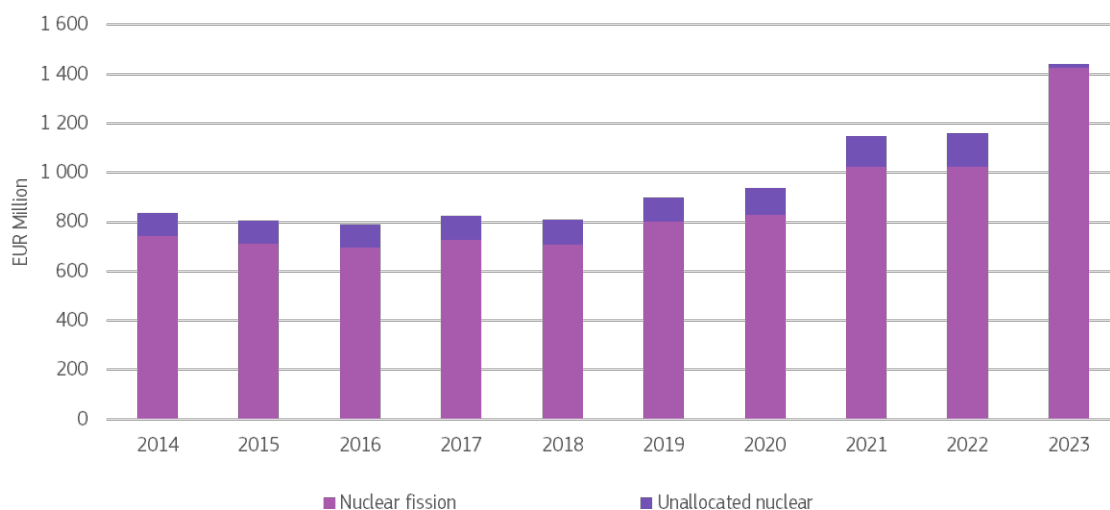
For the United States, the IEA directly reported data for the years 2012-2015 and provided estimated values up until 2020, ceasing these estimates after May 2022. The JRC has relied on these estimates and additional data from sources such as the Fletcher School's Climate Policy Lab, Gallagher and Anadon database, and various documents from the U.S. Department of Energy. There can be differences between the US government's requested budgetary amounts and the final sums enacted by Congress, necessitating a review that often spans two fiscal years to ensure accuracy.

**Figure 8:** Global public R&D investment over the last decade, data for 2023 are provisional



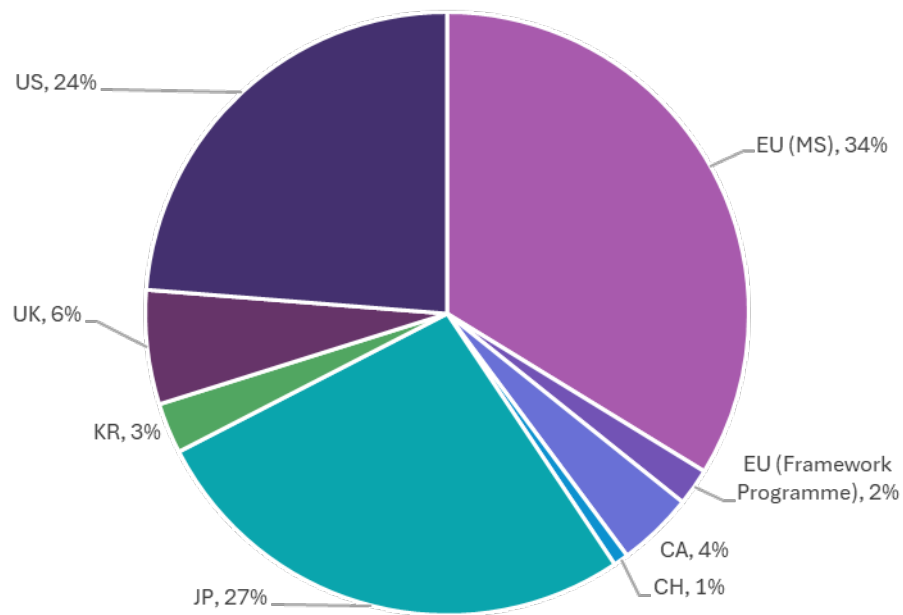
Source: JRC, 2025 based on IEA

**Figure 9:** Public R&D investment in Europe over the last decade, data for 2023 are provisional



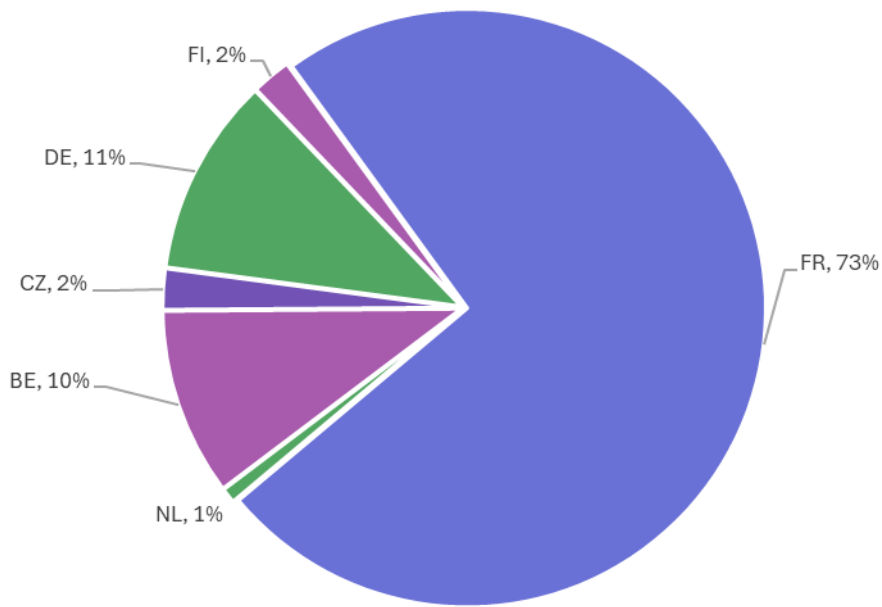
Source: JRC, 2025 based on IEA

**Figure 10:** Distribution of global public spending on nuclear fission research (2013-2023), only countries with a contribution above 1% are included in the figure



Source: JRC, 2025 based on IEA

**Figure 11:** Distribution of public spending in EU MS on nuclear fission research (2013-2023), only MS with a contribution above 1% are included in the figure



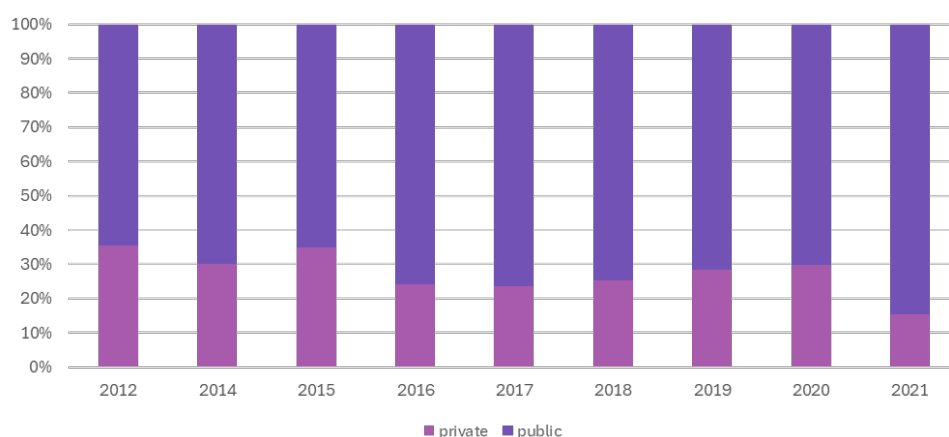
Source: JRC, 2025



## 2.8. Private RD&I funding

We had no access to reliable data for the last years, however for data up to 2021 we can refer to the JCR SETIS data [94], the privates R&I investments where approximately 25% of the public investments. Although some fluctuations could be observed, no trend in the spending level can be observed. Within the EU, almost 50% of the private R&I investments were in France, with Germany the second biggest contributor.

**Figure 12:** Ratio between public and private R&I in Europe



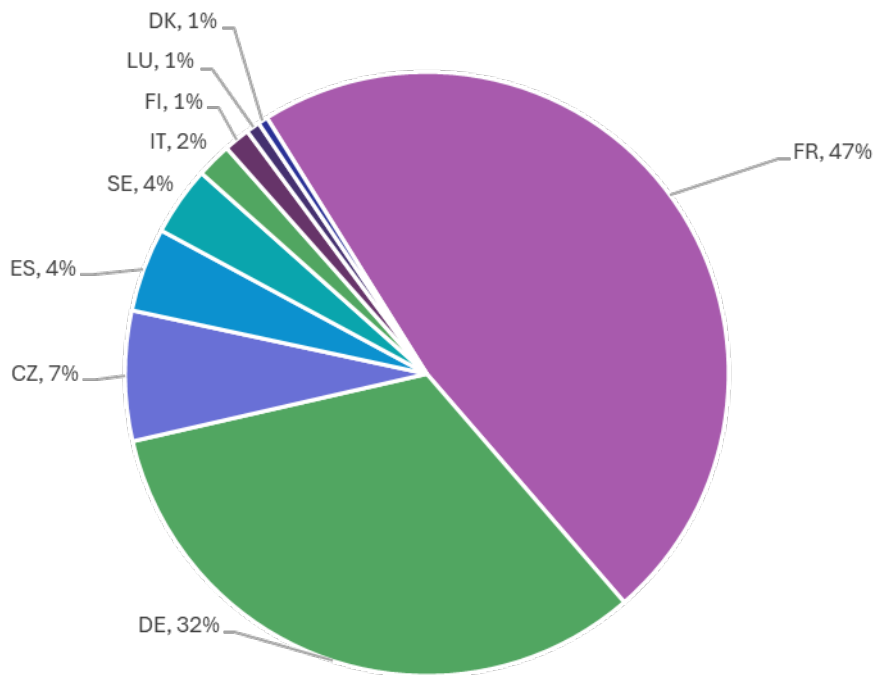
Source: JRC SETIS. 2025

**Figure 13:** Private spending in the EU



Source: JRC SETIS 2025

**Figure 14:** Distribution of private spending in Europe, only MS with a contribution above 1% are included in the figure



Source, JRC SETIS, 2025

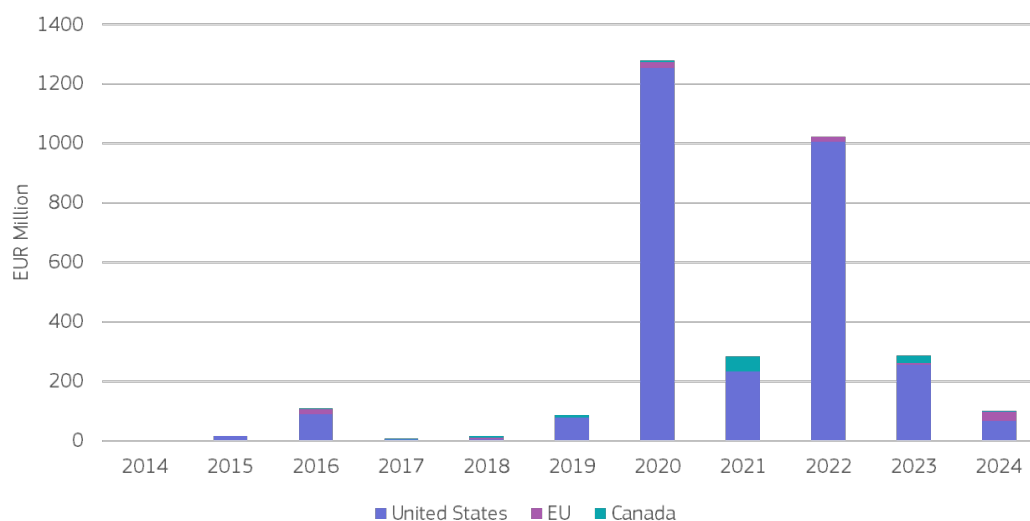
### 2.8.1. Venture capital and early and later-stage investments

Recent analyses of venture capital investment trends reveal intriguing patterns at different investment stages within the European Union and internationally. It appears that in both 2020 and 2022, significant peaks in investments occurred in the US. In 2020, X Energy received EUR 1.08 billion in grant funding from the U.S. Department of Energy to support the delivery of a first-of-a-kind commercial advanced nuclear plant. In 2022, TerraPower raised EUR 843 million in venture funds. These two funding events dominated the landscape. In contrast, financing levels in the EU are significantly lower, with a noticeable delay. While early-stage investments seem to be decreasing in the US, they are increasing in the EU. In 2024, early-stage investment in the EU reached EUR 12 million, with no early-stage funding reported in the US. Later-stage investments are predominantly dominated by the US market. Determining the implications of these observations for the SMR market is challenging, but it seems fair to conclude that Europe is lagging in attracting financing the development of SMR projects compared to the US<sup>12</sup>.

This disparity might be seen as reinforcing the observations of the Draghi Report, which underscores the need for a more dynamic and integrated European venture capital market to drive innovation and sustainability.

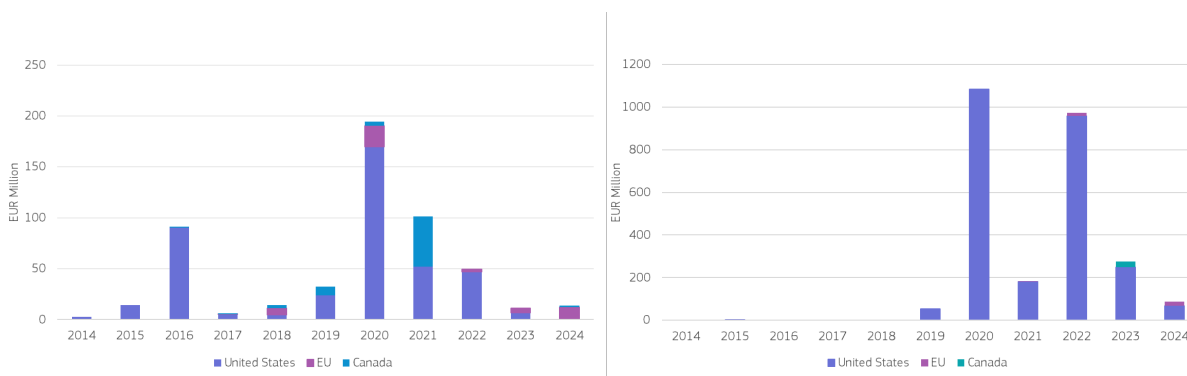
<sup>12</sup> The reported numbers should be taken with some reservations, as the sources seem provisional

**Figure 15:** Global VC/PE EU investment in the small modular reactors (SMRs) sector, by region for all deals.



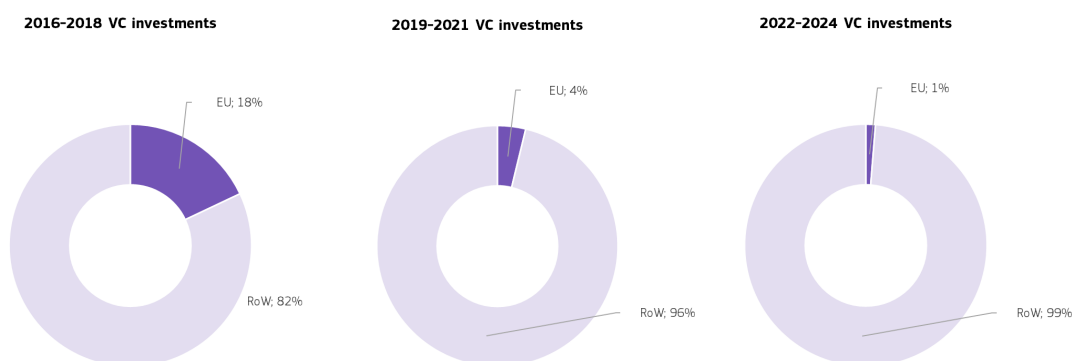
Source: JRC based on Pitchbook, 2025.

**Figure 16:** Global VC/PE EU investment in the small modular reactors (SMRs) sector, by region for early stage (left) and later stage (right) VC deals.



Source: JRC based on Pitchbook, 2025

**Figure 17:** VC/PE investment in the EU and in the ROW for all deals over the 9 years, by period of 3 years.

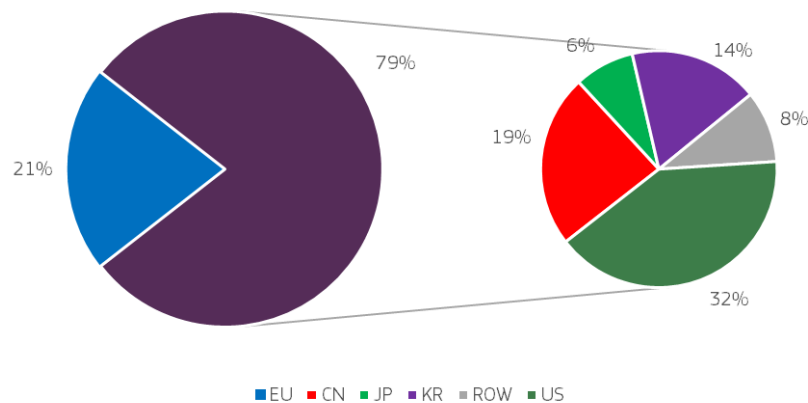


Source: JRC based on Pitchbook, 2024.

## 2.9. Patenting trends

We have access to analyses derived from two distinct methodologies, both conducted within the JRC. The first methodology aligns with that used for all CETO reports and has been specifically developed to generate indicators reflecting global inventive activity [95]. This approach employs patent families as a representation of invention production, with the filing date serving as the closest approximation to the timing of inventive activity and related research and development expenditures. However, it is important to note that statistics based on the earliest filing date have a lag of approximately 3.5 years. Due to the absence of a consistent pattern in historical data, there is no reliable technique available to estimate the most recent years using contemporary data analysis methods. In terms of data on nuclear fission technology, patents within the CPC (Cooperative Patent Classification) category of G21C (nuclear reactors) were extracted from the PATSTAT database. For this methodology 21% of the global high-value inventions were registered in the EU, 79% in the rest of the world, the pie-chart on the right (Figure 18) give a more detailed picture for this part.

**Figure 18:** share of global high-value inventions, using first methodology



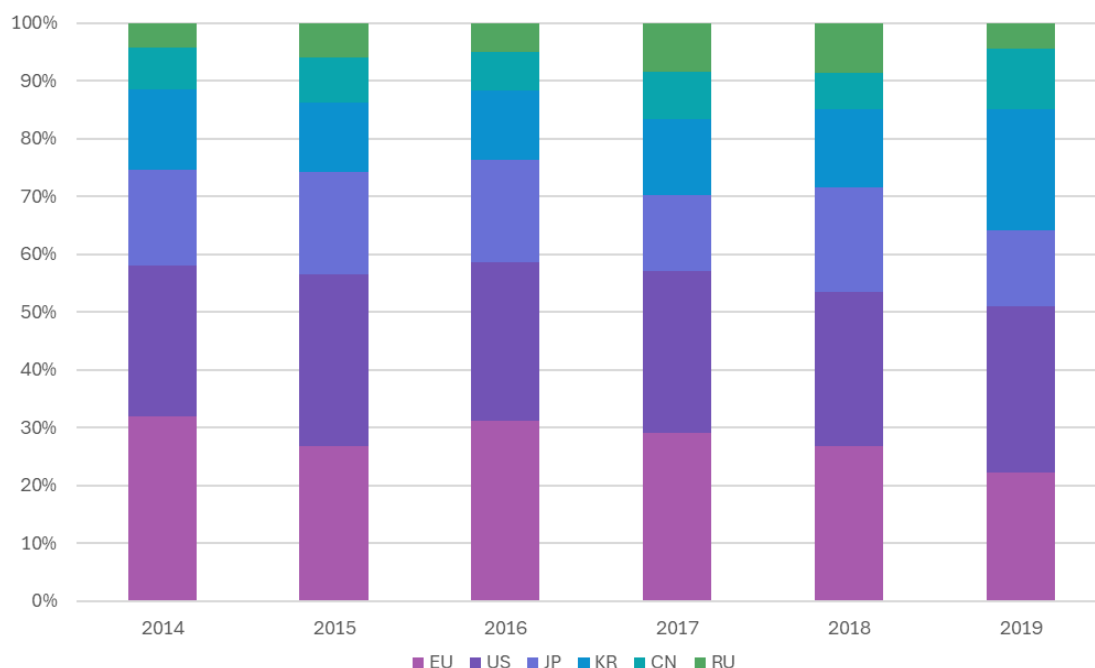
Source: JRC, 2025

The second procedure for obtaining the data is described in more detail in [96]. The dataset focuses on patents belonging to families filed in at least two different offices, with one being an IP5 office (EPO, JPO, KIPO, CNIPA, or USPTO). This approach is used to identify high-value patents. To determine patents relevant to nuclear fission, a language model compared patent abstracts against a standardised reference text. In total, 15 128 patents were marked as relevant to the field.

The chart illustrates the percentage of patents filed from 2014 to 2019, with data progressing from the inner to the outer layers. During this period, the European Union's (EU) contribution remains relatively stable at around 25% of total patents. It remains to be seen whether the increase in Korean patent contributions in 2019 indicates a real growth in research and development activities within Korea or if it is merely an artefact of the data collection process. Despite these variations, the United States continues to lead as the country with the highest number of patent contributions.

When comparing these two methodologies and accounting for the different time frames they analyse, one can observe a consistent identification of 20-25% of high-value patents as originating from Europe.

**Figure 19:** Global share of high value patent, using second methodology, for the application years 2014-2019



Source: JRC, 2025,

## 2.10. Assessment of R&I project developments

The SET Plan, established in 2007, aims to advance the EU's energy and climate objectives by positioning Europe as a global leader in low-carbon energy and energy efficiency technologies. It aligns with the fifth dimension of the Energy Union: 'Research, Innovation, and Competitiveness.' This alignment is reflected in the SET Plan's 10 key actions, each focusing on different aspects of advancing energy technology.

To advance nuclear safety within the SET Plan, several essential actions and outcomes have been identified. These include creating financial incentives for new nuclear plant projects, initiating demonstration and deployment projects by forming consortia and securing stakeholder commitment, and harmonising regulatory frameworks across the EU to accelerate licensing processes. The plan also emphasises developing human resources by promoting relevant technical disciplines in educational institutions and revitalising supply chains and infrastructure for necessary R&D.

European Technology and Innovation Platforms (ETIPs) are open groups comprised of stakeholders from industry and research organisations, endorsed by the European Commission and the SET Plan Steering Group. These platforms play a vital advisory role in the SET Plan by contributing to Implementation Working Groups and helping create their implementation plans. In the context of Action 10, which emphasises nuclear safety, the Sustainable Nuclear Energy Technology Platform (SNETP) serves as the ETIP. SNETP assists the European Commission in structuring research programmes that enhance nuclear safety and foster innovation in nuclear technologies. The funding of such research projects is achieved through the Euratom research and training programme.

The Euratom Work Programme for 2021–2025 is a key driver of research and innovation, with a total budget of around EUR 1.38 billion, of which EUR 266 million is assigned to indirect actions in nuclear fission, safety, and radiation protection. In June 2025, the extension of the work programme for 2026–2027 was adopted by the Council, assigning EUR 115 million for the two additional years for nuclear fission research.

Several projects are ongoing, dedicated to research on SMRs, ranging from safety-related issues to optimizing the integration of SMRs into energy systems—a list of such projects is provided in the annex. With the creation of the European Alliance on Small Modular Reactors, a platform supporting and focusing all research on this domain was launched in 2024.

Euratom is also launching a co-funded European Partnership bringing together European entities having a national mandate for research in materials science and dealing with nuclear materials. It builds upon the success of the Euratom project ORIENT-NM on the Organisation of the European Research Community on Nuclear Materials.

In the research and training programme, there is also funding for joint European programming on the management of radioactive waste. This includes EURAD and its successor EURAD2. These platforms have developed a strategic research agenda that complements national research programmes by identifying joint interests among Waste Management Organisations, Technical Support Organisations, and Research Entities. They also identify activities offering added value at the European level. The European Union contributes EUR 40 million to these research projects over four years.

The Euratom research and training programme also supports work on safeguards research for SMRs, both through indirect and direct actions, i.e. the research activities of the JRC. This includes addressing challenges related to safeguards for new and innovative reactors, fuels, and fuel cycles, with a particular focus on SMRs and AMRs. The research covers areas like new fuel types or waste forms. The European Safeguards Research and Development Association (ESARDA) plays a key role in this area by bringing together experts involved in safeguards to ensure progress and continuous improvement in international safeguards, with the main goals to coordinate research efforts, share information, and collaborate on R&D programmes.

### **3. Value chain analysis**

This chapter provides an overview of the nuclear industry within the European Union, structured around key components of the front end of the nuclear value chain—from uranium mining and fuel processing to reactor technology, a detailed analysis of the back end of the fuel cycle is not in scope of this edition. It also describes factors enabling the deployment of nuclear energy.

#### **3.1. Nuclear Fuel**

Under the Euratom Treaty, the supply of nuclear materials to Community users, for all civilian uses, is ensured by a common supply policy, based on the principle of equal access to sources of supply. The Euratom Supply Agency, established as a separate entity and endowed with specific prerogatives by the Treaty, takes responsibility, under the supervision of the Commission, for the common policy and aims to ensure that, as per the Treaty, all Community users receive a regular and equitable supply of nuclear materials and fuel.

By using its powers (notably, its exclusive right to conclude contracts for the supply of nuclear materials both from inside and outside the Community) in accordance with its statutory mission, by deploying its market-monitoring role and by providing information, expertise and advice on any subject related to the nuclear market, the Agency fosters the security of supply of nuclear materials and services for European users, including in the medium and long term and strongly supports the overall EU policies to that end – such as the European Energy Security Strategy and, most recently, the REPowerEU Communication and Roadmap.

Despite the limited availability of natural resources for uranium extraction in the territories of its Member States, the Community, taking advantage also of the possibilities offered by the Euratom Treaty, has developed a significant nuclear industry. This includes a robust infrastructure, high-level expertise and capacities for the provision of services (including uranium enrichment, the fabrication of fuel assemblies and reprocessing) throughout the whole nuclear fuel cycle. By focusing on these capabilities, Europe can maintain a degree of energy independence, even with limited natural uranium resources.

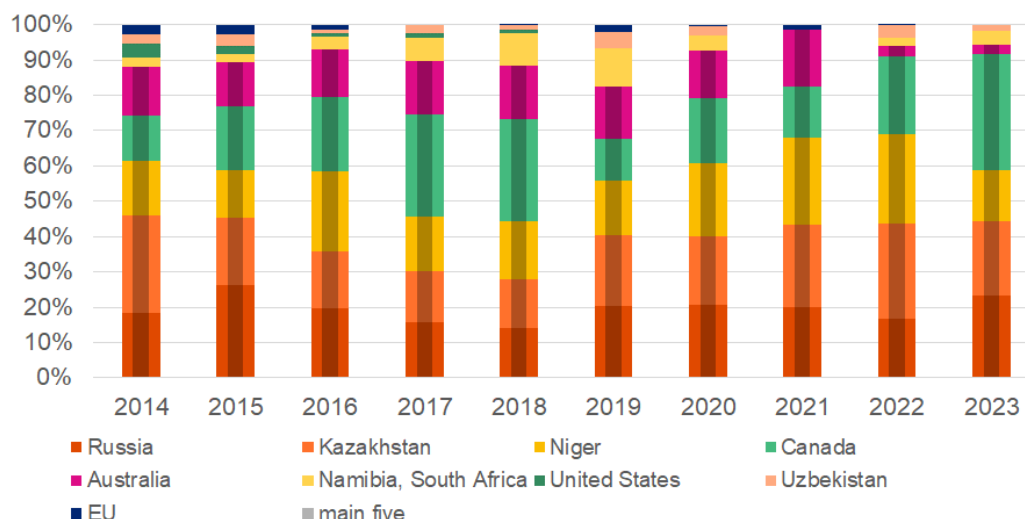
In the following chapters, we will delve into more detailed discussions of processing facilities available in Europe, as well as the strategies in place to ensure steady, reliable nuclear fuel supplies. The next sub-chapters will rely extensively on data from the European Supply Agency's annual report and the staff working documents of the nuclear indicative programme. All data sourced from external references will be clearly attributed to their original sources.

##### **3.1.1. Uranium Mining and Supply**

Between 2020 and 2023, uranium exploration and mine development activities in the European Union accounted for only 0.1% or less of global expenditure in this sector [97]. The primary focus has been on the remediation of closed uranium mines. Several potential mining projects have been blocked by local jurisdictions—such as those in Denmark (Greenland) and Hungary—or due to safety concerns, as seen in Spain, and uranium resources within EU territory remain largely unexploited. As a result, uranium production within the EU currently remains minimal, with only small quantities extracted as a by-product of remediation activities (e.g., in the Czech Republic), though these amounts are declining. However, some European companies are actively engaged in international uranium mining activities, demonstrating some access to the necessary resource base.

Beginning of November 2025, Sweden has opted to lift the ban on uranium mining, effective January 2026. Meanwhile, Finland is developing a project for uranium recovery by extracting it as by-product from the Ni-Zn-Cu-Co Talvivaara deposit in Sotkamo. In June 2024, the Finnish Radiation and Safety Authority granted permission to initiate extraction. Once full operation is achieved in 2026, it is anticipated that the production capacity will reach 200 tonnes of uranium (tU) per year [98].

**Figure 20:** Distribution of origin of uranium imported to the EU,



Source: Euratom Supply Agency, 2025 [99]

According to the Euratom Supply Agency, the EU now has a diversified range of suppliers for nuclear materials and products. In 2024, approximately 14 000 tonnes were imported. As in the previous year, roughly one-third of these imports came from Canada, while deliveries from Kazakhstan accounted for one-fourth. In 2024, Russia's contribution further decreased, while the shares from Australia and China increased [99].

In the longer term, uranium mining could become a major issue. The Euratom Supply Agency has been drawing attention to the matter in its annual reports. To enable nuclear expansion, the Agency finds that it becomes increasingly crucial to reduce to the extent possible uranium import as well as to promote circular nuclear economy. It recommends increasing indigenous resources of supply including by launching uranium prospecting and mining programmes in the territories of the Community Member States and recalls the Commission powers, under the Euratom Treaty, in this respect.

### 3.1.2. Conversion

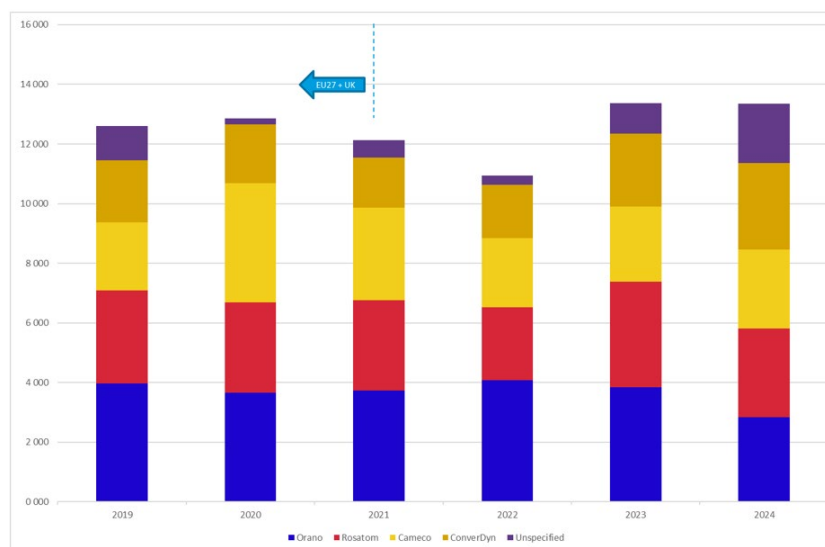
The Russian attack on Ukraine and the ensuing discussions about energy security have led to significant shifts in the uranium conversion market. Increased demand from Western suppliers, has caused a notable rise in prices from February 2022 to December 2023.

Current global conversion capacities include Orano's "Philippe Coste" plant in the EU, with a nameplate capacity of 15 000 tU annually, presently operating at an estimated level of 75% is still



ramping up towards full output. ConverDyn's Metropolis plant in the US is currently producing 9 000 tU per year and aims to reach its 2007 nameplate capacity of 15 000 tU following a temporary shutdown. China's conversion capacity is estimated at 10 500 tU, with further expansions expected to meet its extensive nuclear development plans. In Russia, Rosatom's capacity fluctuates due to upgrades, with estimates ranging between 12 500 and 16 000 tU annually.

**Figure 21:** Sources of converted uranium in Europe [tU]



Source: Euratom Supply Agency, 2025 [99]

The forecasted global demand for nuclear reactors is expected to reach up to 150 000 tU by 2050 [16], necessitating a near tripling of current conversion capacity. In Europe, the reduction of Russian conversion services from 27% to 23%, primarily substituted by Chinese supply, highlights potential vulnerabilities in the EU's strategy for nuclear fuel independence. There is a projected shortfall of at least 10 000 tU in conversion services across Western Europe, Ukraine, North America, Japan, and South Korea. Despite planned capacity expansions that might temporarily reduce this gap, without further investments, this shortfall could exceed 10 000 tU annually by 2050.

### 3.1.3. Enrichment

Uranium enrichment is a crucial step in the nuclear fuel cycle, as it increases the concentration of the fissile isotope U-235 to a level suitable for use in most nuclear reactors. Natural uranium contains only about 0.7% U-235, requiring enrichment to around 5% to meet the needs of the LWRs currently prevailing in the EU.

Urenco, with enrichment plants in the EU, plans to expand capacity to approximately 31 500 tonnes of separative work units (tSWU) per year by 2040. The largest global enricher, Rosatom/TVEL from Russia, currently has a capacity of 27 100 tSWU per year. Brazil's INB and Japan's JNFL have a combined capacity of under 200 tSWU annually, with Japan's Rokkasho-mura plant currently inactive. Both countries plan modest capacity increases before 2030.

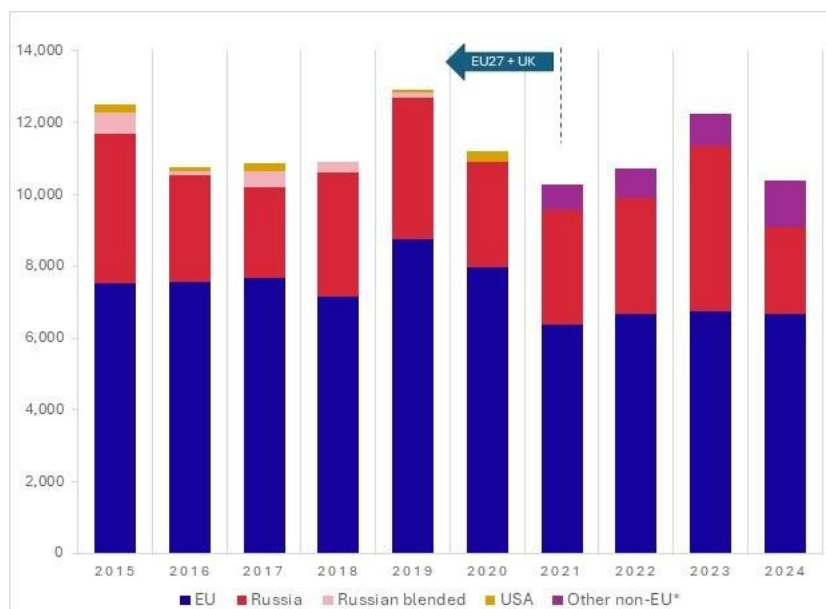
Increased geopolitical tensions and resulting shifts in global energy markets have highlighted the need for strategic self-sufficiency in enrichment services, catalysing market changes. Although enrichment service deliveries to EU utilities increased by 12% in 2023 compared to 2022, there has

been no clear upward trend in demand between 2016 and 2023, with annual deliveries ranging between 10 290 and 12 912 tSWU.

Significant investments in expanding enrichment capacity are underway. For instance, Urenco plans to expand its US and Dutch facilities by adding centrifuge technology. Orano in France intends to increase its facility's capacity by over 30%, supported by a EUR 1.7 billion investment. The US sees developments with Orano's Project IKE in Tennessee and Global Laser Enrichment's plans for the Paducah facility. China has also expanded its capacity from 6 500-7 000 tSWU in 2018 to an estimated 10 500 tSWU currently.

Enrichment technology is tightly controlled and proliferation-sensitive, limited to certain governments and a few commercial operators. The market for enrichment services is responding to geopolitical factors and is poised for significant growth as nations aim for higher strategic independence. Since February 2022, enrichment prices have increased significantly, multiplying by 2.5 times by December 2023

**Figure 22:** Origin of enriched uranium in Europe [tU]



Source: Euratom Supply Agency, 2025 [99]

Many concepts for advanced SMRs require a higher U-235 enrichment than conventional LWRs, known as High Assay Low-Enriched Uranium (HALEU), which is below 20% U-235. Currently, there is no such enrichment capacity in Europe, which may present a bottleneck for the demonstration and deployment of advanced SMRs. The United States is currently building enrichment capability, but at a capacity that will likely be insufficient for even the US domestic market. Also, the UK invests in a HALEU initiative, including a GBP 196 million award to URENCO for a Capenhurst uranium enrichment facility capable of producing 10 tonnes of HALEU annually by 2031. Within the Union the above mentioned ORANO project, supported by an loan of EUR 400 million from the European Investment Bank, aims to lay the foundation for HALEU production within the Union.

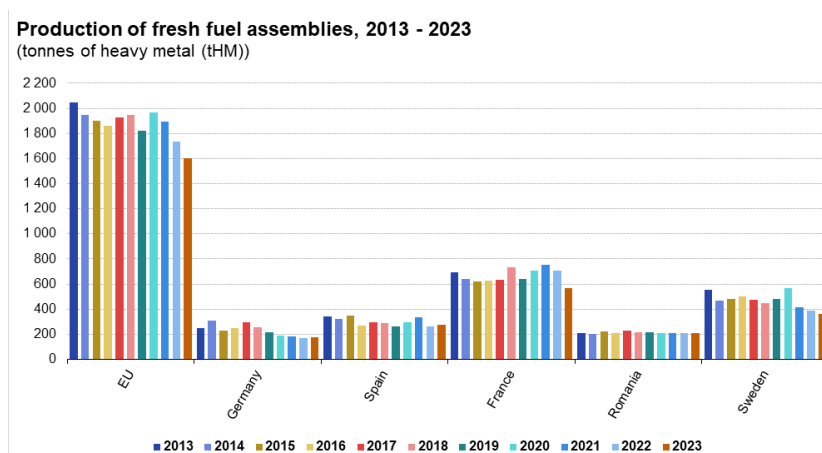
The projects outlined above are capable of supplying LEU+ fuel (enriched to 5–10% U-235), which is compatible with traditional reactor designs. By facilitating longer fuel cycles and power uprates,

this fuel lowers operational costs through reduced refuelling outages. Regulatory approvals for its production and deployment are underway, with initial test assemblies now in use within commercial reactors.

### 3.1.4. Fuel Element production

The production of nuclear fuel elements, or cartridges, occurs in several European countries, including Germany, Spain, France, Romania, Sweden, and the UK. In the past, the European Union has promoted diversified supply, resulting in a highly competitive market for nuclear fuel elements. Most nuclear reactors in Europe now have at least two qualified and licensed alternatives for the supply of their fuel elements. The facilities in Romania produce fuel elements for CANDU reactors, catering almost exclusively to the domestic market. In Sweden fuel elements for Russian-built VVER reactors are also produced. In its Lingen plant Framatome applied for a license to start producing VVER-fuel under licences from TVEL, before further developing its own design. After their qualification by safety authorities, non-Russian fuel designs can serve as an alternative to imports from Russia.

**Figure 23:** Production of fresh fuel in Europe



Source: Eurostat 2025

According to the WITS data bank, in 2024, the EU imported nuclear fuel, primarily from the Russian Federation (USD 686.05 million), the United Kingdom (USD 121.49 million), and the United States (USD 57.63 million). Simultaneously, the EU exported nuclear fuel to several countries, with the main export partners being Ukraine (USD 229.46 million), Switzerland (USD 152.63 million), China (USD 145.47 million), South Africa (USD 60.73 million), and the United Kingdom (USD 50.27 million) [100].

ORANO operates the only MOX fuel production facility in Europe at Marcoule (France), with an annual output of 82 tHM in 2023 [101]. This fuel is primarily used in France (currently, the only reactors licensed to operate with MOX fuel in the European Union are located in France, Belgium and the Netherlands). Overall, MOX fuel accounts for approximately 5% of the nuclear fuel used in the European fleet. According to the Euratom Supply Agency, the 2023 production saved slightly more than 400 tonnes of uranium and reduced the need for enrichment by 300 tSWU [99]. The base material for the MOX fuel is supplied by ORANO's reprocessing plant in La Hague (France), where approximately 1000 tonnes of used fuel are treated annually [101].

Advanced SMRs require different types of fuels in various physical and chemical forms, with higher enrichment levels compared to those used in LWRs. These also include MOX fuel, minor actinide-bearing fuel, fuel containing burnable poisons, TRISO fuel, and molten salts. In the EU, aside from MOX fuel production for LWRs, there is currently no commercial-scale manufacturing capability for these advanced fuels. Therefore, to successfully deploy advanced SMRs, significant investments to develop this fuel element production capacity would be necessary.

### **3.1.5. Nuclear Technology**

The construction and operation of a nuclear power plant involve several key cost components that contribute to the overall capital expenditure. Construction costs typically account for 40-45% of the total capital investment, ranging from approximately USD 1 billion to USD 3.8 billion. These expenses encompass site development, design, architecture, and engineering. In regions with limited local expertise, foreign Engineering, Procurement, and Construction (EPC) companies often manage 75-80% of the construction work, while local firms contribute 20-25%, primarily through on-site labour and project management. Costs can vary based on factors such as labour availability, construction standards, and site-specific conditions, particularly in areas with high living costs or significant environmental requirements.

Equipment costs constitute 50-55% of the total capital expenses, ranging from USD 1.3 billion to USD 4.6 billion. These costs are distributed among the plant's primary circuits. The Primary Circuit, or Nuclear Island, which includes critical components such as the reactor pressure vessel and steam generators, accounts for about 25% of the equipment costs and must adhere to strict safety standards. The Secondary Circuit, or Turbine Island which converts thermal energy into electricity, also represents roughly 25% of these costs. The Balance of Plant (BoP), which includes essential systems such as cooling towers and HVAC, accounts for 25-30% of the equipment costs. Finally, the Switchyard and Grid Connection, which are crucial for integrating electricity into the grid, comprise 10-25% of equipment costs. Additionally, other costs such as transportation, commissioning, and fuel load contribute 5-10% of the total capital cost, ensuring the plant's operational readiness once construction is complete.

The nuclear supply chain is structured in tiers. At the top (Tier 1), contractors like EPC firms or technology vendors deliver the Nuclear Steam Supply System (NSSS) or the entire Nuclear Power Plant (NPP) in turnkey projects. Tier 2 consists of system integrators, which may include reactor vendors. Tier 3 features original equipment manufacturers (OEMs) providing major components like steam generators and reactor pressure vessels (RPVs), relying on sub-components from Tier 4 suppliers. Tier 5 involves processors and fabricators of specialised materials, while Tier 6 supplies raw materials. Vertically integrated companies may streamline this process by handling more production in-house, reducing subcontracting needs. This tiered structure reflects the interdependence and complexity of the nuclear supply chain, which demands specialised manufacturing and technical expertise. Barriers to market entry rise with specialisation, leading to fewer participants. This can lead to a vertically segmented supply chain, with a low number of suppliers and each layer, increasing risk of double marginalisation, i.e. a stacking of cost mark-ups at various stages of the supply chain leading to a final price exceeding what a vertically integrated monopoly would charge, therefore a vertically integrated supply chain could in theory reduce this risk. New nuclear projects focus on how Tier 1 vendors manage their supply chains, while operating plants deal directly with Tier 3 and below for spare parts. Decisions made by technology vendors impact the entire supply chain throughout a plant's lifecycle.

In Europe, France is the only country with a fully integrated nuclear value chain, capable of supplying almost all necessary components for a nuclear construction project. Following France are countries like Germany, which possess extensive experience and capability in nuclear energy. The next tier in Europe includes countries such as Italy and Spain, which have some advanced elements of the nuclear value chain elements. Italy primarily exports, as no nuclear power plants are currently in operation in the country.

Historically focussed on domestic markets, the nuclear industry now benefits from increased global competition. However, domestic supplier preferences, supported by regulatory and technical barriers, particularly for the NSSS, persist in several countries.

#### **3.1.5.1. Nuclear Island**

The Primary Circuit, also known as the Nuclear Island, is a crucial component of a nuclear power plant, responsible for housing and controlling the nuclear reaction. For a PWR, the most common reactor type presently operating in Europe, it functions as a closed-loop system, enclosing a coolant that can become radioactive during operations to efficiently transfer heat from the reactor core for steam generation. Key components of the Primary Circuit include the reactor pressure vessel, which holds the reactor core; steam generators that use the heat from the reactor to produce steam; and coolant pumps that circulate the coolant to regulate temperature effectively.

Among the most critical components in modern nuclear power plants are large-scale or heavy forgings, which are essential for constructing key structures such as containment vessels, reactor vessels, turbine rotors, and steam generators. A single large reactor can require hundreds of these forgings, totalling over 4 000 tons of steel, with individual pieces weighing between 500 and 600 tons. The production of such massive components relies on a limited number of ultra-large forging presses capable of processing these enormous hot steel ingots. While the largest number of heavy forges are in Russia and Asia, within the European Union, Germany, France, and Italy have large forges operating. In Europe only one can produce most of the large size requirements for large-scale nuclear power plants. Further investments are ongoing to ensure that all parts of a large EPR and EPR2 can be produced domestically [102].

In the past, a significant number of suppliers of forged reactor pressure vessels existed in the EU (such as Klöckner-Werke, Kraftwerke Union, the Rotterdam Dry Dock Company, Škoda, and Uddcomb, which also fabricated large RPVs). Additionally, ENSA of Spain has been involved in the recent past in the production of large RPV, using forgings from outside sources. Forging smaller RPVs as required for most SMRs is comparatively more straightforward, and more suppliers are available. This leads to more competition, thus in principle shortening delivery times, accelerating deployment, and reducing costs.

Worldwide, the number of active steam generator providers for PWRs is larger than for pressure vessels. Similarly, a larger number of providers is in Russia and Asia. However, within Europe, significant suppliers and experience exist in a small number of countries—France, Italy, Germany, Spain, and the Czech Republic.

Original equipment materials are a major concern not only for new builds but also for the long-term operation of existing power plants. Over the last few decades, the business environment for the nuclear industry in Europe has slowed down for most due to a lack of new builds. This has led some nuclear suppliers diversifying and seeking new industrial markets, with some opting to leave the nuclear market altogether. This trend is apparent among European nuclear operators who are

witnessing a declining number of appropriately qualified nuclear suppliers as several companies exit the market.

For example, the number of equipment supplier companies holding ASME N-Type certificates has declined in both Europe and North America, with an observed decrease of around 5% per year in the period 2013-2020 [103]. Although not all nuclear suppliers are ASME certified, this trend suggests that the nuclear-grade supply chain is shrinking.

A 2017 study commissioned by Finnish utility TVO [104] found that the quality assurance (QA) and quality management (QM) requirements for safety-related systems, structures, and components (SSCs) in Finland's nuclear facilities are nearly equivalent to those in the oil and gas industry. Consequently, high-quality and reliable SSCs can be achieved using well-established non-nuclear industry standards, such as those from the American Petroleum Institute (API), International Electrotechnical Commission (IEC), and American National Standards Institute (ANSI)/International Society of Automation (ISA). Companies that adhere to these standards may be encouraged to enter the nuclear market if a sufficiently large number of reactors are constructed.

The European SMR pre-Partnership Workstream 4 – Supply chain adaption [105] sought to identify potential suppliers and assess their status in Europe. While the overall findings were positive, the study highlighted potential bottlenecks in the supply chain. It found that certain sectors, such as nuclear-grade pumps and valves, may face a constrained supply chain in Europe due to stringent quality requirements, especially for body castings and forgings. The generator/turbo generator sector could also face challenges due to a restricted market. However, this may be mitigated in the context of SMRs, as their power range allows more suppliers to comply without significant additional effort. Additionally, critical activities like the production of control rods, reactor cores, grids, fuel elements, and criticality monitors/systems might have limited suppliers.

Similar to the findings of the 2017 Finnish study, it was concluded that products and skills from other technologically advanced sectors could be utilised in the nuclear energy supply chain, including items like pumps and valves under the concept of "Commercial Grade Dedication." The smaller overall size of SMRs will enable a broader range of suppliers to compete for the necessary scope of work, as smaller forgings will allow more forges and steelworks to participate in the SMR supply chain. This could help mitigate some supply chain concerns and create opportunities for new suppliers to enter the market. Supporting this view, a study of the Italian nuclear supply chain [106] estimated that it could provide up to nine reactor pressure vessels for SMRs per year.

### ***3.1.5.2. Turbine Island and Balance of Plant***

The Secondary Circuit, or Turbine Island, plays a crucial role in a nuclear power plant by using heat from the Primary Circuit to produce steam, which drives turbines connected to generators for electricity production. In PWRs, this circuit is designed to strongly limit the possibility of radioactive contamination as the steam should be generated through heat transfer without direct contact with the reactor coolant. The key components of this circuit include turbines, which convert thermal energy into mechanical energy; generators, which then convert mechanical energy into electrical energy; and condensers, which condense the steam back into water for reuse in the cycle.

The BoP comprises the supporting and auxiliary systems necessary for the plant's operation, ensuring the smooth functioning of processes beyond the reactor core. This includes systems for emergency power supplies, emergency cooling, safety measures, and a tertiary cooling circuit that removes heat from the Secondary Circuit, often dissipating it through cooling towers.

The Switchyard and Grid Connection represent the final stage of the power generation process, where electricity is transferred from the plant to the grid for public and industrial use. This stage involves managing the transmission of power from the generator and ensuring it is converted to the appropriate voltage levels for the electrical grid. Key components include transformers, which adjust voltage levels; switchgear, which controls the flow of electricity and protects circuits from overloads; and transmission lines, which deliver electricity from the switchyard to the broader electrical grid.

In Europe, certain countries demonstrate significant expertise in areas related to nuclear activities, particularly mechanical engineering that is applicable to the nuclear sector, though not exclusively nuclear-specific. France, Germany, and Italy are leaders in this field, showcasing substantial experience and projects. Additionally, the Czech Republic and Sweden are actively engaged in similar endeavours, focusing on mechanical engineering that supports the nuclear industry.

### **3.1.6. Nuclear Services**

The nuclear industry involves a comprehensive range of services essential at various stages of a nuclear facility's lifecycle. Beginning with the supply chain, services include supplier assessments, quality audits, material and equipment testing, and non-destructive testing to ensure compliance with the highest standards. During construction and commissioning, services extend to civil work supervision, mechanical and electrical system testing, welding process qualifications, and rigorous safety coordination, all aimed at minimising risks and ensuring projects progress smoothly and within budget.

In the operational phase, facilities regularly undergo planned outages for refuelling, maintenance and upgrades. Specialised services during these times focus on minimising downtime and maintaining regulatory compliance through in-service inspections, statutory inspections, health and safety audits, and energy management. Additionally, careful planning and execution of maintenance tasks ensure the longevity and reliability of nuclear operations.

As facilities move towards decommissioning, specialised services ensure safe and efficient dismantling, reducing residual radioactivity to levels that allow for the release of property. This involves practical radiation protection measures, comprehensive nuclide analysis, waste management, and strategic handling of nuclear fuel storage. In parallel, comprehensive training programmes equip personnel with the necessary technical skills and qualifications, ensuring a skilled workforce.

The nuclear industry relies on specialised transport companies to safely move radioactive materials, and that's especially true for spent nuclear fuel, in compliance with relevant EU directives [107]. Currently, there are over 20 million shipments of radioactive materials annually, this number covers also materials for e.g. nuclear medical applications. Transport remains a critical enabler of nuclear energy, both in countries with nuclear energy and those embarking on new nuclear energy programmes.

### **3.1.7. Licensing Capabilities**

In Europe, protection of the health of the population (including the workers) against the dangers arising from ionising radiations is, pursuant to the Euratom Treaty, a competence of the Community. The Community's duty is to set and regularly review the relevant basic safety standards [108], aligning them to the highest level globally (in due account of the IAEA recommendations), while its Member States must transpose them in their internal legal and administrative orders.

As the Court of Justice of the EU found, however, there is a significant overlap between radiation protection and the safety of nuclear installations, and therefore nuclear safety is also a Community competence. The Nuclear Safety Directive 2009/71/EURATOM [109], and its amendment [110], creates a framework in which independent national regulatory authorities are tasked with overseeing licensing, enforcing safety requirements, and conducting inspections for nuclear installations. Member States must conduct periodic self-assessments of their national framework and competent regulatory authorities and invite an international peer review of relevant aspects of their national framework and/or authorities. The results of any peer review must be reported to the Member States and the Commission.

Smooth and rapid development of SMRs within the Community would call for more harmonisation of the applicable technical standards, as well as in the practices of the licensing authorities. This could be the way forward, provided that the approach fully respects the Community competence in the field of nuclear safety, as well as the applicable Community legal framework.

An OECD-NEA report suggests, indeed that greater global harmonisation could resolve some issues. By aligning technical standards across borders, while abiding by the Euratom acquis, regulators could reduce duplication, expedite assessments, and enhance investor confidence. Within the European Union, national regulatory authorities have been working towards these goals. Interaction between the nuclear industry and regulatory bodies on new developments can provide input to setting new Euratom rules, and if necessary, address design constraints early in the process. To strengthen such process the European Nuclear Safety Regulators Group (ENSREG), an expert advisory body to the European Commission, consisting of officials from national nuclear safety authorities across EU Member States has established a Task Force on Small Modular Reactors. This Task Force aims to gather lessons and best practices, assess specific SMR designs, interact with the European Industrial Alliance on SMRs, and identify regulatory challenges to be addressed.

Another consideration is the challenge posed by transporting prefabricated SMRs, including their fuel, which may challenge the regulatory framework. Within the Alliance, Member State authorities are collaborating to propose responses to challenges posed by new reactor designs and unconventional fuels to current regulations.

Some of the new designs and unconventional fuels can present challenges in determining the most appropriate safeguards approach and methodology. For this reason, early communication and collaboration between designers and the safeguards research community are both essential and already taking place. This is particularly important for concepts that incorporate 'safeguards by design,' where safeguards practices and requirements are best integrated from the earliest stages of the design process as proactively embedding safeguards ensures alignment with regulatory expectations and enhances the feasibility of future verification and monitoring efforts.

### **3.1.8. Waste management**

Under the Radioactive Waste and Spent Fuel Management Directive (2011/70/EURATOM) [111], each member state is required to establish a national strategy for managing spent nuclear fuel and radioactive waste. These obligations are rooted in the Euratom Treaty. Member states must develop and carry out national programmes to ensure the safe and responsible oversight of these materials. Inventory data on radioactive waste and spent fuel within the EU is published periodically, in line with the Euratom framework for waste management.

Radioactive waste production is a universal concern across all EU countries, with 17 member states also managing spent fuel within their borders. Given the radiological properties and potential risks



of these materials, their safe management is essential throughout all phases of the nuclear lifecycle. This material must be regulated as radioactive waste under the appropriate legislative and regulatory frameworks by relevant authorities.

Annually, the EU generates approximately 40 000 cubic meters of radioactive waste and around 1 000 tonnes of heavy metal in spent nuclear fuel, corresponding to an electricity supply of 620 TWh, based on 2023 data. Of this, over 90% is categorised as very-low-level or low-level waste, while high-level waste comprises 0.2% [23]. The rest is classified as intermediate-level waste.

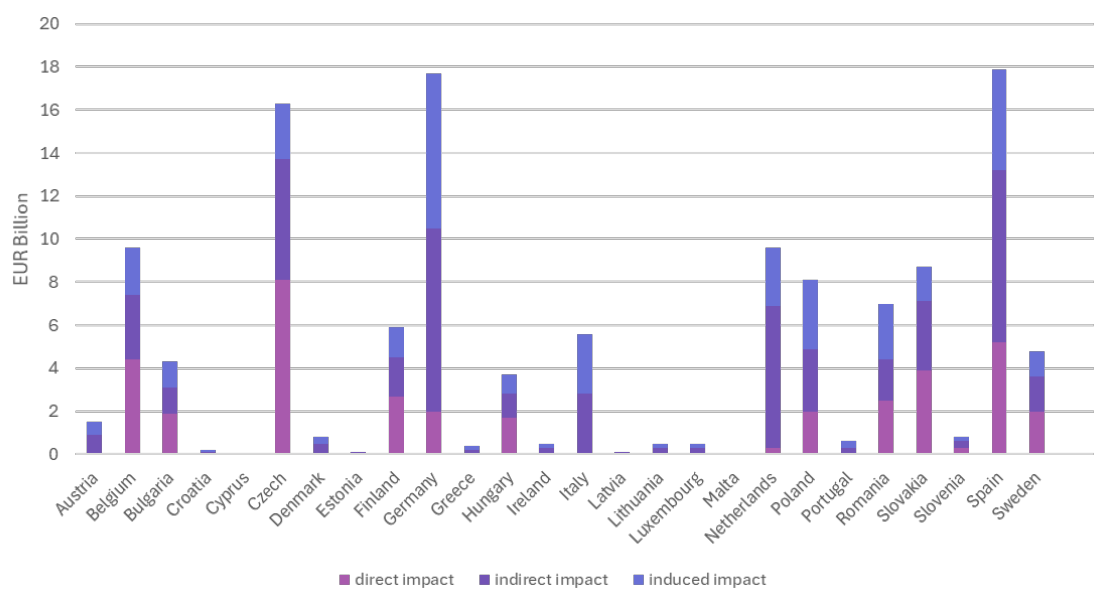
Progress has been made in safely disposing of very-low-level and low-level waste in the EU and regarding final repositories, for spent nuclear fuel, Finland and Sweden have advanced farthest. Currently, the Finnish regulator has a deadline of 31 December 2025 to complete its assessment of the license application [112]. In Sweden, an agreement for the construction of an underground final repository for spent nuclear fuel was signed in early 2025, with disposal planned to start in the 2030s [113].

3.2. Turnover and Gross Value Added

According to a study [11] the economic impact of the nuclear sector in Europe was approximately EUR 250 billion, and if the nuclear capacity were to be kept and the 100 GW level, the impact would remain more or less stable.

The report also highlights the economic impact across all EU member states, showing that even "non-nuclear" countries can significantly benefit from nuclear energy sector's contributions. France has been omitted from the plot as its contributions of EUR 57, 56.8 and 31.5 billion are on a different scale.

Figure 24: Estimate of the economic output of the nuclear industry in Europe, for clarity of the plot, the data for France are not included in the figure



Source: Deloitte, 2025 [11]

### 3.3. Environmental and socio-economic sustainability

To summarize some findings from the analysis of the environmental impacts of nuclear power compared to other electricity generation technologies [12]:

- Nuclear energy's average lifecycle greenhouse gas (GHG) emissions for electricity production are on par with those of hydropower and wind.
- Emissions of nitrous oxides (NO<sub>x</sub>), sulphur dioxide (SO<sub>2</sub>), particulate matter (PM), and non-methane volatile organic compounds (NMVOC) from nuclear energy are very low, comparable to or even better than emissions from solar photovoltaic (PV) and wind energy.
- Concerning acidification and eutrophication potentials, nuclear energy performs similarly to or better than solar PV and wind.
- In terms of freshwater and marine eco-toxicity, as well as ozone depletion and photochemical oxidant creation potential, nuclear energy compares favourably to solar PV and wind.
- The land required for nuclear energy generation is roughly the same as that needed for a gas-fired power plant with equivalent capacity. It is significantly less than that required for wind or solar PV installations.

In examining the areas requiring special attention when utilizing nuclear energy, several key issues have been identified:

**Thermal Pollution of Freshwater Bodies:** Large inland nuclear power plants that use once-through cooling systems withdraw a substantial amount of water from rivers or lakes for normal operations. The return of heated cooling water can cause significant thermal pollution, requiring careful management. Strict controls must be in place to monitor the maximum discharge temperature of the condenser cooling water and the resultant temperature of the freshwater body after mixing. Thorough analysis of water withdrawal options and thermal pollution risks should be integral to the site selection process.

**Water Consumption:** Power plants that use a thermal cycle to convert heat to mechanical energy require continuous cooling. While once-through cooling involves low water consumption, technologies such as recirculation cooling, evaporative cooling towers, or pond cooling consume considerable amounts of water due to evaporation. Water usage for these cooling methods is comparable to that of concentrating solar power and coal. Careful assessment of water resource availability and the potential environmental impacts of high-water consumption is essential during site selection to ensure the implementation of optimal solutions.

The discussions on the impact of nuclear energy on human health and the environment, along with the potential consequences of severe accidents, yield several important findings:

The average annual radiation exposure to the public from nuclear energy-based electricity generation is approximately 0.2 microsieverts. This level is ten thousand times lower than the average annual dose from natural background radiation.

In terms of public exposure in the event of accidents, the fatality rates from severe accidents in current Western Gen II nuclear power plants are around  $5 \times 10^{-7}$  fatalities per GWh. This rate is significantly lower than that of any fossil fuel-based electricity production and is comparable to

hydropower in OECD countries and to wind power, with only solar power having a markedly lower fatality rate.

According to Life Cycle Impact Analysis studies, the overall human health impact from both radiological and non-radiological emissions in the nuclear energy chain is on par with the impacts from offshore wind energy.

Overall, while nuclear energy presents some potential risks to human health and the environment, implementing specific safety measures—such as careful site selection, appropriate facility design and construction, and strict operational and waste management practices—ensures that these risks remain within established regulatory limits. Impacts during key lifecycle phases, including mining and milling, nuclear power plant operation, and others, are addressed through rigorous regulatory and legislative compliance.

### **3.4. Role of EU companies**

The European nuclear industry is an intricate and multifaceted sector, encompassing a variety of specialised fields, such as reactor design and construction, fuel production and enrichment, reprocessing and waste management. Its robust value chain is supported by interdependent areas such as reactor technology, fuel cycle services, component manufacturing, digital solutions, decommissioning, and nuclear research. A diverse network of companies, research institutions, and technological innovators underpins the industry's capabilities, particularly in tackling the pressing challenges of decarbonisation and energy security.

While large corporations often take centre stage in discussions about the nuclear sector, the contributions of smaller, niche players are equally vital. These enterprises bring essential expertise and innovation to the table, ensuring the industry remains cohesive and dynamic. This report highlights some of the more prominent companies active in the sector, yet it remains mindful of the indispensable role played by the lesser-known entities which, together, help maintain the innovative spirit of Europe's nuclear industry.

1. Électricité de France (EDF) is a major nuclear operator, central to designing, building, operating, and decommissioning reactors. EDF's nuclear operations are a cornerstone of the French energy mix, contributing to national energy security and supporting the country's transition to low-carbon energy. EDF is also involved in the development of small modular reactors (SMRs) and next-generation European Pressurised Reactors (EPRs) to meet future energy needs. Framatome, a subsidiary of EDF, is a global leader in reactor design, construction, and maintenance. Specialising in pressurised water reactor (PWR) technology—employed in over 60% of the world's nuclear reactors—Framatome supports EDF's nuclear fleet in France and key projects like the UK's Hinkley Point C power station, covering the full lifecycle of nuclear plants from supplying components like steam generators and control rods to offering digital solutions for plant optimisation.
2. Orano, formerly Areva, plays a pivotal role in the nuclear fuel cycle both in Europe and globally. The company handles uranium enrichment, MOX fuel fabrication, spent fuel reprocessing, and radioactive waste management. The Georges Besse II enrichment plant in France is a significant part of the European and even global supply chain, while the La Hague reprocessing facility possess a notable portion of the world's spent nuclear fuel reprocessing capabilities. Ongoing investments in the Georges Besse II site aim to increase production and enable HALEU production. In 2024, Orano was among the top three global uranium producers, with a market share of approximately 10%. The company operates

uranium mining sites in Canada, Kazakhstan, and Niger<sup>13</sup>, with additional projects under investigation to assess their viability [114]. These initiatives support the long-term sustainability of nuclear energy by ensuring a stable supply of raw materials and advancing closed fuel cycle technologies.

3. Urenco, a joint venture between the UK, Germany, and the Netherlands, is a leading global provider of uranium enrichment services. Employing centrifuge technology, the company supplies low-enriched uranium to more than 50 reactors worldwide, holding a substantial share of the global enrichment market. Its facilities in the UK, Germany, and the Netherlands ensure a stable and secure fuel supply for Europe, reducing dependence on external sources. Urenco has recently announced plans to expand its capacity in the UK and US, including the development of HALEU (High-Assay Low-Enriched Uranium) enrichment facilities, reflecting increasing demand for nuclear fuel driven by the need for low-carbon energy and the evolution of advanced reactor technologies
4. Siemens Energy plays an important role in integrating nuclear power in modern energy systems. Its nuclear division focuses on reactor systems, safety instrumentation, and digital solutions that improve plant efficiency and safety. The company supplies steam turbines, transformers, and control systems for nuclear plants, while developing digital platforms for real-time monitoring and predictive maintenance. Siemens Energy collaborates with industry partners for the long-term operations of reactors and supports the deployment of next-generation technologies, contributing to the transition to a low-carbon energy mix by ensuring nuclear power remains a flexible and reliable part of the grid. Its expertise in grid integration and digitalisation assists in addressing the challenges of balancing nuclear energy with renewable sources.

The situation for small modular reactors is particularly complex, with numerous startups emerging in Europe and globally. A previous chapter provides an overview of existing activities, projects, and companies based on the literature [49] [45].

### 3.5. Employment

According to established frameworks [11] [115] direct employment in the nuclear industry involves employees working directly at the plants, including permanent subcontractors such as security personnel and full-time equivalent outage personnel. Indirect employment encompasses roles within the nuclear supply chain in industries that supply products and services to it. Induced employment involves jobs created in the broader economy due to direct and indirect spending. It is estimated that each gigawatt of nuclear power generates 50 000 labour years of direct employment [115].

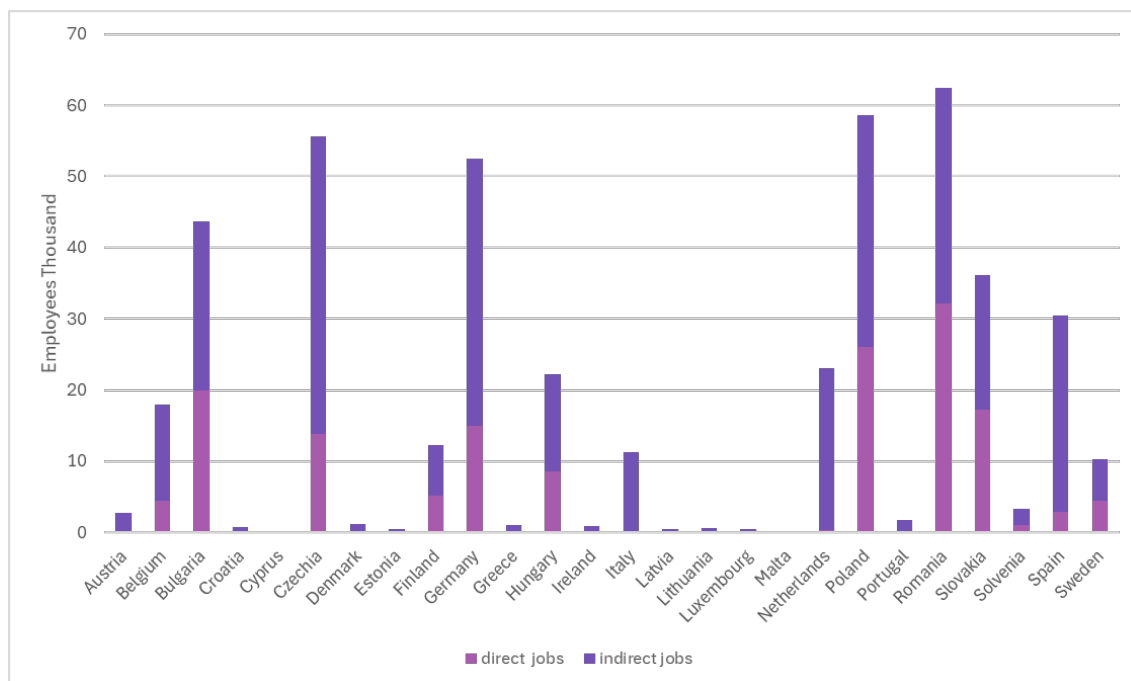
In other frameworks [116] [117] direct employment includes roles in the nuclear supply chain. This variation can complicate the comparison of individual components; however, the sum of direct and indirect employment remains relatively stable. Comparing various sources shows that, over a decade, the sum of direct and indirect employment for the nuclear industry in France ranges from approximately 196 000 to 240 000 [11] [116] [117]. The higher figures might include some

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<sup>13</sup> The coup of 2023 in Niger resulted in a dispute between the new government and ORANO over the uranium mines (see e.g. [143])

double-counting in the indirect sector [115]. A figure illustrates the sum of direct and indirect jobs in the nuclear sector in Europe in 2023. To enhance data clarity, France, with 60 000 direct and 136 000 indirect jobs, is omitted from the plot.

**Figure 25:** Employment in the nuclear sector, numbers of France are omitted from the plot to enhance clarity



Source: Deloitte, 2025 [11]

The same report predicts an increase in total employment (direct, indirect, and induced) from approximately 900 000 to 1 000 000 by 2050. This job growth is driven by the need to replace and decommission existing power plants in the coming decades, requiring additional jobs for new builds and decommissioning, even if current production capacity is only maintained.

Many skilled professionals in the nuclear industry were recruited in the 1970s and 1980s, during a period of rapid industry growth. The declining interest for nuclear energy in the following decades led to limited recruitment and resulted in a significant portion of workers approaching retirement age today. The ageing workforce and the decline in the number of nuclear professionals under the age of 35 present a risk for the future construction and operation of nuclear power plants and the decommissioning of old ones [23]. The necessary skills and competencies span a wide range, from nuclear physics and nuclear engineering to various technical, legal, and administrative roles. In addition, the nuclear industry has to compete with other attractive industries for a shrinking number of STEM graduates and suitable workers with vocational training.

An overview of the taxonomy of required profiles across different frameworks is provided in following study [118].

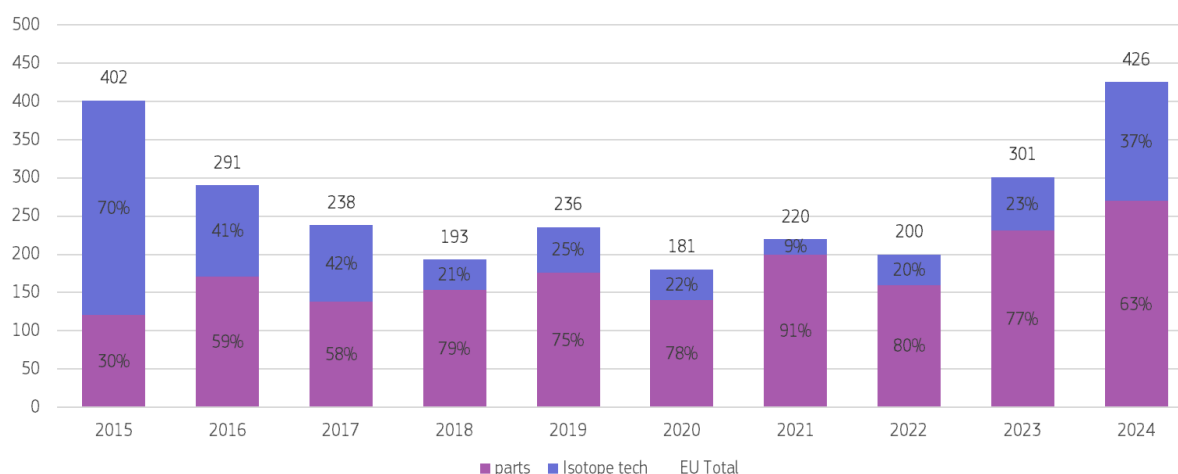
### 3.6. EU production data

The selection of Prodcom codes for monitoring production in the nuclear sector is based on the Net-Zero Industry Act (NZIA) list of components [119] with code 28993910 (Machinery and apparatus for isotopic separation, and parts thereof) and the European Climate Neutral Industry Competitiveness Scoreboard (CINDECS) [120] with code 25302200 (Parts of nuclear reactors).

In 2024, the EU production value of these components increased by 40%, reaching EUR 426 million. This growth was driven by a 17% rise in parts production value and a more than doubling of machinery for isotopic separation. Over the past decade (2015-2024), the overall EU production value grew by 6%, with an annual compound growth rate of 1% and an average annual value around EUR 270 million. Parts of nuclear reactors accounted for over 70% of EU production, while machinery for isotopic separation made up the remaining 30%.

Most Member States keep their production data confidential, making it impossible to identify leading EU producers.

**Figure 26:** EU production value per component [EUR Million]



Source: JRC, 2025

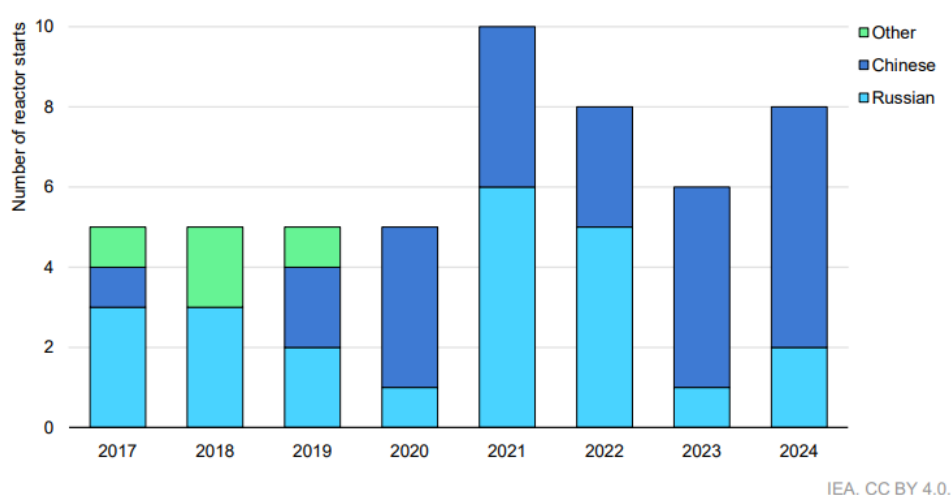
## 4. EU market position and global competitiveness

### 4.1. Global markets and growth prospects

#### 4.1.1. Current EU positioning in the global market

In recent years, the nuclear industry has undergone a significant shift, with the construction of large-scale reactors increasingly dominated by emerging markets and developing economies, particularly China and Russia. By the end of 2024, 63 reactors were under construction globally, with a total power capacity of 71 GW [17]. A substantial portion of this capacity comes from China's efforts, with 29 reactors under construction offering a combined 33 GW, accounting for nearly half of global new builds. Most of these reactors are of Chinese design. Meanwhile, Russia has established itself as the leading exporter of nuclear technology, with 23 GW of reactors being built in six countries, highlighting its influence in the nuclear market. Other significant projects are underway in India, Turkey, Egypt, Bangladesh, and Ukraine, largely utilising Russian designs.

**Figure 27:** Nuclear power plant construction starts by national origin of technology, 2017-2024



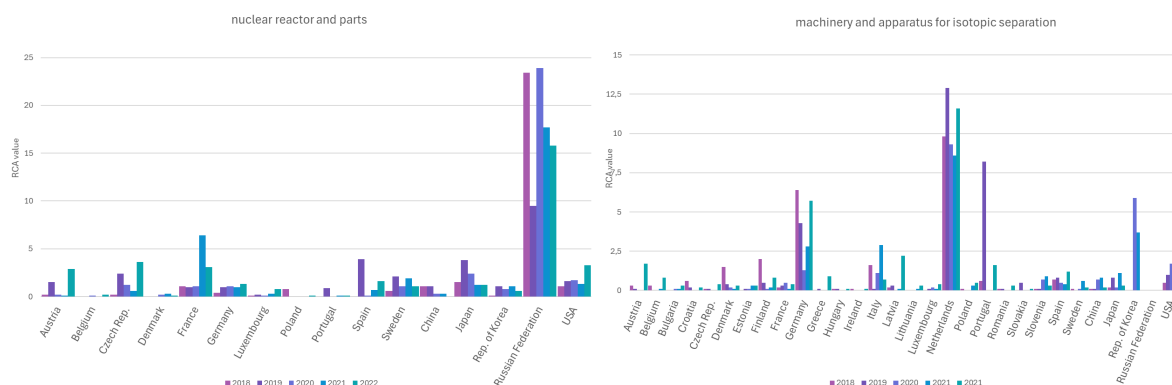
Source: IEA analysis based on IAEA PRIS database (Accessed 10 January 2025)

The European vendor EDF has been involved in two new builds in China over the last decade and is currently constructing two reactors in the UK. However, given the new builds signed in Europe, non-European reactor designs have been chosen. In many of these projects, a significant contribution will come from local suppliers.

As a measure for the positioning of the EU industry in the global market, we will use the Revealed Comparative Advantage (RCA) in USD, as provided by [121]. This metric compares a country's share of a particular product in its total exports with that product's share in total world exports. A value greater than one indicates the country has a competitive advantage in that product, while a value less than one suggests a disadvantage. We will look at four categories: nuclear reactors and parts, fuel elements (non-irradiated), enriched uranium and plutonium, and machinery and apparatus for isotopic separation (HS 8401.20). Russia's dominance in the global nuclear market is evident. Nonetheless, even in this sector, industries in some European countries have carved out a

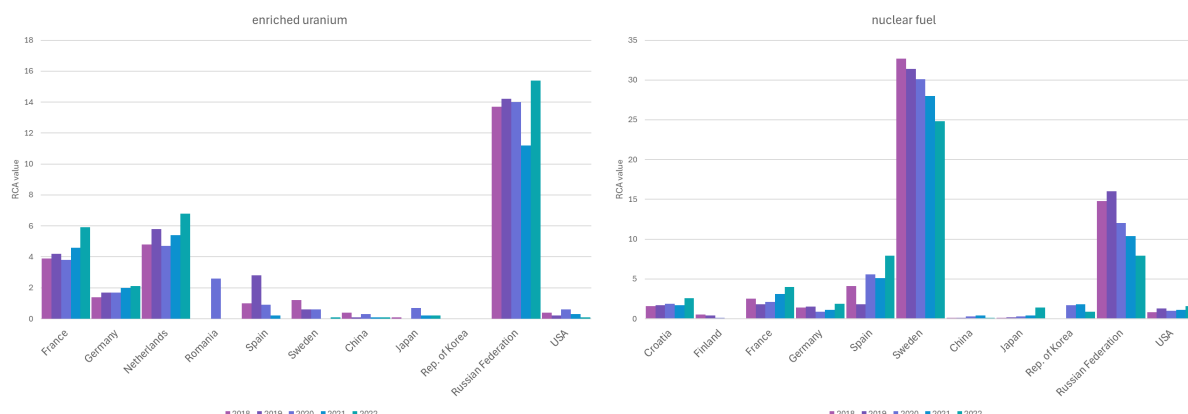
significant presence on the global stage. In contrast, the situation differs for the remaining three categories, where the competitive positioning of European countries varies.

**Figure 28:** RCA for nuclear commodities for selected countries



Source JRC 2024

**Figure 29:** RCA for nuclear commodities for selected countries



Source JRC 2024

For a more detailed assessment of the market, we refer to existing reports and the concept of Technology Progression Potential<sup>14</sup> (TPP) from the Economic Complexity framework. This metric estimates the probability that a country will be active in a given technology five years after the latest available data. "Active" is defined as the country's share in that technology exceeding its overall share of worldwide patents. TPP reflects a country's underlying technological capability. An analysis [96] indicates that nuclear fission capabilities are particularly strong in France and Germany, with Germany being the global leader. It further suggests that Italy, France, and Germany have the potential to maintain and strengthen their competitive positions in nuclear fission technology. We should emphasise the report's caveat regarding potential differences in subsets of nuclear fission, such as SMRs. (cf. [122]).

<sup>14</sup> For more details on the concept [140] and references therein



### 4.1.2. Market prospects

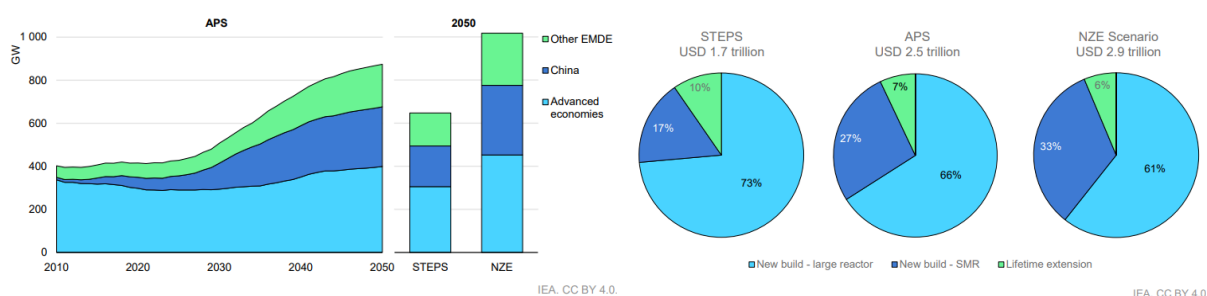
The European Union's nuclear power capacity is projected to grow from around 98 GW today to 109 GW by 2050, [23] an 11% increase, based on the intentions stated in Member States' National Energy and Climate Plans (NECPs). This projection assumes that existing reactors will either operate within their current design lifetimes or receive lifetime extensions, while new build projects outlined in the NECPs are completed as planned.

The European Union acknowledges the considerable potential for SMRs, even though none are currently operational within its territory. It is anticipated that the first SMRs will begin operation in the early 2030s. Projections suggest that deployed capacity within the Union could range from 17 GWe to 53 GWe by 2050, depending on the development scenarios, with half of this capacity dedicated to heat and hydrogen production. All scenarios highlight a rapid deployment acceleration after 2040, aligning with the broader rollout of AMRs. However, this growth depends on several critical factors, including a strong nuclear supply chain, a skilled workforce, large-scale public and private investments, and robust research capabilities.

Considering the "market value" of these plans up to 2050, it is estimated that investments totalling approximately EUR 241 billion, in present value terms, will be necessary. Within this scenario, new-build large-scale reactors are projected to require EUR 205 billion, while lifetime extensions will account for EUR 36 billion of these investments [23].

At global level, IEA [25] and IAEA [16] have developed various scenarios outlining expected installed capacities and investments in new-build large reactors and small modular reactors. The lower-bound scenario anticipate approximately 600 GW of installed capacity by 2050, requiring investments of around USD 1.7 trillion. Whereas the upper bound scenarios project installed capacities around 1000 GW with investment needs of USD 2.9 trillion ( cf Figure 30)

**Figure 30:** Estimated installed capacities and investment needs for three global scenarios<sup>15</sup>



Source: IEA,2025 [25]

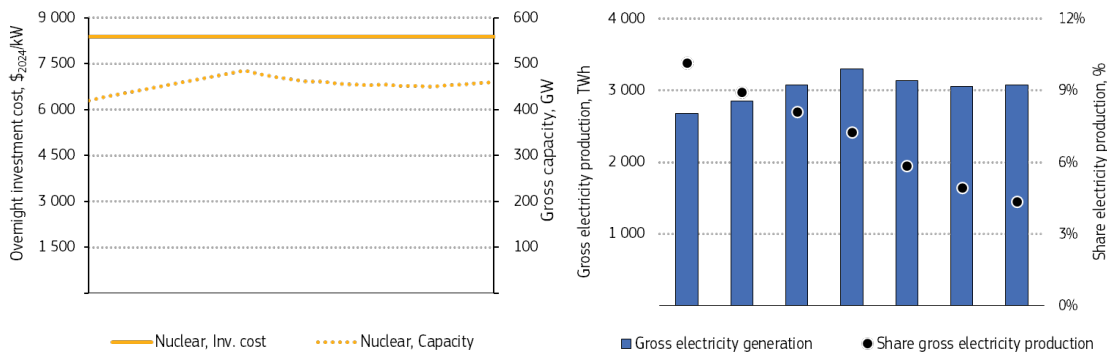
In contrast, JRC's *Global CETO 2°C scenario 2025*<sup>16</sup> generated by the POLES-JRC [123] model projects rather stagnating nuclear capacities until 2050 (Figure 31, left). The global energy model

<sup>15</sup> Details on the three scenarios STEPS (reflecting today's policy settings), APS (all climate target are met fully and in time) and NZE (a normative scenario reaching net zero emission by 2050) can be found in [25]

<sup>16</sup> A short description of the *Global CETO 2°C scenario 2025* and the POLES-JRC model can be found in Annex 3.

POLES-JRC considers two nuclear technologies: current uranium reactor technology (GEN II, GEN III) and GEN IV technology, assumed to become available from 2050 onwards. Nuclear power is used for both electricity generation and hydrogen production, powering low-temperature electrolysis with GEN II/III and high-temperature electrolysis with GEN IV. Small Modular Reactors (SMRs) are implicitly represented in POLES-JRC by the aforementioned two nuclear technologies. Moreover, POLES-JRC accounts for the uranium/plutonium fuel cycle including mining and reprocessing. Current nuclear capacities by country were updated in 2025, considering both existing and planned capacities. Regarding planned capacities, national nuclear energy policies, ranging from supportive to phase-out strategies, were taken into account. In terms of investment costs, POLES-JRC applies the same overnight costs for all nuclear power projects across countries. These investment costs are based on the International Energy Agency’s World Energy Outlook 2022 [124] for Europe and the USA, adjusted for inflation to USD2024. This cost assumption aligns closely with the PINC report’s [23] figure of 8 000 EURO/kW. Notably, the projected evolution of nuclear investment costs (Figure 31, left) remains constant until 2050 as POLES-JRC does not apply learning effects to nuclear technologies, unlike for other clean technologies [125]. The approach to keep future investment costs constant stems from recent cost escalations and the mixed views in literature regarding potential future cost reductions for nuclear energy.

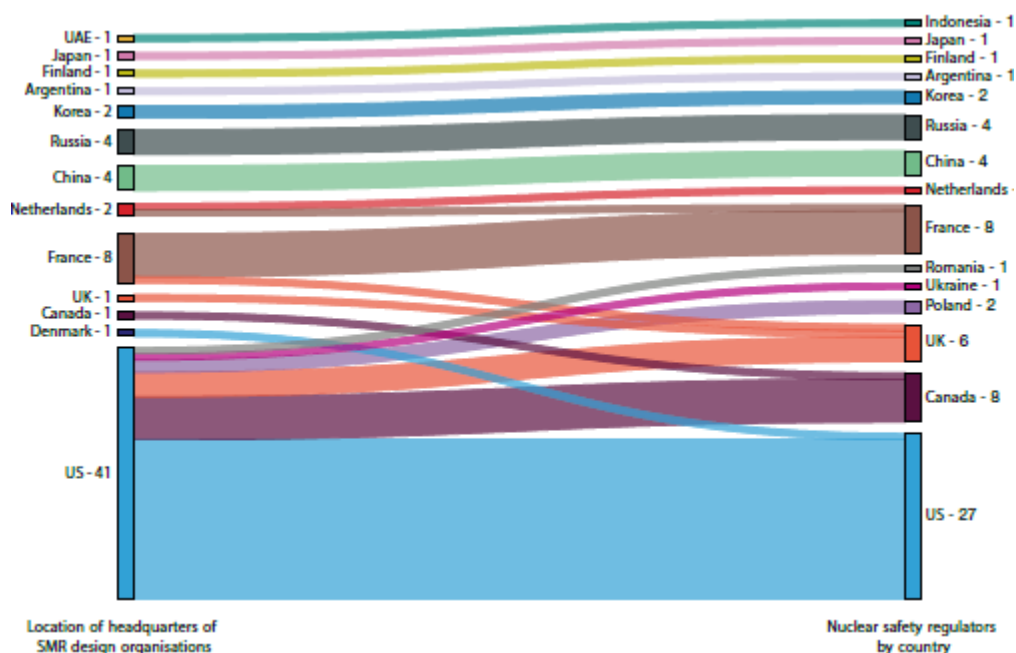
**Figure 31:** Global overnight investment cost and gross capacity (left) and Global electricity generation (right)



Source: POLES-JRC, 2025

A future model will include regional cost differentiation. Considering that in the present policy scenarios China and emerging and developing economies (EMDE) are driving the future global nuclear installed capacity, and as recent number suggest the building costs of nuclear power plants in those regions is lower than in the West, one could expect that the overall nuclear proportion in the energy mix will be increased, compared to the present model prediction.

**Figure 32:** Relation between country of headquarters of SMR design organisation and country of national safety regulators handling the pre-licensing request



Source: IEA, 2025, [49]

As commercial SMRs are not really deployed yet, an indicator for competitiveness can be the number of locations of pre-licensing request compared to the location of the headquarters of the SMR design organisations [49]. As depicted in Figure 32, presently a strong domestic orientation seems to be in place. However, US companies are actively seeking deployment opportunities outside their domestic market.

## 4.2. Trade (Import/export) and trade balance

As the global nuclear landscape continues to evolve, European providers encounter significant opportunities to participate in international nuclear projects. The globalisation of nuclear supply chains has made it easier for these providers to participate in such projects; however, export controls and regulatory restrictions can sometimes act as non-tariff trade barriers. Despite these challenges, there is currently no comprehensive set of indicators to monitor the nuclear industry's status regarding export and import balances, because the separation of components between nuclear and non-nuclear use is not always clearly defined.

In the first analysis we use the Nuclear Trade Atlas Data 2018-2002 [121]. We employ two distinct approaches in evaluating this data to gain a better understanding of the nuclear trade landscape. First, we examine the general dataset, focusing on items marked with high and highest relevance for the nuclear trade [126]. This approach provides an overview of the industry's broader trade activities. The data were aggregated to account for the trade flow in and out of the Union. In Figure 33, the left axis gives the trade volume and the right, the trade balance.

**Table 5:** trading commodities marked with this relevance for nuclear power applications

| HS relevance | HS code                   | Nuclear commodity             |
|--------------|---------------------------|-------------------------------|
| highest      | 2612.10                   | Uranium ores and concentrates |
| highest      | 2844.10                   | Natural uranium               |
| highest      | 2844.20                   | Enriched uranium; plutonium   |
| highest      | 8401.10, 8401.40          | Nuclear reactors and parts    |
| highest      | 8401.30                   | Fuel elements, non-irradiated |
| high         | 2612.20                   | Thorium ores and concentrates |
| high         | 2845.10                   | Heavy water                   |
| high         | 8112.12, 8112.13, 8112.19 | Beryllium                     |

Source: JRC 2023 [126]

**Figure 33:** Trade volume and balance of nuclear commodities ranked as “highest” and “high”, the right is excluding natural resources



Source JRC 2024

To better gauge the performance of the nuclear industry by excluding natural resources from the analysis. This more targeted dataset concentrates on specific HS codes: HS 8401.10 and HS 8401.40 for nuclear reactors and parts, HS 2844.20 for enriched uranium, and HS 8401.30 for non-irradiated fuel elements. By concentrating on these key components, we aim to provide a clearer picture of the industry's intrinsic capabilities, independent of its dependency on external raw materials.

A more detailed analysis is based on the Comext and Comtrade databases, where export values encompass all customs procedures, including re-exports, inward or outward processing, etc. The most recent data available is from 2024, which is subject to potential future revisions and should therefore be regarded as incomplete.

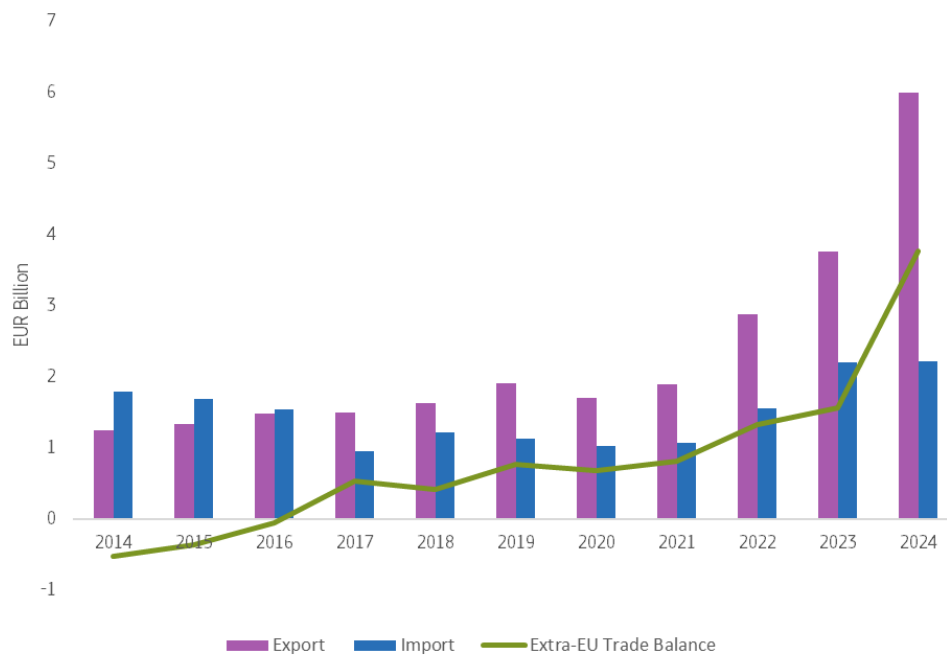
For monitoring trade in nuclear, the selection of trading goods is organised into two primary categories:

- Fuel: 284420 (uranium enriched in U 235 and its compounds: plutonium and its compounds; alloys, dispersions, including cermets, ceramic products, and mixtures containing uranium enriched in U 235, plutonium, or compounds of these products [Euratom]), 840130 (Fuel elements "cartridges," non-irradiated, in casing with handling fixtures, for nuclear reactors [Euratom])

- Equipment: 840120 (Machinery and apparatus for isotopic separation and parts thereof, not elsewhere specified [Euratom]), 840140 (Parts of nuclear reactors, not elsewhere specified [Euratom])

Figure 34 illustrates that since 2017 the EU maintains a trade surplus in nuclear fuel. In 2024, extra-EU imports remained stable at EUR 2.2 billion compared to 2023, while exports increased by 59% reaching almost EUR 6 billion. Notably, EU imports were composed by 68% of U235 and EU exports by 87%.

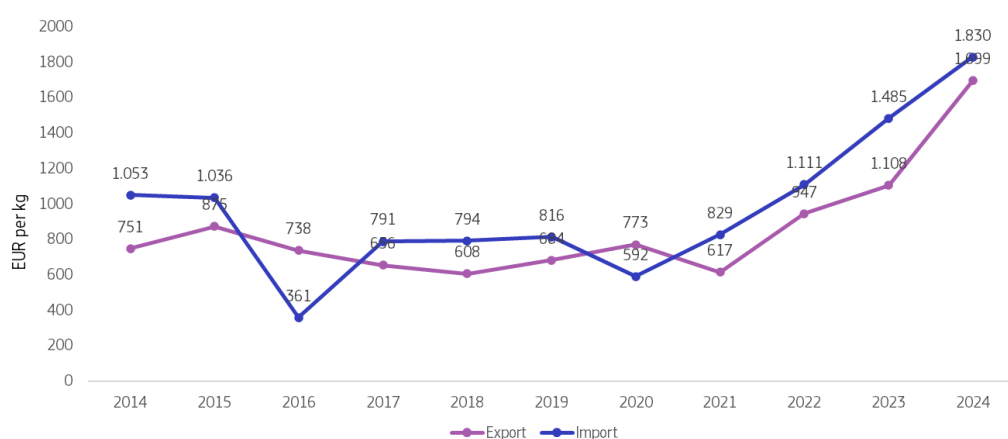
**Figure 34** Extra-EU trade of nuclear fuel for 2014-2024 [EUR Billion].



Source: JRC based on COMEXT data, 2025.

Figure 35 shows that the value per quantity of EU trade flows has an increasing trend since 2021. In 2024, Extra EU imports increased by 23%, with a value of EUR 1 830 per kilo, while exports increased by 53%, with a value nearly EUR 1 700 per kilo.

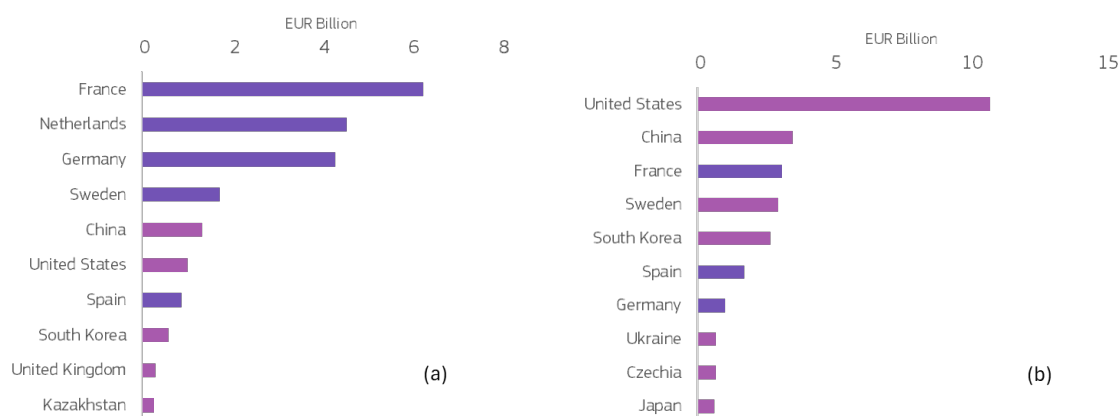
**Figure 35** Value per quantity in nuclear fuel trade flows for 2014-2024 [EUR per kilo].



Source: JRC based on COMEXT data, 2025.

Global exports of nuclear fuel increased from EUR 5.2 billion in 2022 to EUR 7.1 billion in 2023 to EUR 9.2 billion in 2024. Over the 2022–2024 period, the EU accounted for 83% of global exports (including intra-EU trade), while extra-EU exports (excluding intra-EU trade) represented 74% of global export transactions. The EU covered 44% of its importing needs with intra-EU trade. France was the largest global exporter for the same period, accounting for 29% of global exports, followed by the Netherlands (21%) and Germany (20%). The US, China and France were the leading global importers.

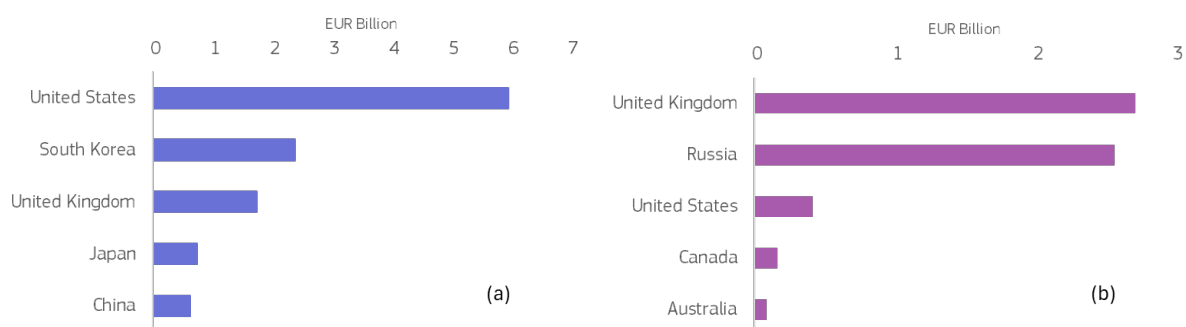
**Figure 36** Top ten global (a) exporters and (b) importers of nuclear fuel for 2022-2024 [EUR Billion]



Source: JRC, 2025, based on COMTRADE data

During the same period, the US was the largest importer from the EU, receiving 47% of the extra-EU exports, followed by South Korea and the UK with 19% and 14%, respectively. The UK was the largest exporter to the EU, accounting for 45% of extra-EU imports, followed by Russia (43%) (see Figure 37).

**Figure 37** Top five countries (a) importing from and (b) exporting to the EU for 2022-2024 [EUR Billion].



Source: JRC 2025, based on COMEXT data

Table 6 shows the growing markets<sup>17</sup> of nuclear fuel during 2021-2023<sup>18</sup>. The US had the largest net import increase, followed by China and South Korea, where the EU captured 48%, 31% and 11% of each country's growing market, respectively. The EU also secured expanding markets in South Africa (99%), Switzerland (92%), the UK (69%) and Saudi Arabia (49%).

**Table 6:** Growing markets of nuclear fuel based on a 2-year average of net import change

| Country        | Total import (2021-2023)<br>[EUR Million] | % import from<br>the EU | 2-year average of<br>net import change |
|----------------|-------------------------------------------|-------------------------|----------------------------------------|
| United States  | 8 360                                     | 48%                     | 1 393                                  |
| China          | 2 262                                     | 31%                     | 733                                    |
| South Korea    | 2 075                                     | 11%                     | 316                                    |
| Kazakhstan     | 513                                       | 3%                      | 251                                    |
| Türkiye        | 108                                       | 0%                      | 108                                    |
| United Kingdom | 152                                       | 69%                     | 37                                     |
| United Arab    | 111                                       | 0%                      | 30                                     |
| South Africa   | 106                                       | 99%                     | 26                                     |
| Switzerland    | 312                                       | 92%                     | 25                                     |
| Saudi Arabia   | 2                                         | 49%                     | 1                                      |

Source: JRC based on COMTRADE data

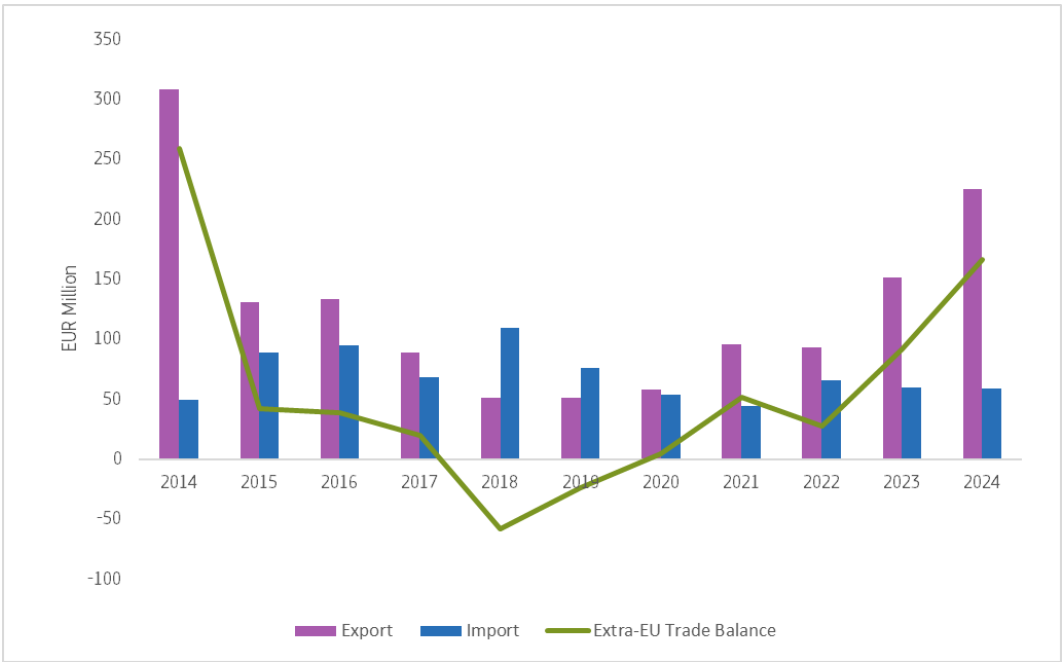
In 2024, the EU nearly doubled its nuclear equipment trade surplus, reaching EUR 166 million (Figure 39). This increase was driven by a 48% increase in extra-EU exports that reached

<sup>17</sup> Calculated as  $net\ import\ change = [(import_{2022} - import_{2021}) + (import_{2023} - import_{2022})]/2$

<sup>18</sup> Latest year data (2024) is incomplete for comtrade, because it does not provide estimates for the missing values as comext does.

EUR 225 million while extra-EU imports remained stable compared to 2023 at EU 59 million. Notably, EU imports composed 98% of parts (HS 840140) while exports 87%.

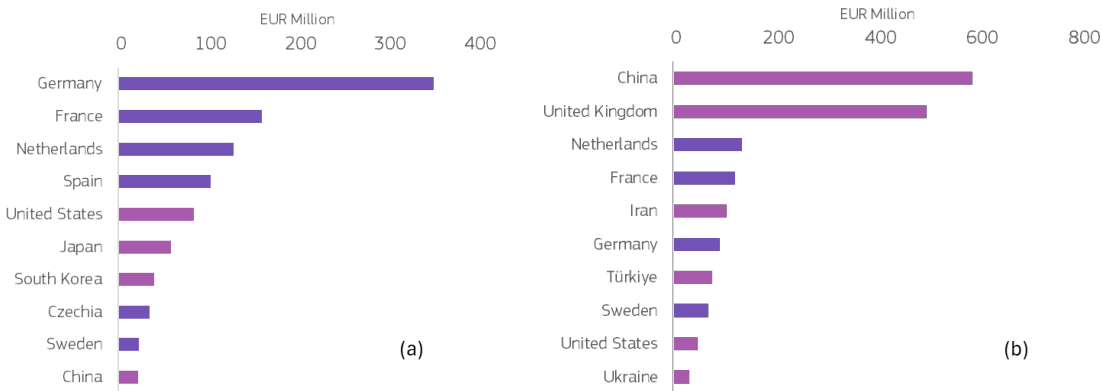
**Figure 38** Extra-EU trade of nuclear equipment for 2014-2024 [EUR Million]



Source: JRC based on COMEXT data, 2025.

Global exports of nuclear equipment increased from EUR 225 million in 2022 to EUR 341 million in 2023 to EUR 477 million in 2024. Over the 2022–2024 period, the EU accounted for 79% of global exports (including intra-EU trade), while extra-EU exports (excluding intra-EU trade) represented 65% of global export transactions. The EU covered 64% of its importing needs with intra-EU trade. Germany was the largest global exporter for the same period, accounting for 34% of global exports, followed by France with 15% and Netherlands with 12%. China, the US and Netherlands were the leading global importers Figure 39.

**Figure 39:**Top ten global (a) exporters and (b) importers of nuclear equipment for 2022-2024 [EUR Million].

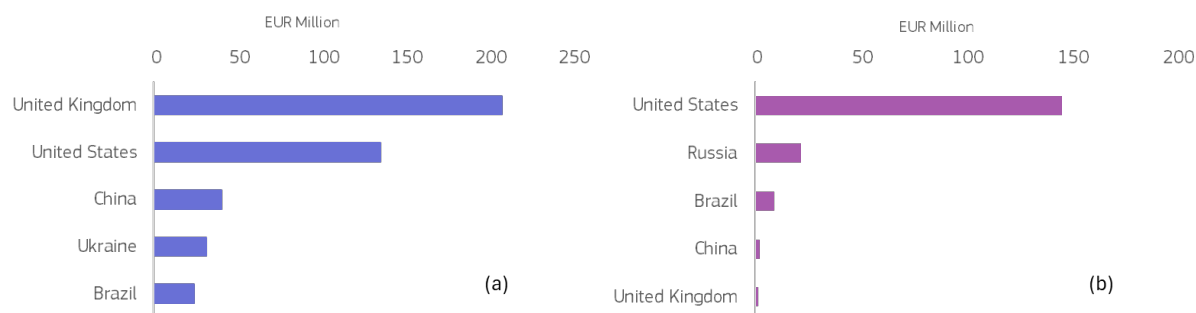


Source: JRC based on COMTRADE data, 2025.



During the same period, the UK was the largest importer from the EU, receiving 44% of the extra-EU exports, followed by the US and China with 29% and 9%, respectively. The US was the largest exporter to the EU, accounting for 79% of extra-EU imports, followed by Russia with 12% (Figure 40).

**Figure 40:**Top five countries (a) importing from and (b) exporting to the EU for 2022-2024 [EUR Million].



Source: JRC, 2025, based on COMEXT data

Table 7 shows the growing markets<sup>19</sup> of nuclear equipment during 2021-2023<sup>20</sup>, China had the largest net import increase, followed by the UK and Brazil, where the EU captured 5%, 75% and 85% of each country's growing market, respectively. The EU also secured expanding markets in Kazakhstan (100%) and Thailand (99%).

**Table 7:**Growing markets of nuclear equipment based on a 2-year average of net import change

| Country        | Total import (2021-2023) [EUR Million] | % import from the EU | 2-year average of net import change |
|----------------|----------------------------------------|----------------------|-------------------------------------|
| China          | 303                                    | 5%                   | 108                                 |
| United Kingdom | 372                                    | 75%                  | 43                                  |
| Brazil         | 11                                     | 85%                  | 3                                   |
| Kazakhstan     | 5                                      | 100%                 | 3                                   |
| Thailand       | 7                                      | 39%                  | 1                                   |
| Indonesia      | 7                                      | 1%                   | 1                                   |

Source: JRC based on COMTRADE data

<sup>19</sup> Calculated as  $net\ import\ change = [(import_{2022} - import_{2021}) + (import_{2023} - import_{2022})]/2$

<sup>20</sup> Latest year data (2024) is incomplete for comtrade, because it does not provide estimates for the missing values as comext does.

### **4.3. Resource efficiency and dependence in relation to EU competitiveness**

The nuclear supply chain in Europe is generally well-developed; however, there are potential bottlenecks, particularly in areas like large-scale forging. If the demand for SMRs increases rapidly, there could also be challenges in obtaining mechanical equipment such as nuclear-grade valves. With those exceptions, Europe's nuclear industry does not critically depend on external countries for the production of essential nuclear materials.

Nuclear energy projects utilise a wide range of materials, from common ones like concrete, steel, and copper to specialised materials essential for specific functions. These specialised materials include certain rare earth elements, boron, zirconium, and specific alloys. In fact, concrete and steel account for the majority of the materials needed for constructing a nuclear plant [23]. Nuclear-grade concrete requires a rare, specific type of sand. At the same time, Europe is losing its steel manufacturing and processing capabilities against Chinese competition. The price of copper and other key materials is subject to high volatility. This illustrates that even common materials require attention to ensure their availability in a secure European value chain.

While nuclear energy is known for its lower mineral demands compared to other low-carbon technologies, similar to hydropower [127], it still requires specific minerals such as chromium, copper, nickel, hafnium, and yttrium [128]. Rare earth elements are primarily used in reactor control applications. For example, samarium-149 is an effective neutron absorber for control rods; europium is valued for its low melting point and neutron absorption capabilities; and gadolinium is used for its effectiveness in neutron radiography and for enhancing the durability of reactor shielding. However, the reliance on rare earth elements in nuclear reactors is relatively minor compared to other clean energy technologies [129].

The use of high-alloy steels in nuclear plants significantly influences the demand for chromium and nickel. Still, precise data on the quantities of different steel alloys used is rarely available. Given the maturity of nuclear technology, a significant reduction in mineral demand is unlikely in the near future. . Notably, only one of these critical elements, hafnium, is classified as critical under the 2017 Critical Raw Materials Act, with France being the primary global producer. Some additional demands might be created by new concepts and new designs

There are no clear indicators that key materials used within the nuclear supply chain will be inaccessible in the near term. Nonetheless, challenges may arise in accessing some materials in a cost-competitive and timely manner at required volumes, due to geopolitical and other factors.

Despite the scarcity of its domestically available natural uranium, the EU has developed a significant nuclear industry, thanks also to the resources of the Euratom Treaty. The common supply policy can be a precious tool to address the challenges lying ahead on the way to a clean-energy future.

## 5. Conclusions

Nuclear energy remains a vital component of the energy mix in Europe today, providing a significant share of the continent's electricity. As of recent data, approximately 98 GW of nuclear power capacity is installed in the European Union. This capacity plays a crucial role in delivering low-carbon electricity, contributing significantly to Europe's climate goals. The nuclear industry is a well-established sector in the EU, contributing approximately EUR 250 billion annually to the economy and supporting around 500 000 jobs. This mature supply chain encompasses construction, operation, maintenance, decommissioning, and waste management for existing nuclear plants, ensuring a reliable energy source and economic stability.

Looking ahead, the EU is focused on several strategic areas to maximise the potential of nuclear energy. Long-term operation of existing reactors is crucial to maintaining a low-carbon supply while meeting climate objectives. But so are new reactor builds, using established technologies, to maintain and even expand the EU's nuclear capacity. Member States are actively planning and proposing new large nuclear power plant projects, with varying degrees of progress. A strategic priority for the EU is the development and deployment of next-generation nuclear technologies, such as Small Modular Reactors (SMRs) and Advances Modular Reactors (AMRs). These innovative reactors are expected to begin operations in the early 2030s, with a projected capacity ranging from 17 GWe to 53 GWe by 2050. SMRs particularly offer advantages in integrated energy systems with high renewable penetration, providing flexible generation and enhancing system efficiency.

However, transitioning to these new technologies is not without challenges. While nuclear energy production costs are competitive, new plant builds require significant initial capital investments, and the supply chain for these projects needs strengthening. Fuel supply sourcing, notably natural uranium, is another area of focus, with efforts to establish secure supply chains through the Euratom Supply Agency. Human resources are critical; a skilled workforce is essential to realise future nuclear projects, and training programmes must keep pace with technological advancements.

Europe is actively advancing SMR research and development, with numerous start-ups and established companies contributing to this effort. To support this, policymakers are creating a supportive ecosystem for SMR innovation and deployment. Key initiatives include the formation of an industrial alliance for SMRs, which aims to streamline collaboration between stakeholders. This alliance is also working with national regulators towards harmonised licensing requirements early in the design process, reducing delays and costs. Additionally, new policies to simplify access to venture capital for European start-ups, helping them scale their technologies are required and envisaged.

Collaboration between traditional nuclear firms and agile start-ups is vital to delivering the clean, affordable, low-carbon energy solutions Europe needs. By combining the expertise of experienced organisations with the innovation of emerging companies, the sector can accelerate the development of SMRs to meet climate goals and energy demands.

However, the research and development needs across all aspects of nuclear fission technology—spanning the front end of the fuel cycle (e.g., advanced nuclear fuels), reactor technology, and the back end (spent fuel reprocessing and long-term waste management, including partitioning and transmutation)—must not be overlooked. While the back end was beyond the scope of this report, it remains a critical area. The safe, long-term storage of nuclear waste is a central concern in any

discussion of the risks and benefits of nuclear energy. Meanwhile, reprocessing spent nuclear fuel could significantly enhance the EU's resource and energy independence.

With strategic investment, robust policy frameworks, and strong collaboration across stakeholders, the nuclear industry can—and will—achieve a strong position in the future. However, companies must navigate intense global competition, not only for traditional nuclear power plants but also for SMRs. This competition will test the EU's ability to maintain, or in some domains establish, technological autonomy in the nuclear sector.

Ultimately, alongside economic and technological considerations, public acceptance will be a decisive factor in determining how significant nuclear energy's contribution to the energy mix can be in addressing energy security and climate change.

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## List of abbreviations and definitions

| Abbreviations | Definitions                                         |
|---------------|-----------------------------------------------------|
| AMR           | advanced modular reactor                            |
| BWR           | boiling water reactor                               |
| EDF           | Électricité de France                               |
| EPC           | Engineering, Procurement, and Construction services |
| ETIP          | European Technology and Innovation Platform         |
| GWe           | Gigawatt electric                                   |
| HALEU         | high-assay low-enriched uranium                     |
| HTGR          | high temperature gas cooled reactor                 |
| iPWR          | Integrated pressurised water reactor                |
| KHNP          | Korea Hydro & Nuclear Power                         |
| LCOE          | Levelised cost of electricity                       |
| LMR           | Liquid metal cooled reactor                         |
| LTO           | Long term operation                                 |
| LWR           | light water reactor                                 |
| MOX           | Mixed oxide                                         |
| MSR           | Molten salt reactor                                 |
| MWe           | Mega Watt electric                                  |
| MWth          | Mega Watt thermal                                   |
| NPP           | nuclear power plant                                 |
| OEM           | original equipment manufacturer                     |
| PINC          | nuclear Illustrative Programme                      |

| <b>Abbreviations</b> | <b>Definitions</b>                               |
|----------------------|--------------------------------------------------|
| PWR                  | pressurised water reactor                        |
| PHWR                 | pressurised heavy wate reactor                   |
| RCA                  | revealed comparative advantage                   |
| RPV                  | reactor pressure vessel                          |
| SET                  | Strategic energy technology                      |
| SMR                  | small modular reactor                            |
| SNETP                | Sustainable nuclear energy technology platform   |
| SSC                  | structures, systems, and components              |
| STEM                 | science, technology, engineering and mathematics |
| TPP                  | technology progression potential                 |
| tSWU                 | Ton separative work unit                         |
| tU                   | tons uranium                                     |

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## **Annex 1: Small Modular Reactors (LW-SMRs and AMRs) European projects Co-funded by the Euratom Research and Training Programme**

### **ELSMOR**

Towards European Licencing of Small Modular Reactors

Duration 48 months, Start date 1 September 2019, End date 31 August 2023

EC contribution: EUR 3.5 million - Overall budget EUR 4.3 million

<https://cordis.europa.eu/project/id/847553>

### **McSAFER**

High-Performance Advanced Methods and Experimental Investigations for the Safety

Evaluation of Generic Small Modular Reactors

Duration 42 months, Start date 1 September 2020 End date 29 February 2024

EC contribution: EUR 4 million - Overall budget EUR 4 million

<https://cordis.europa.eu/project/id/945063>

### **SASPAM-SA**

Safety Analysis of SMR with Passive Mitigation strategies – Severe Accident

Duration 48 months, Start date 1 October 2022, End date 30 September 2026.

EC contribution : EUR 3 million Overall budget EUR 4.3 million

<https://cordis.europa.eu/project/id/101059853>

### **EASI-SMR**

Ensuring Assessment of Safety Innovations for SMR

Duration 48 months, Start date 1 September 2024, End date 31 August 2028

EC contribution: EUR 15 million - Overall budget: EUR 20 million

<https://cordis.europa.eu/project/id/101164810>

## **SANE**

Safety Assessment of Non-Electric uses of nuclear energy

Duration 36 months, Start date 1 September 2024, End date 31 August 2027

EC contribution: EUR 3.5 million - Overall budget: EUR 3.6 million

<https://cordis.europa.eu/project/id/101163929>

## **ECC-SMART**

Joint European-Canadian-Chinese Development of Small Modular Super-Critical  
Water-cooled Reactor Technology

Duration 52 months, Start date 1 September 2020 End date 28 February 2025

EC contribution: EUR 4 million - Overall budget EUR 9 million

<https://cordis.europa.eu/project/id/945234>

## **GEMINI For Zero Emission**

Nuclear cogeneration, polygeneration, decarbonisation, high temperature reactor

Duration 36 months, Start date 1 June 2022, End date 31 May 2025

EC contribution: EUR 2.9 million – Overall budget EUR 3.5 million

<https://cordis.europa.eu/project/id/101059603>

## **TREASURE**

Towards reliable and safe GFR

Duration 48 months, Start date 1 October 2024, End date 30 September 2028

EC contribution: EUR 4 million - Overall budget EUR 4.1 million

<https://cordis.europa.eu/project/id/101164616>

## **LESTO**

Lead fast reactor safety design and tools

Duration 48 months, Start date 1 November 2024, End date 31 October 2028

EC contribution: EUR 4 million -Overall budget EUR 4.7 million

<https://cordis.europa.eu/project/id/101166337>

## **ANSELMUS**

Advanced Nuclear Safety Evaluation of Liquid Metal Using Systems

Duration 48 months, Start date 1 September 2022, End date 31 August 2026

EC contribution: EUR 3.5 million - Overall budget EUR 4.5 million

<https://cordis.europa.eu/project/id/101061185>

## **ESFR-SIMPLE**

European Sodium Fast Reactor - Safety by Innovative Monitoring, Power Level flexibility and Experimental research

Duration 48 months, Start date 1 October 2022, End date 30 September 2026

EC contribution: EUR 3 million -Overall budget: EUR 6.5 million

<https://cordis.europa.eu/project/id/101059543>

## **MIMOSA**

Multi-recycling strategies of LWR SNF focusing on Molten Salt technology

Duration 48 months, Start date 1 June 2022, End date 31 May 2026

EC contribution: EUR 5.7 million - Overall budget: EUR 6.9 million

<https://cordis.europa.eu/project/id/101061142>

## **ENDURANCE**

EU knowledge hub for enabling molten salt reactor safety development and deployment

Duration 48 months, Start date 1 October 2024, End date 30 September 2028

EC contribution: EUR 4 million - Overall budget : EUR 5.2 million

<https://cordis.europa.eu/project/id/101165896>

## **TANDEM**

Small Modular Reactor for a European safe and Decarbonised Energy Mix

Duration 36 months, Start date 1 September 2022, End date 31 August 2025

EC contribution: EUR 3.

4 million Overall budget: EUR 3. 8 million

<https://cordis.europa.eu/project/id/101059479>



## **HARMONISE**

Towards harmonisation in licensing of future nuclear power technologies in Europe

Duration 36 months, Start date 1 June 2022, End date 31 May 2025

EC contribution: EUR 2.5 million - Overall budget: EUR 2.8 million

<https://cordis.europa.eu/project/id/101061643>

## **Annex 2: Energy System Model and Scenario:**

### **Global CETO 2°C Scenario 2025**

The Global CETO 2°C Scenario 2025 has been generated by the global energy model POLES-JRC (Prospective Outlook for the Long-term Energy System).

POLES-JRC covers the entire global energy system with a high level of regional detail, encompassing 66 countries and regions, from primary supply (fossil fuels, renewables) to transformation (power, biofuels, hydrogen, and hydrogen-derived fuels, such as e-fuels and ammonia) and final consumption sectoral demand (industry, buildings, transport). International markets and energy fuel prices are calculated endogenously. The model comprises a comprehensive portfolio of technologies, with technology dynamics and interactions across sectors modelled using endogenous technology learning.

Detailed documentation of the POLES-JRC model is provided in [1]. Techno-economic assumptions used in the current version of the model are provided in [2]. The latter report provides also a comprehensive overview of the dynamics and interaction of various clean energy technologies until the end of the century. POLES-JRC results are published in the annual "Global Climate and Energy Outlooks" (GECO) report, which provides detailed country energy and GHG balances, and an online visualisation interface.

The Global CETO 2°C scenario 2025 is designed to limit global temperature increase to 2°C at the end of the century. It is driven by a single global carbon price. The Global CETO 2°C scenario 2025 builds on the POLES-JRC model version used for GECO 2024 [3] and has been enhanced as following:

- Cost optimisation for electrolyzers powered by renewables (PV, wind) has been implemented considering an over-sizing of renewable capacities and battery storage.
- Electrolyser investment costs have been increased reflecting recent literature revisions.
- Updated investment costs for renewable technologies and utility battery according to [4] as well as updated installed capacities for power generating technologies.
- Revised global wind profiles (off-shore and on-shore).
- Updated data on new vehicles and vehicle stock by transport mode (battery and fuel cell vehicles, ICE, hybrid) and vehicle type (passenger cars, light and heavy trucks, buses).
- Updated hydrogen infrastructure cost related to road transport.

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